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(54) **SEMICONDUCTOR STRUCTURE AND METHOD FOR FORMING THE SAME**

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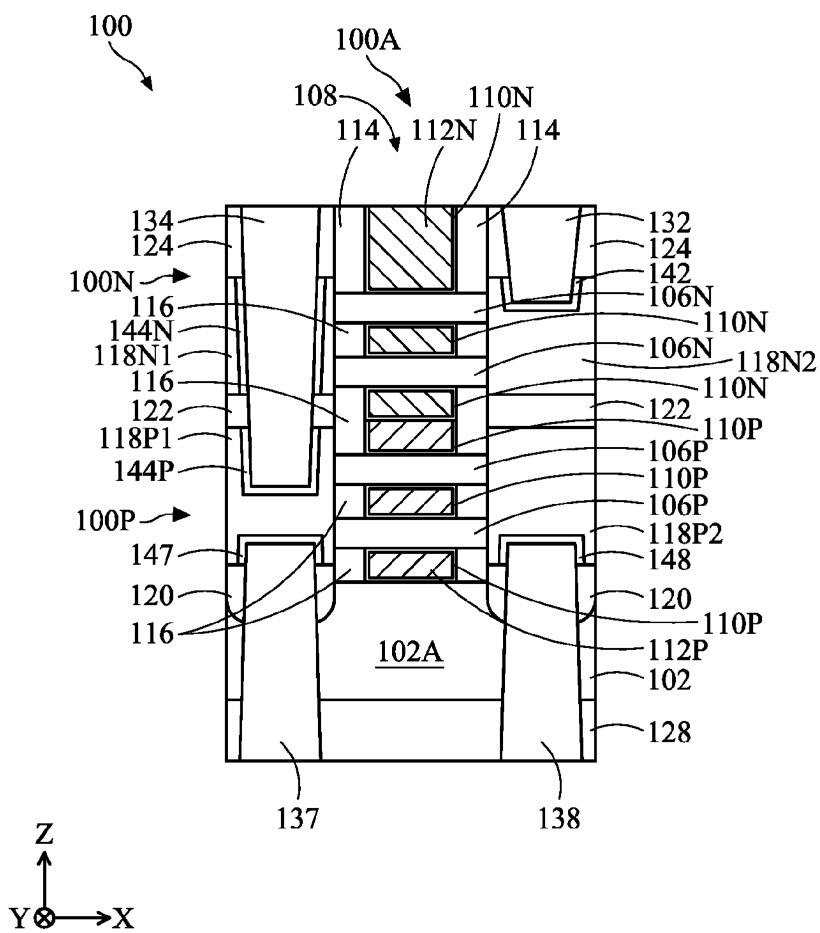
H01L 29/06	(2006.01)
H01L 29/423	(2006.01)
H01L 29/66	(2006.01)
H01L 29/775	(2006.01)
H01L 29/786	(2006.01)

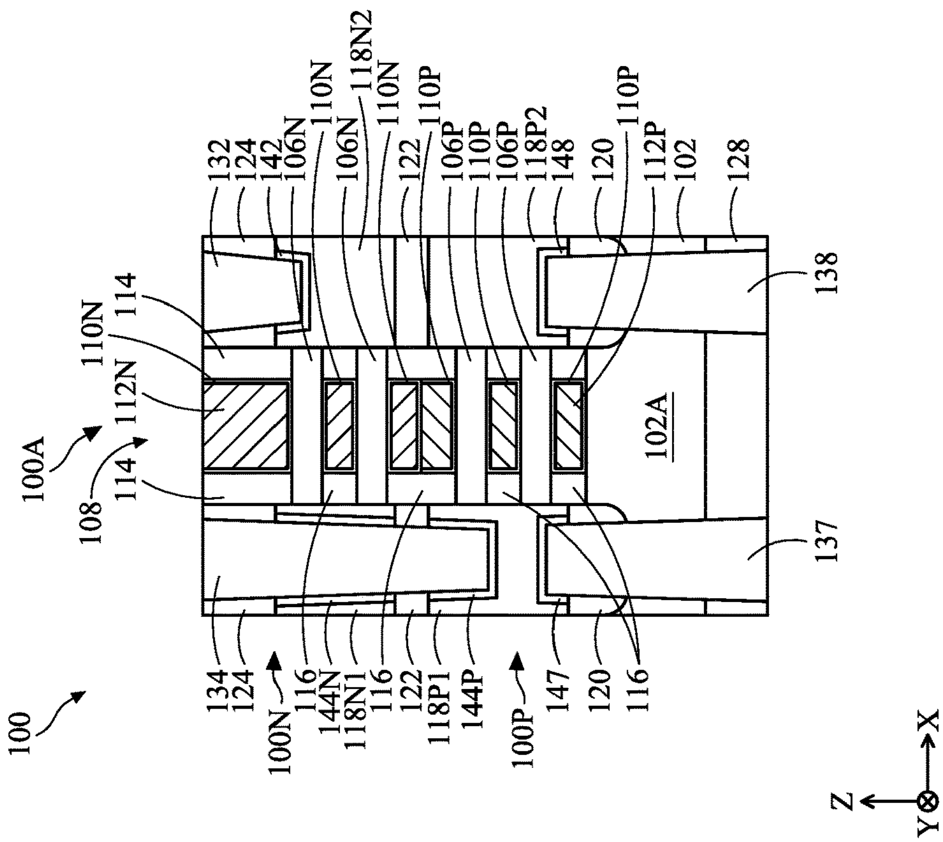
(52) **U.S. Cl.**

CPC ***H10D 30/6729*** (2025.01); ***H10D 30/014*** (2025.01); ***H10D 30/43*** (2025.01); ***H10D 30/6735*** (2025.01); ***H10D 30/6757*** (2025.01); ***H10D 62/121*** (2025.01); ***H10D 64/017*** (2025.01); ***H10D 84/017*** (2025.01); ***H10D 84/038*** (2025.01); ***H10D 84/85*** (2025.01)

(57) **ABSTRACT**

A semiconductor structure includes a first transistor and a second transistor over the first transistor. The first transistor includes first nanostructures spaced apart from each other in a Z-direction, and first second source/drain features attached to opposite sides of the first nanostructures in an X-direction. The second transistor includes second nanostructures over the first nanostructures and spaced apart from each other in the Z-direction; and third and fourth source/drain features, attached to opposite sides of the second nanostructures in the X-direction and vertically overlapping the first and second source/drain features, respectively. The semiconductor structure further includes a gate structure that is wrapped around the first nanostructures and the second nanostructures. The semiconductor structure further includes a first source/drain contact, extending through the third source/drain feature and partially extending into the first source/drain feature; and a second source/drain contact extending into the first source/drain feature.

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110N
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118N2
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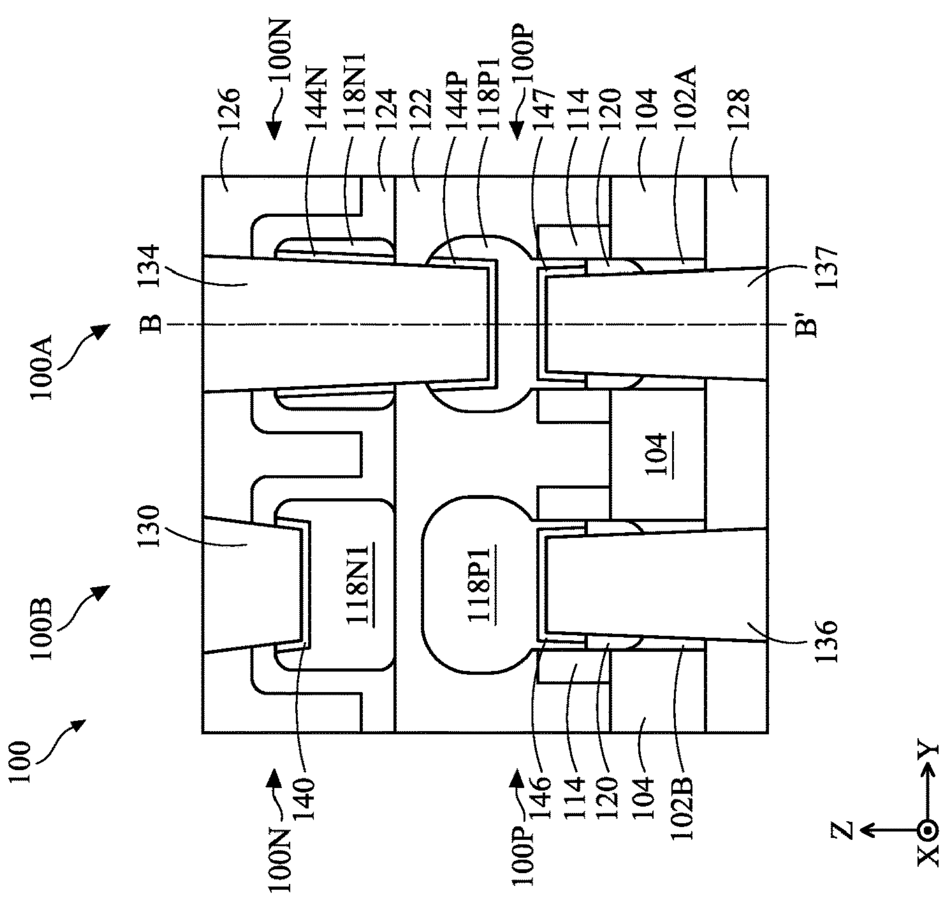
110P
106P
110P
106P
118P2
148
120
110P
112P
102
128
138
102A
137
134
124
100N
116
144N
118N1
116
122
118P1
144P
100P
147
120
116
Z
X
Y

FIG. 1B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100B
100A
B
B'
134
130
126
100N
144N
118N1
124
122
144P
118P1
100P
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114
120
104
102A
128
137
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118N1
118P1
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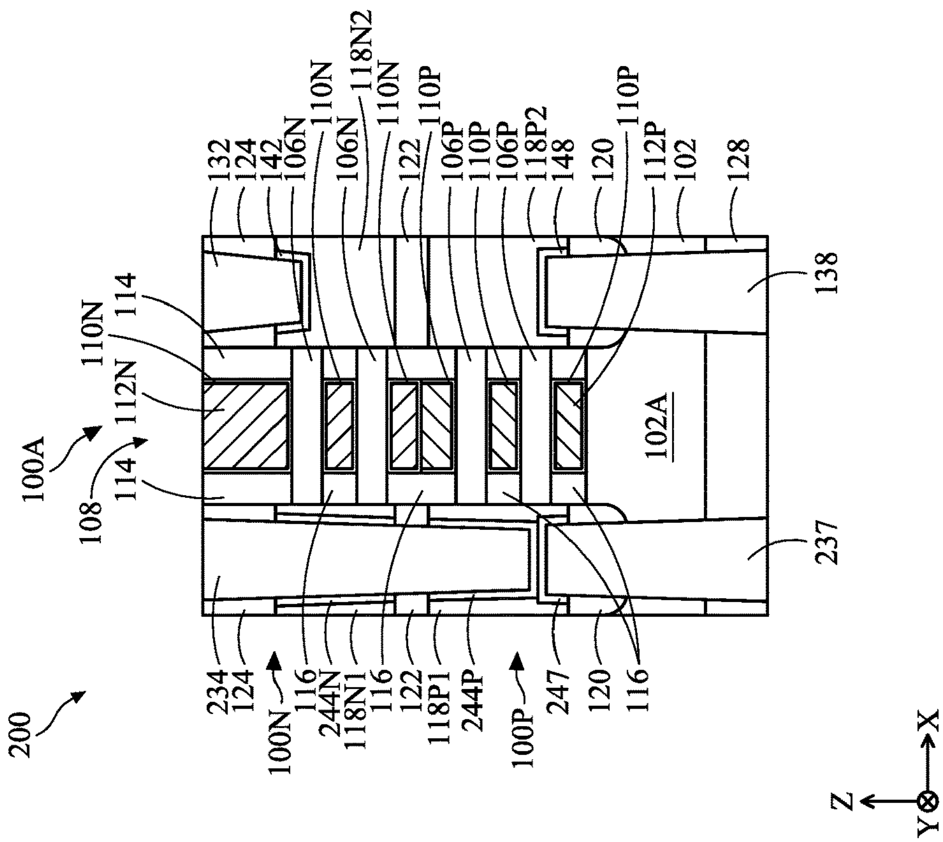
100N
140
100P
146
114
120
104
102B
Z
Y
X

FIG. 1A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100A
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112N
110N
114
132
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142
106N
110N
106N
110N
118N2
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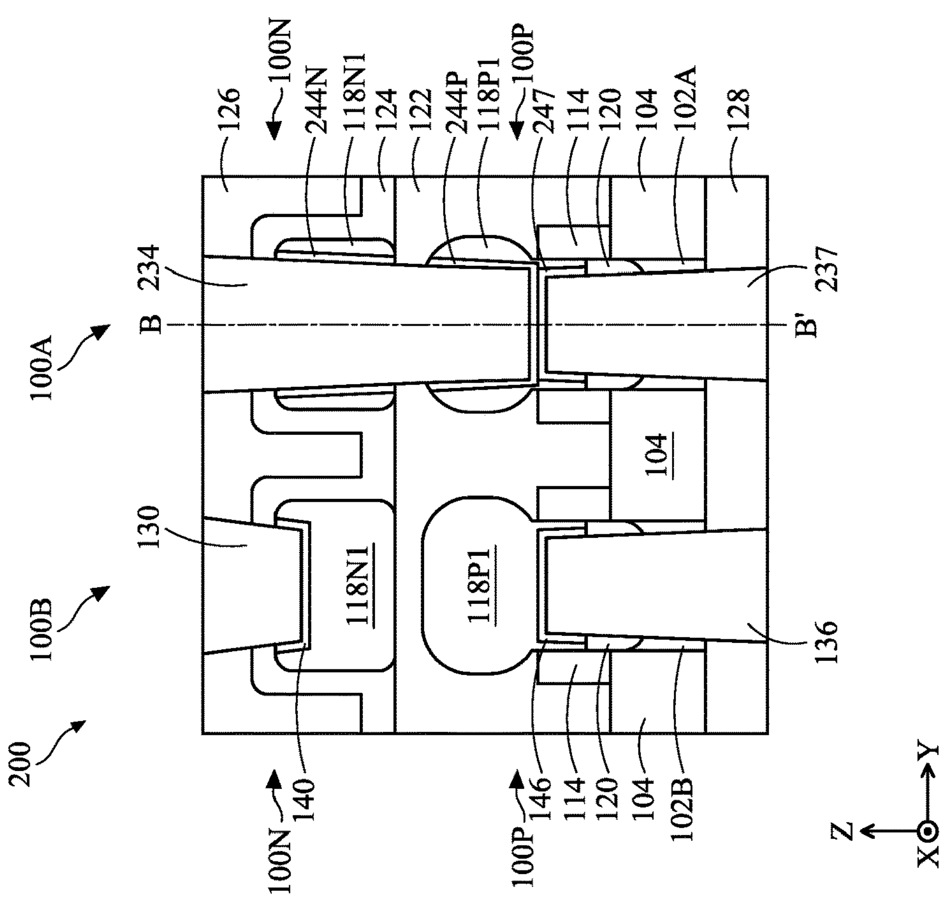
110P
106P
110P
106P
118P2
148
120
110P
112P
102
128
138
102A
237
234
124
100N
116
244N
118N1
116
122
118P1
244P
100P
247
120
116
Z
Y
X

FIG. 2B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130
B
234
B'
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100N
244N
118N1
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244P
118P1
100P
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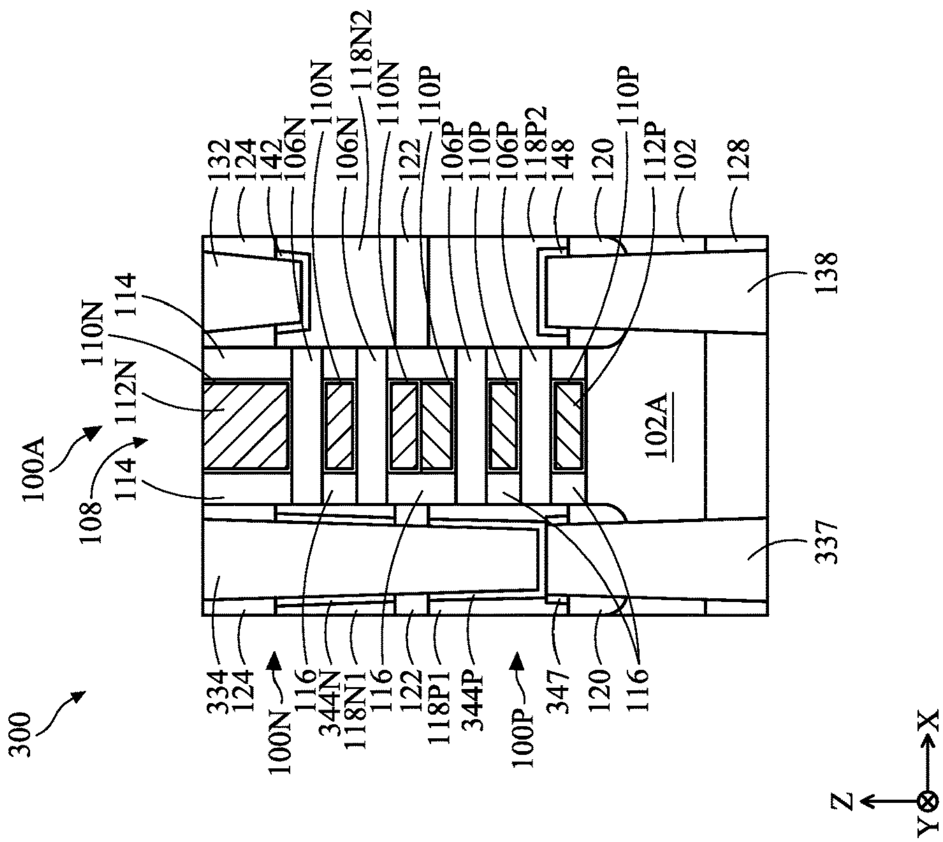
114
120
104
102A
128
237
118N1
118P1
104
136
100N
140
100P
146
114
120
104
102B
Z
Y
X

FIG. 2A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114
132
124
142
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110N
106N
110N
118N2
122
110P
106P
110P
106P
118P2
148
120
110P
112P
102
128
138
102A
337
334
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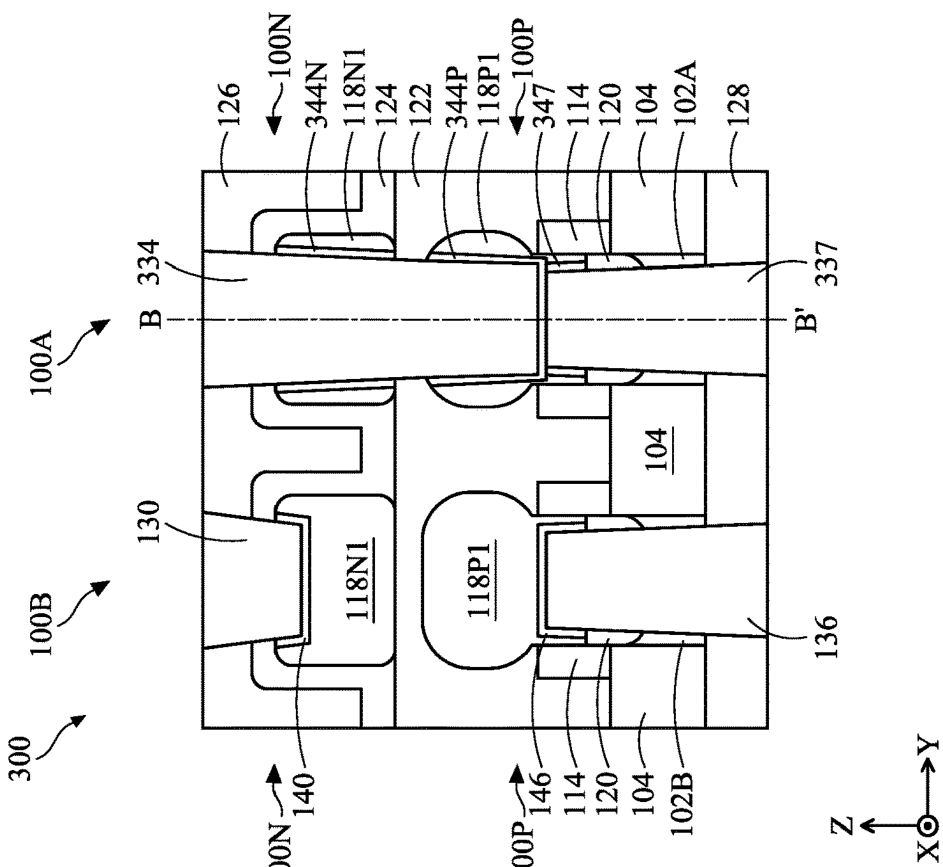
100N
116
344N
118N1
116
122
118P1
344P
100P
347
120
116
Z
Y
X

FIG. 3B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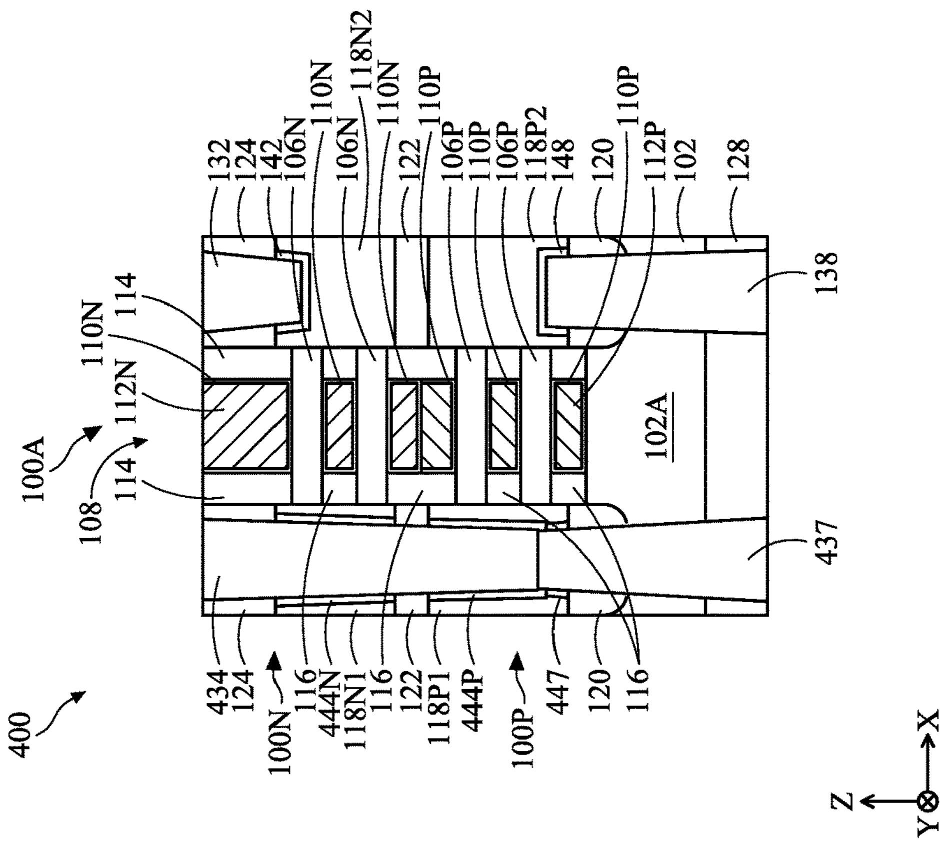
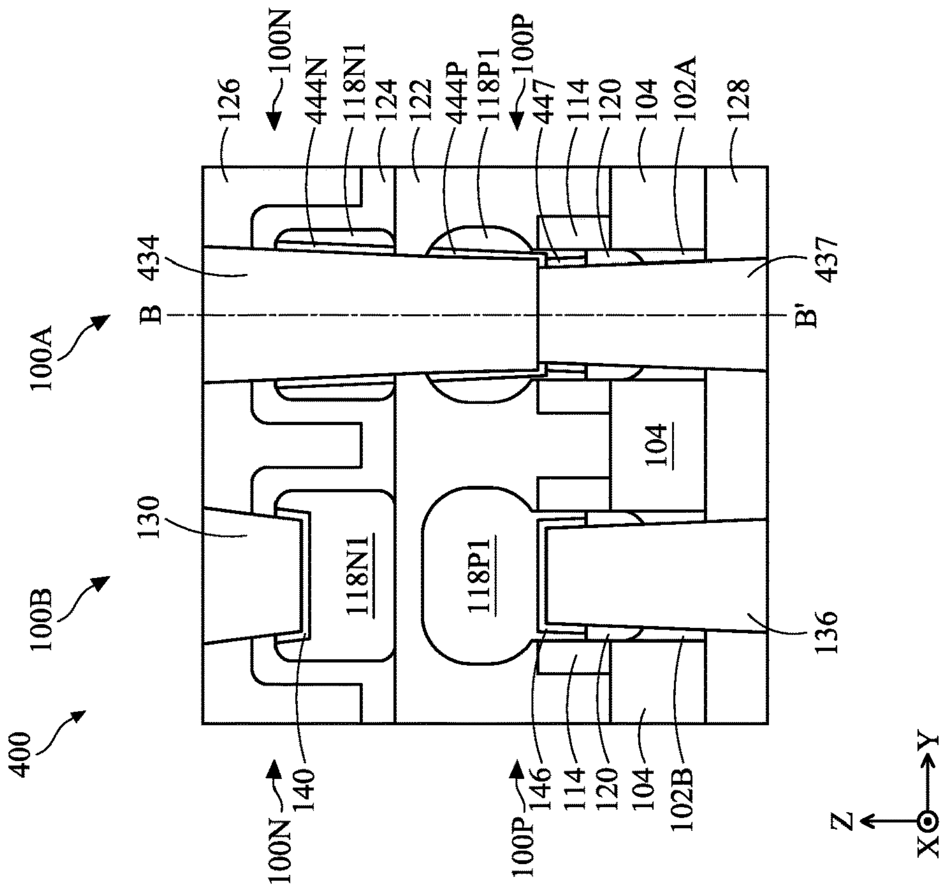
100B
100A
130
334
B
B'
126
100N
344N
118N1
124
122
344P
118P1
100P
347
114
120
104
102A
128
337
104
118N1
118P1
136
100N
140
100P
146
114
120
104
102B
Z
X
Y

FIG. 3A


400
100A
108
114
112N
110N
114
132
124
142
106N
110N
106N
110N
118N2
122
106P
110P
110P
106P
110P
106P
118P2
148
120
110P
112P
102
128
138
102A
437
434
124
100N
116
444N
118N1
116
122
118P1
444P
100P
447
120
116
Z
Y
X
FIG. 4B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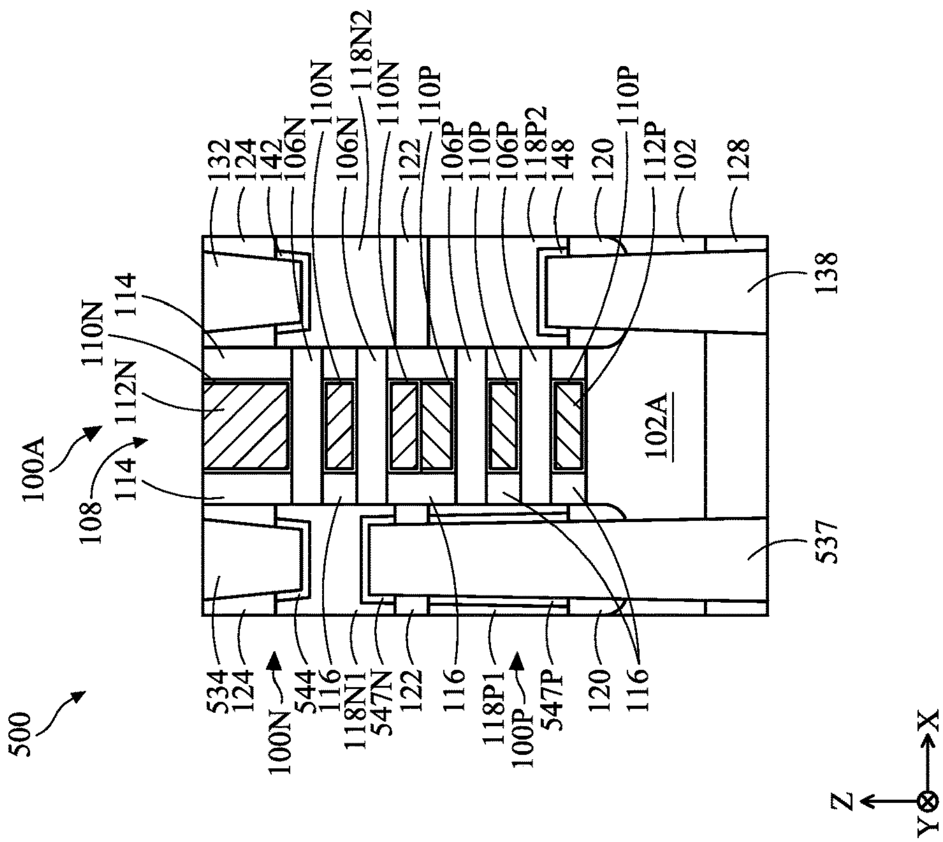
100B
100A
B
434
B'
437
130
118N1
118P1
104
136
126
100N
444N
118N1
124
122
444P
118P1
100P
447
114
120
104
102A
128
100N
140
100P
146
114
120
104
102B
Z
X
Y
FIG. 4A


500
100A
108
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112N
110N
114
132
124
142
106N
110N
106N
118N2
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110P
106P
110P
106P
118P2
148
120
110P
112P
102
128
138
102A
537
534
124
100N
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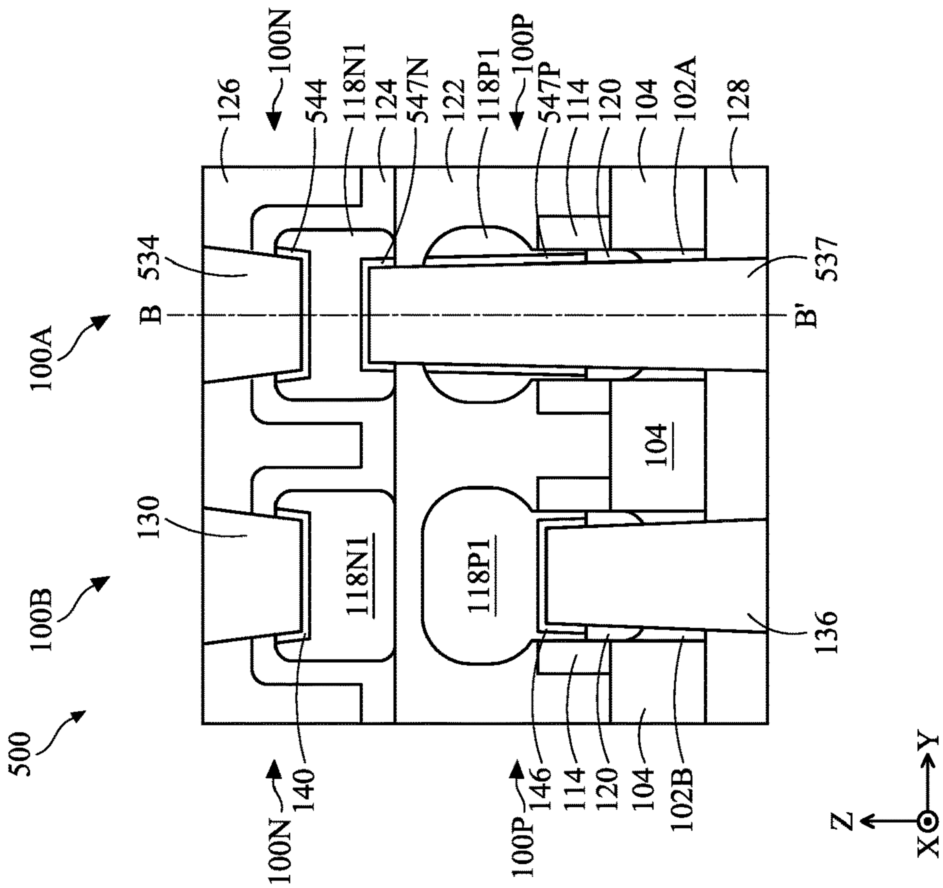
116
118N1
547N
122
116
118P1
100P
547P
120
116
Z
X
Y

FIG. 5B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100A
B
B'
100B
126
100N
544
118N1
124
547N
122
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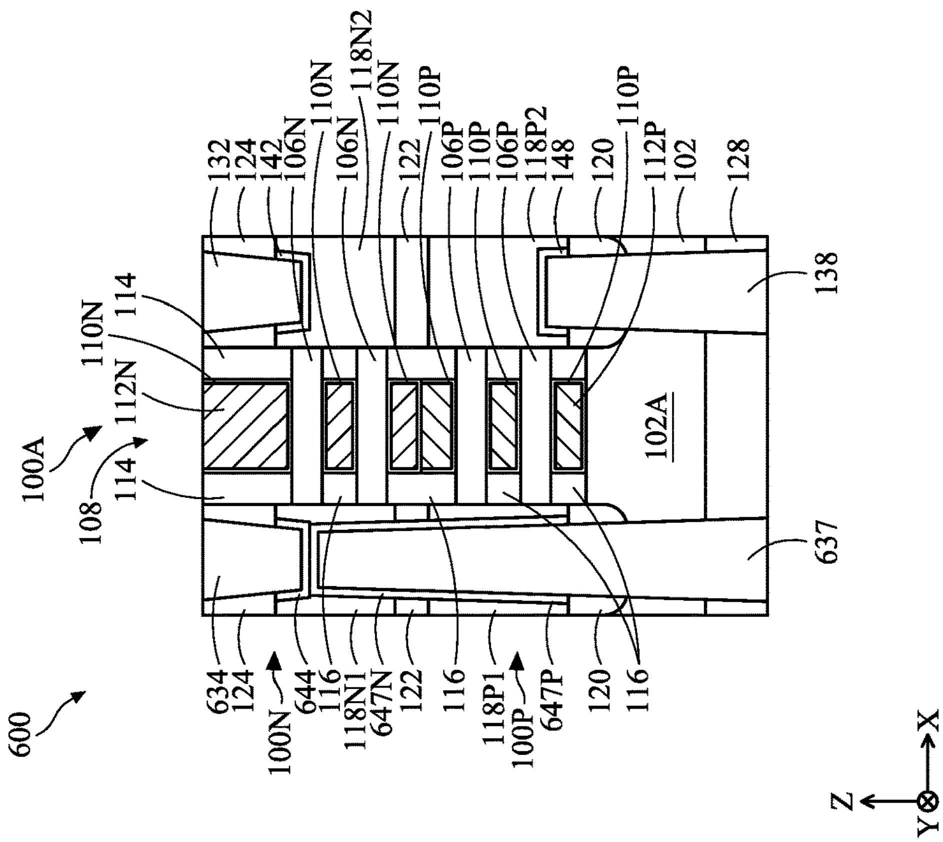
100P
547P
114
120
104
102A
128
534
537
130
118N1
118P1
104
136
100N
140
100P
146
114
120
104
102B
Z
Y
X

FIG. 5A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110P
106P
118P2
148
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110P
112P
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128
138
102A
637
634
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100N
644
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647P
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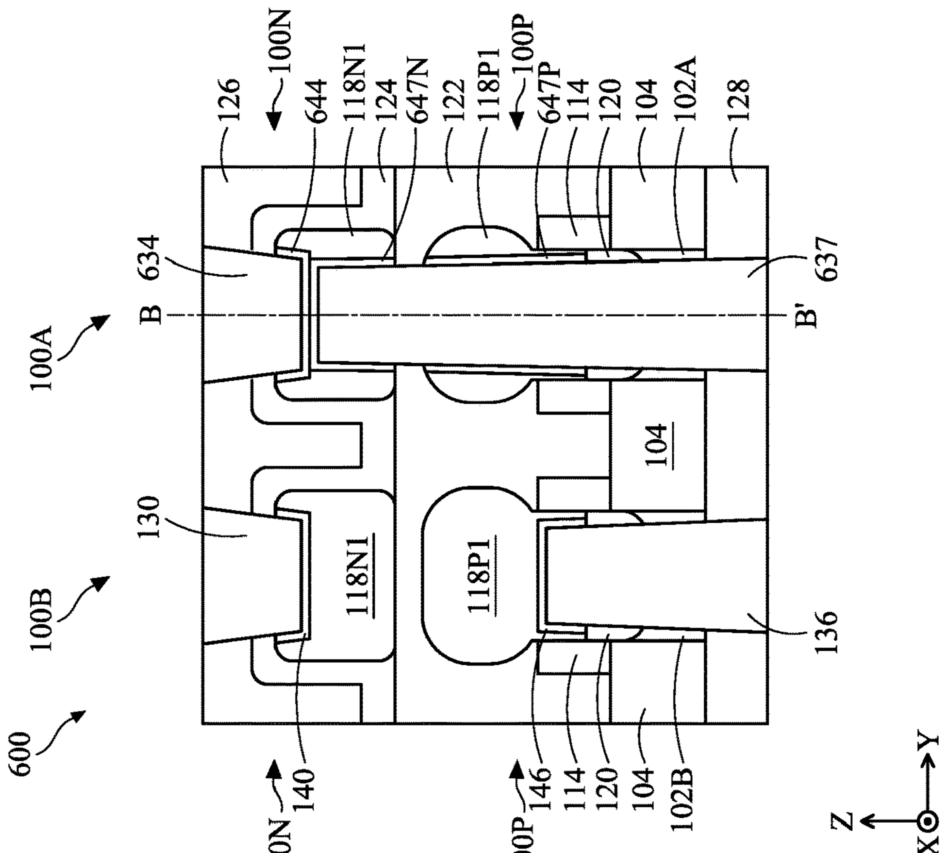
116
Z
Y
X

FIG. 6B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120
104
102A
128
637
B'
118N1
118P1
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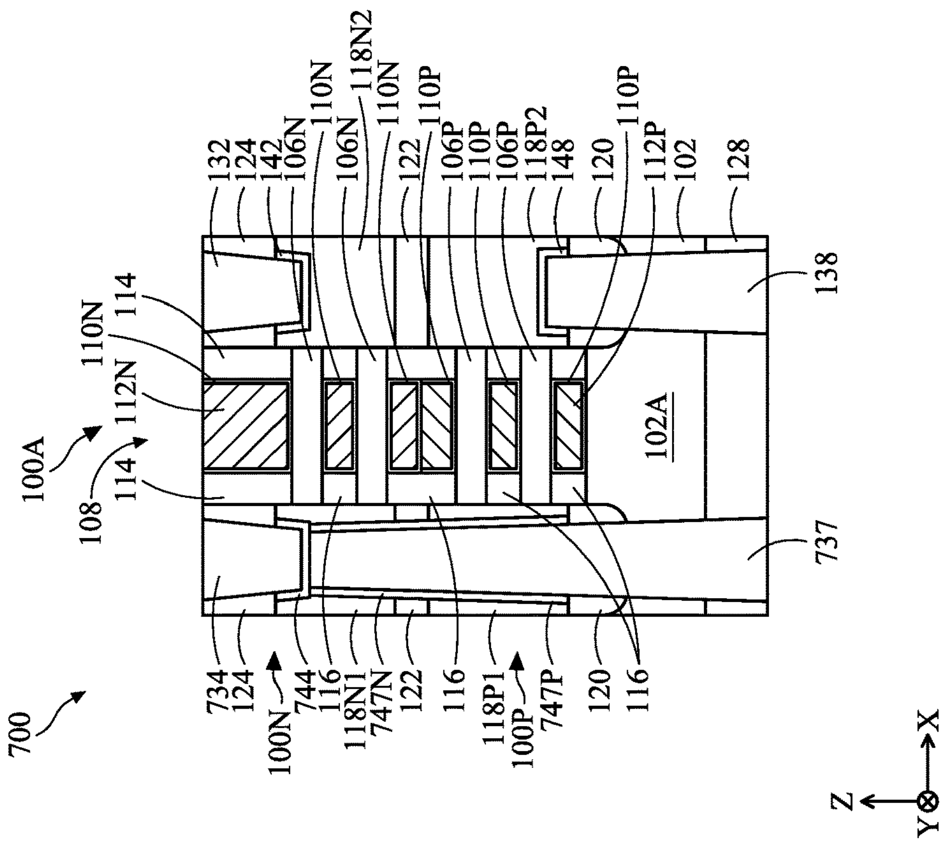
136
100N
140
100P
146
114
120
104
102B
Z
X
Y

FIG. 6A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106N
118N2
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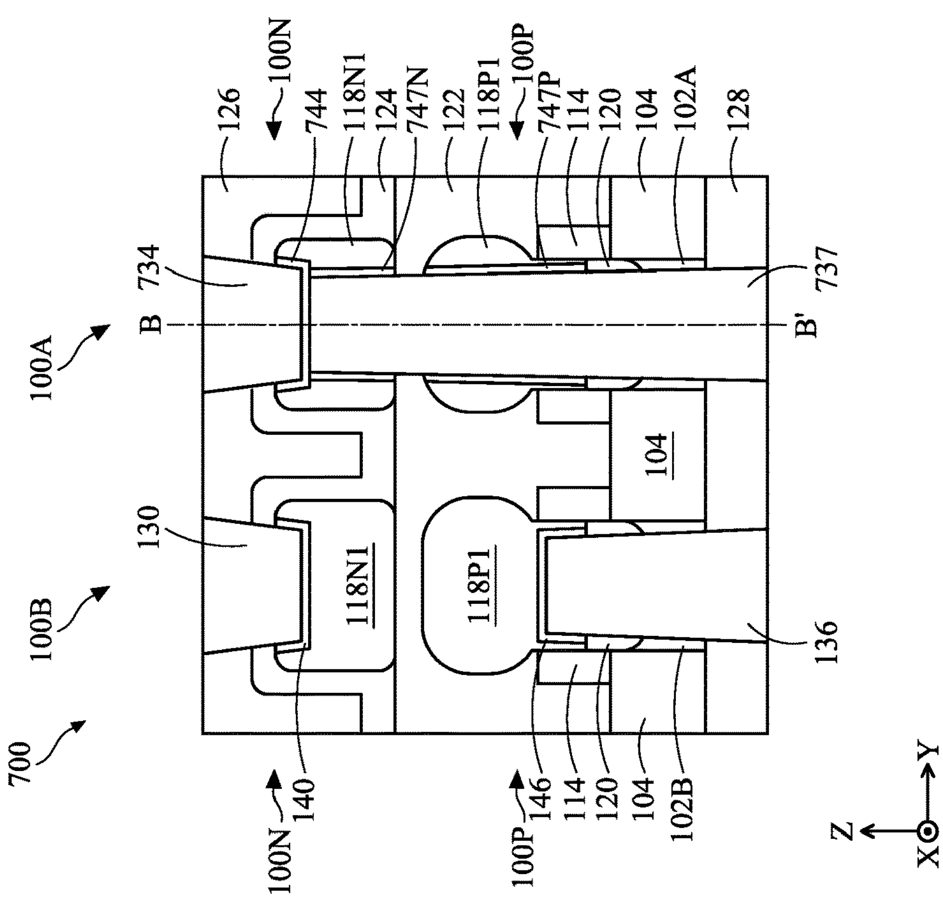
110P
106P
110P
106P
118P2
148
120
110P
112P
102
128
138
102A
737
734
124
744
116
118N1
747N
122
116
118P1
747P
120
116
Z
X
Y

FIG. 7B


700
100B
100A
100N
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118P1
104
140
146
114
120
104
102B
136
B
B'
734
737
126
744
118N1
124
747N
122
118P1
747P
114
120
104
102A
128
Z
Y
X

FIG. 7A

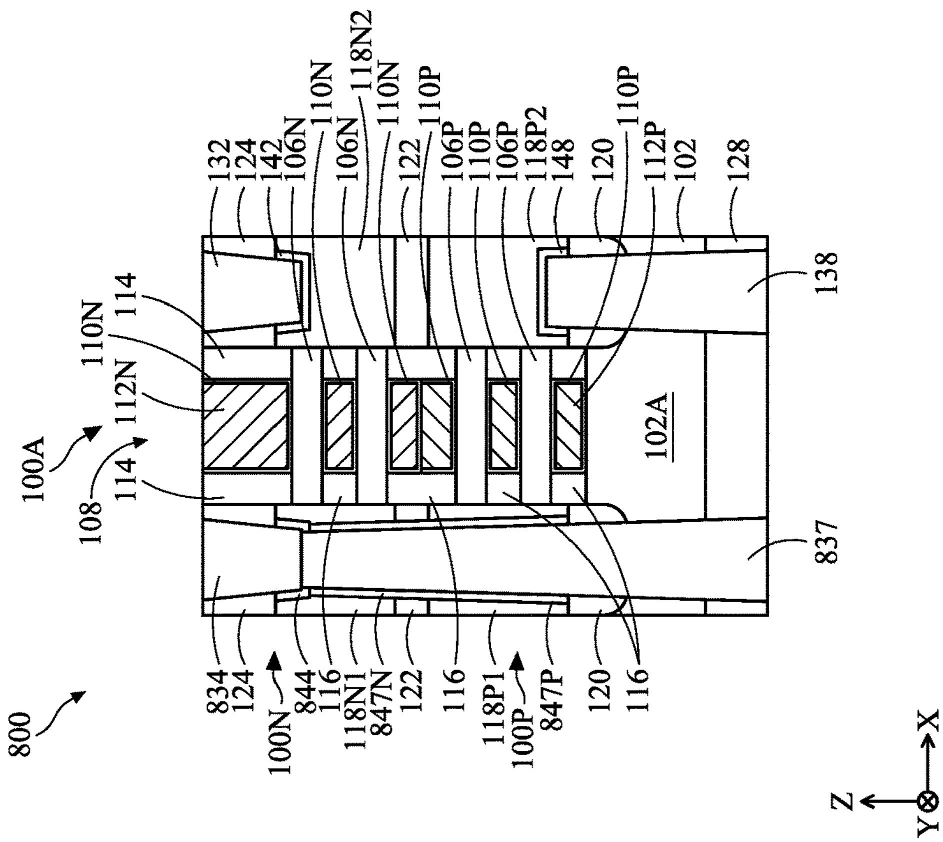
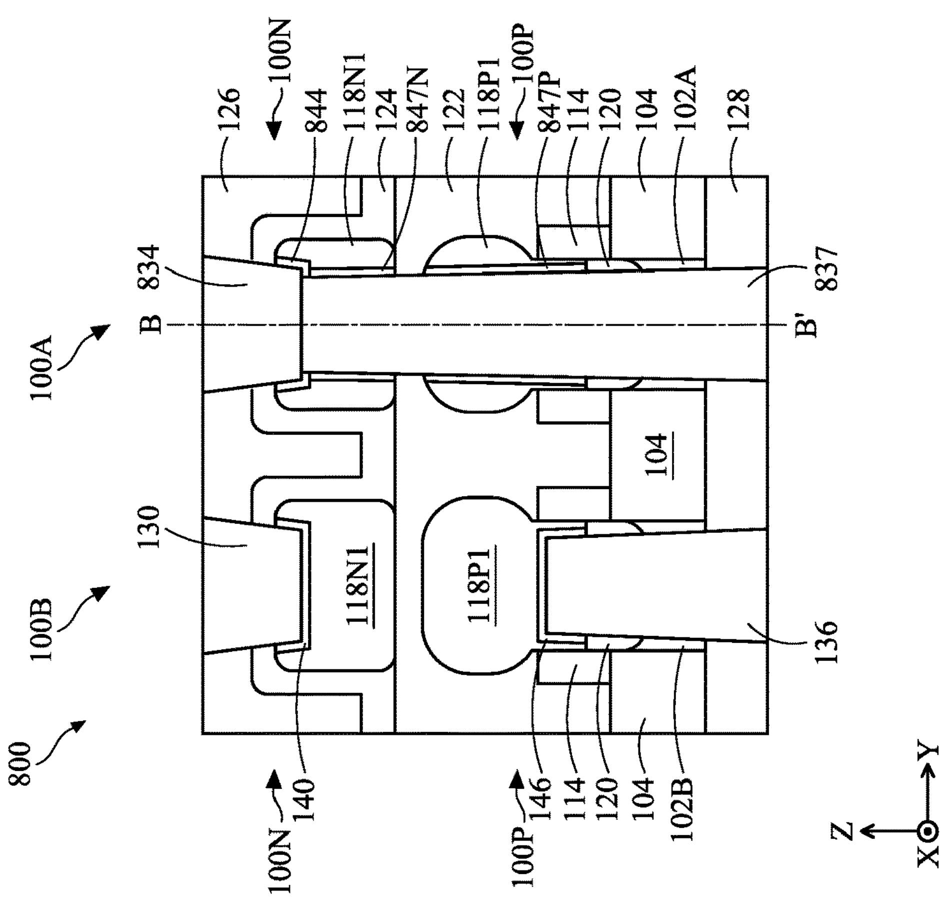

800
100A
108
114
112N
110N
114
132
124
142
106N
110N
106N
118N2
110N
122
110P
106P
110P
106P
118P2
148
120
110P
112P
102
128
138
102A
837
834
124
100N
844
116
118N1
847N
122
116
118P1
100P
847P
120
116
Z
X
Y

FIG. 8B


800
100B
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B
834
126
100N
844
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124
847N
122
118P1
100P
847P
114
120
104
102A
128
837
B'
118N1
118P1
104
136
100N
140
100P
146
114
120
104
102B
Z
Y
X

FIG. 8A

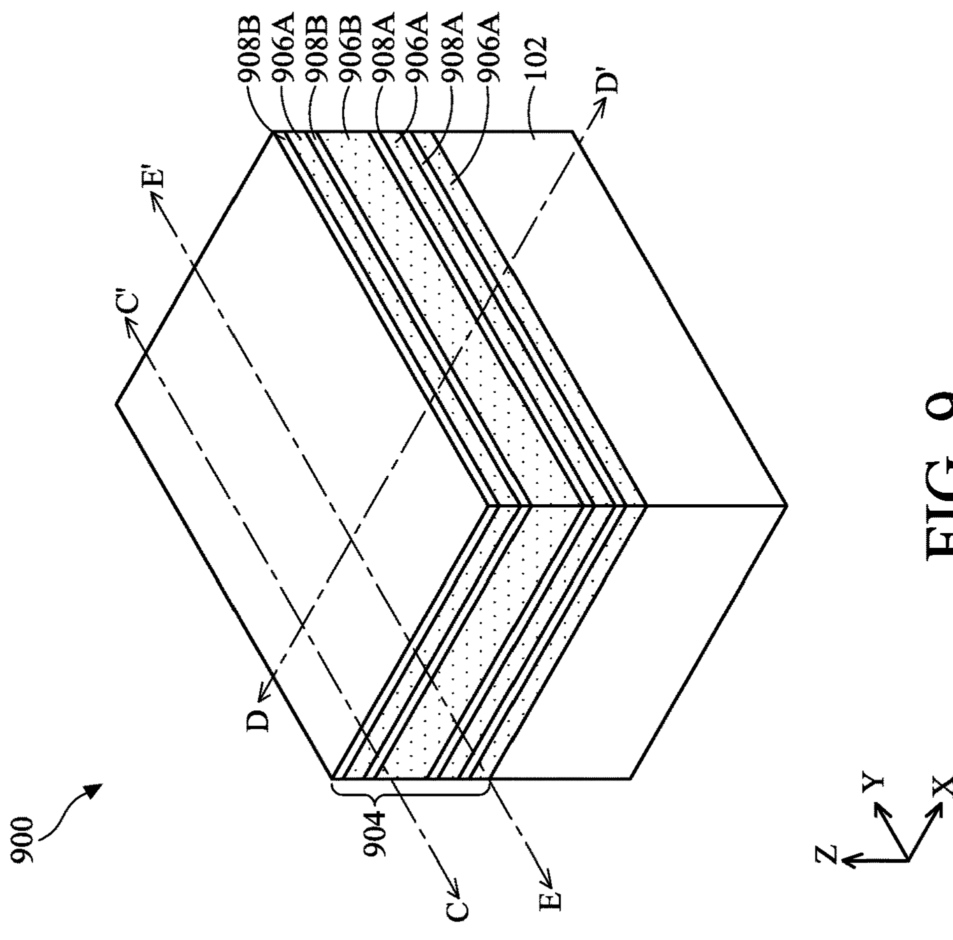
900
904
908B
906A
908B
906B
908A
906A
908A
906A
102
C
C'
D
D'
E
E'
Z
Y
X

FIG. 9

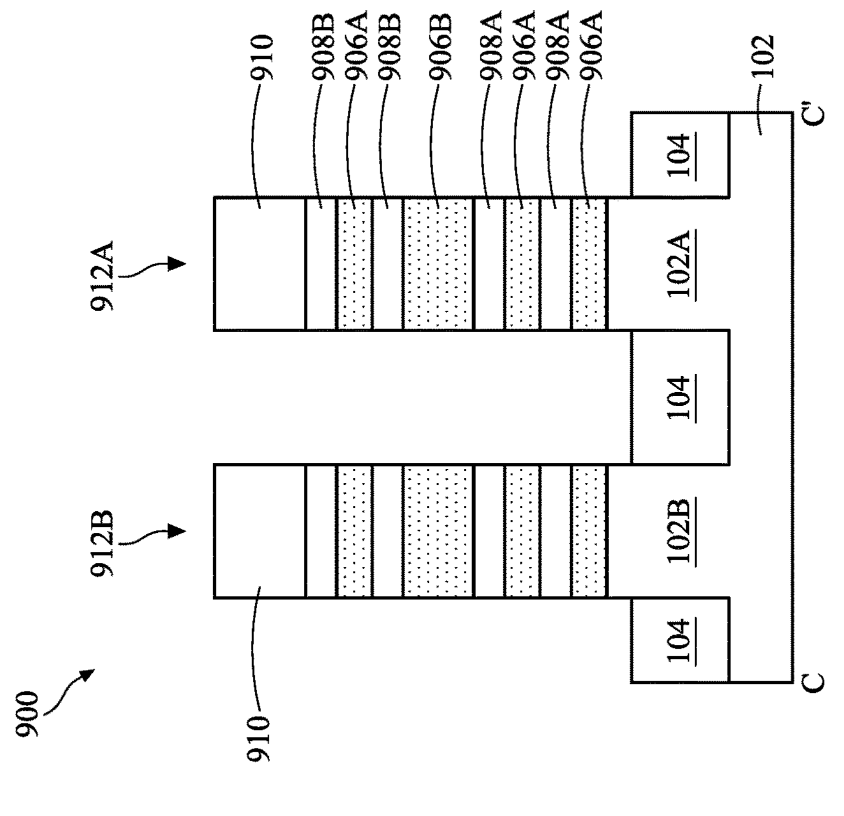
900
912A
912B
910
908B
906A
908B
906B
908A
906A
908A
906A
102
102A
102B
104
C
C'
Z
Y
X

FIG. 10

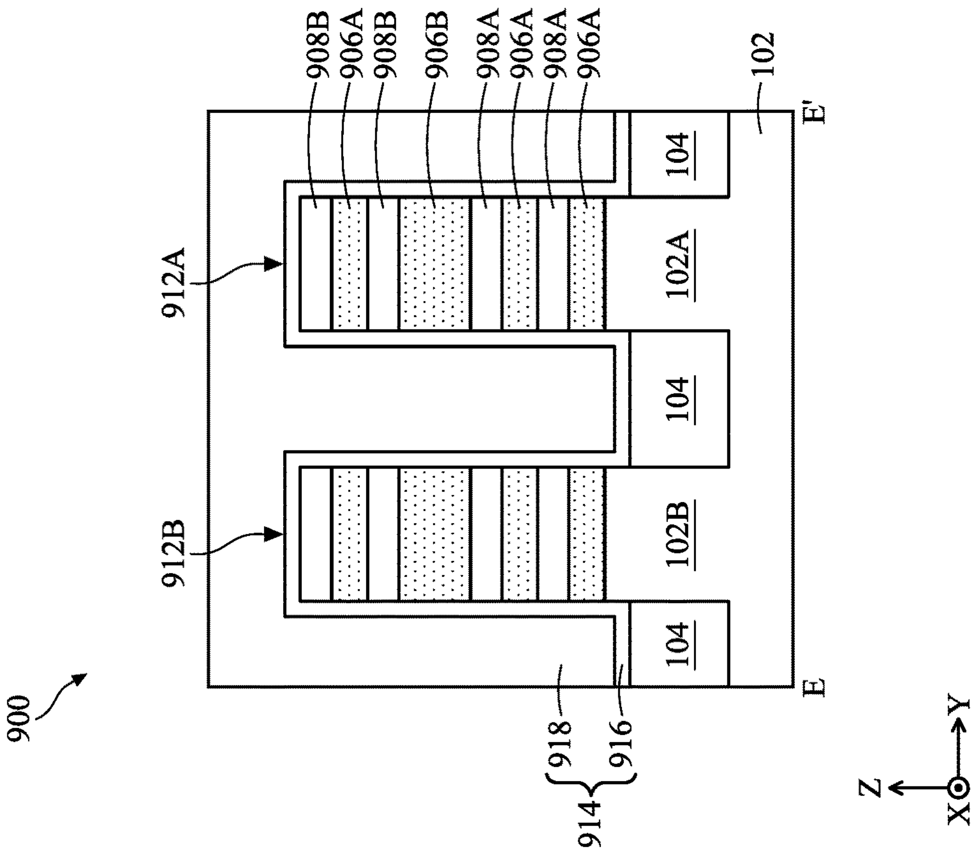

FIG. 11B

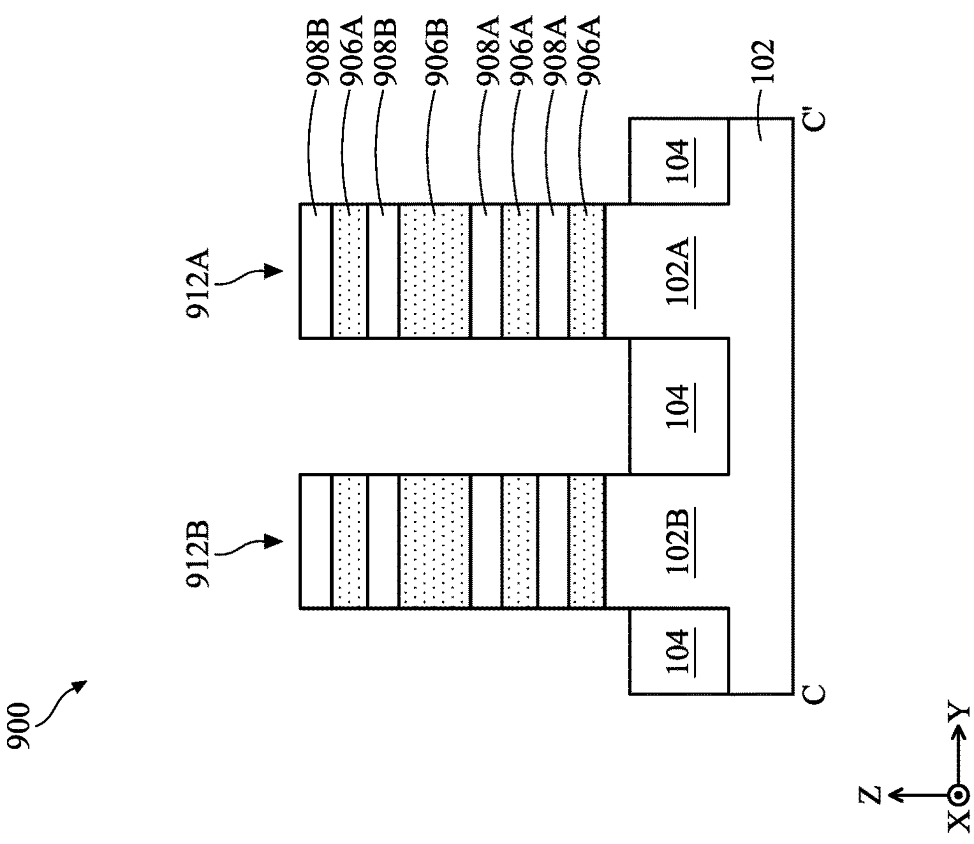

FIG. 11A

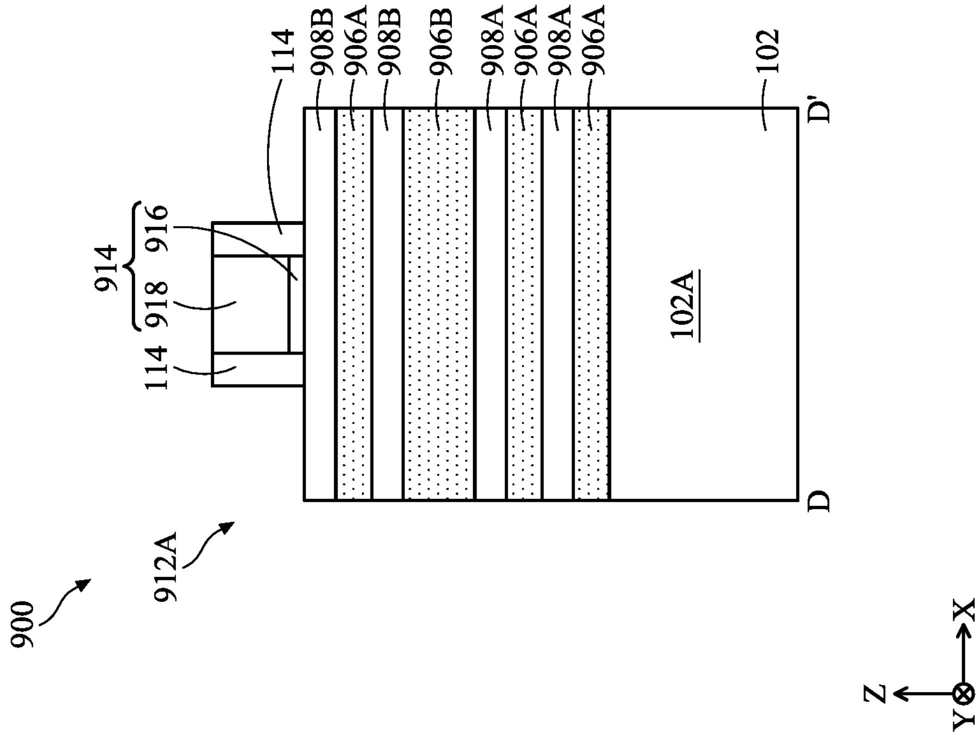

FIG. 12B

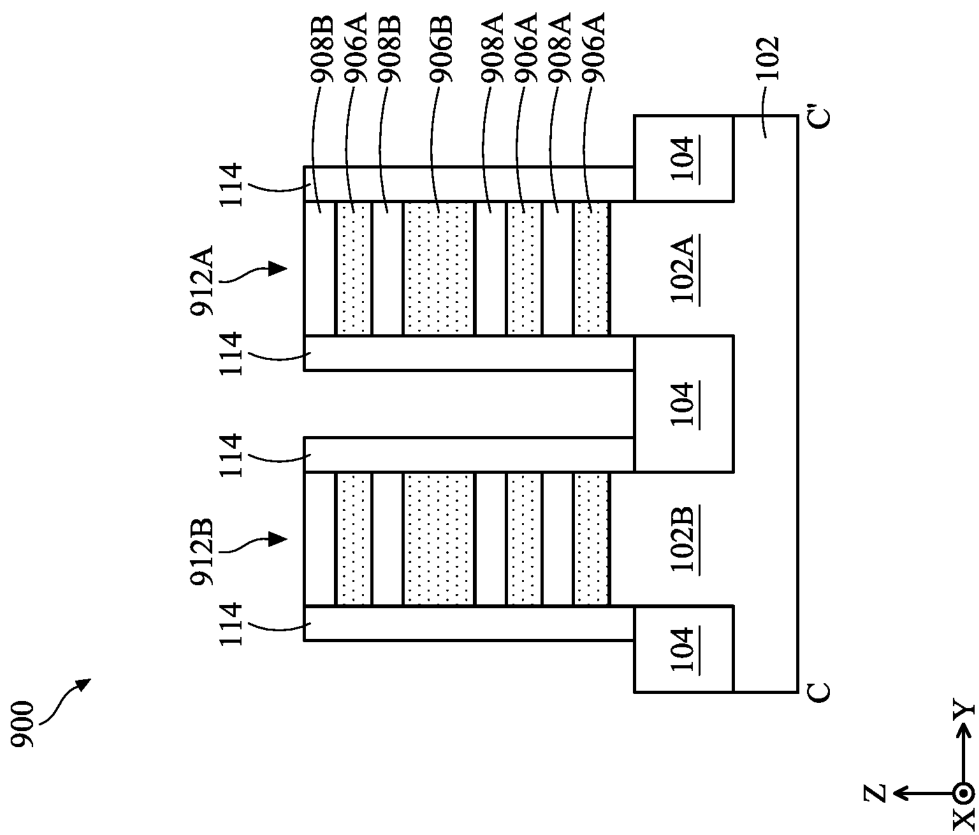

FIG. 12A

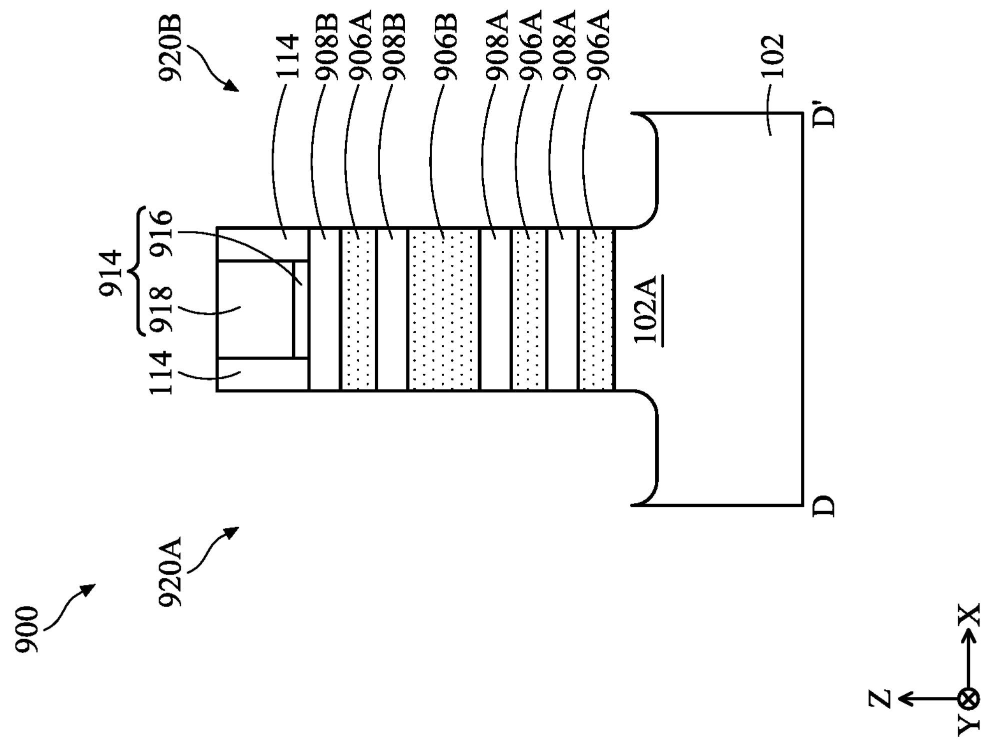

FIG. 13B

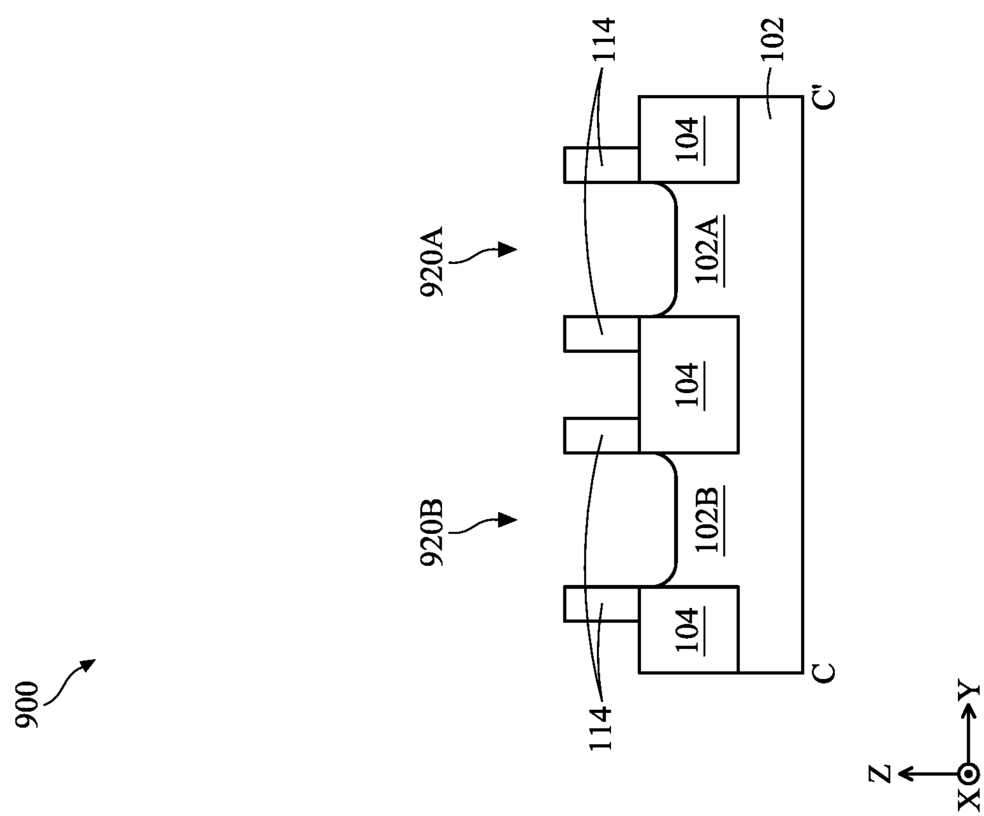

FIG. 13A

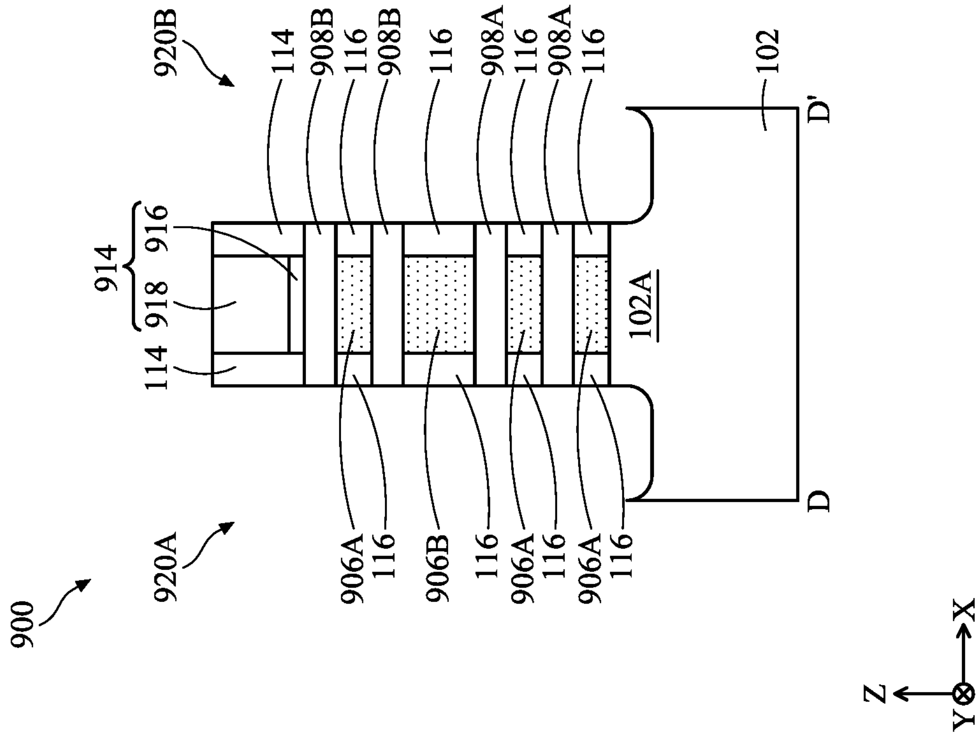

FIG. 14B

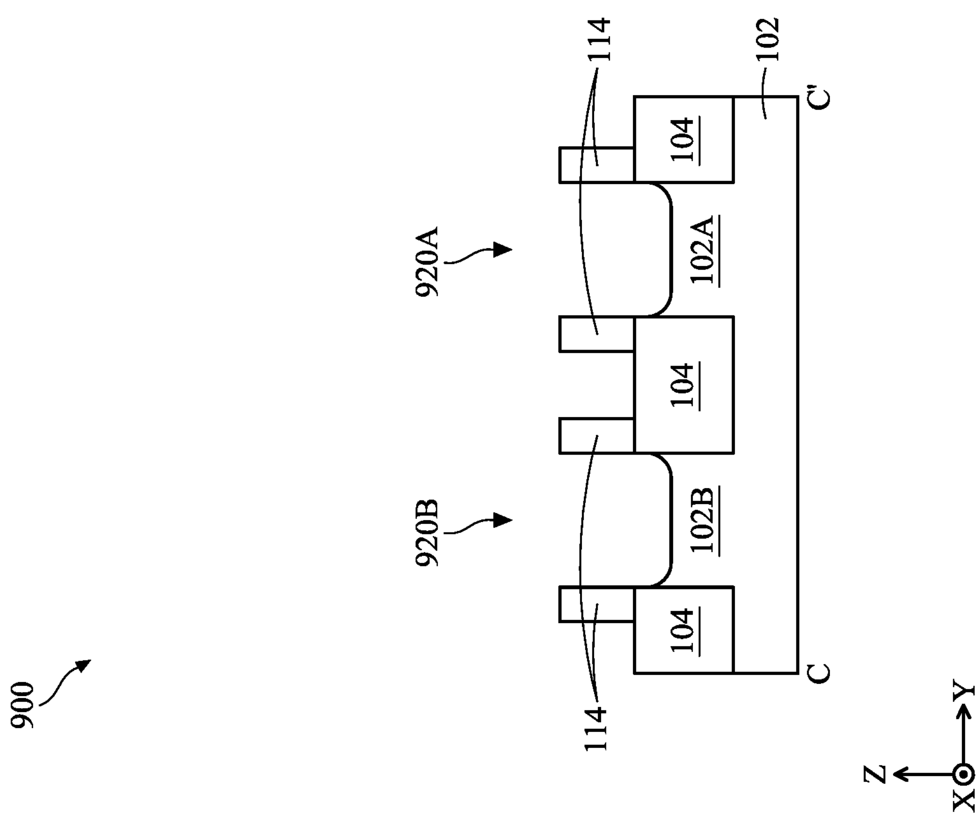

FIG. 14A

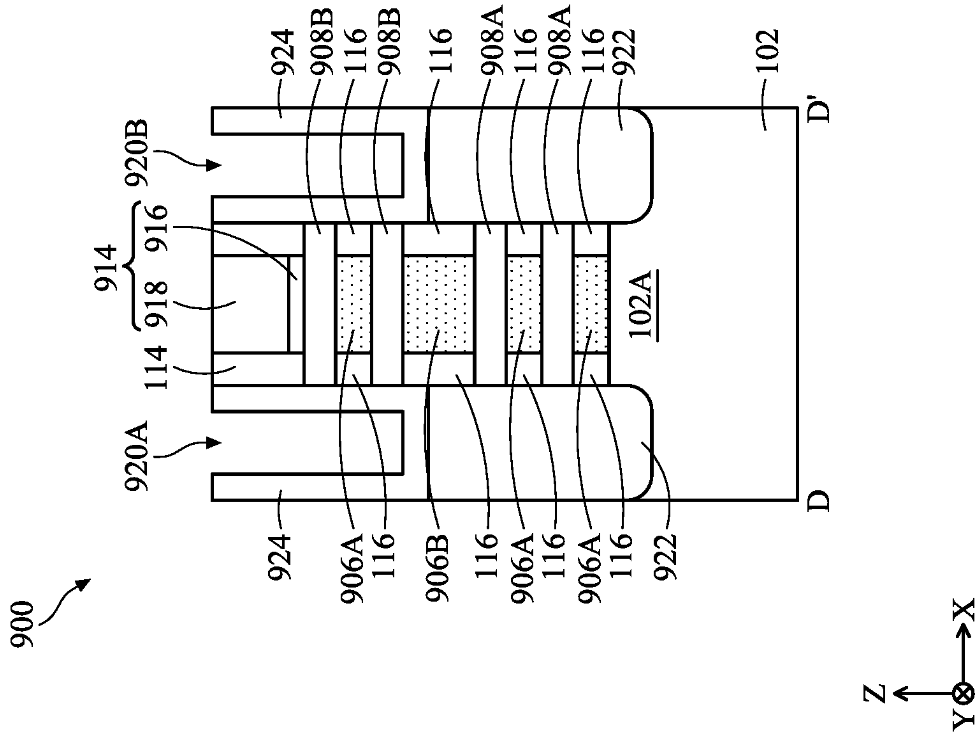

FIG. 15B

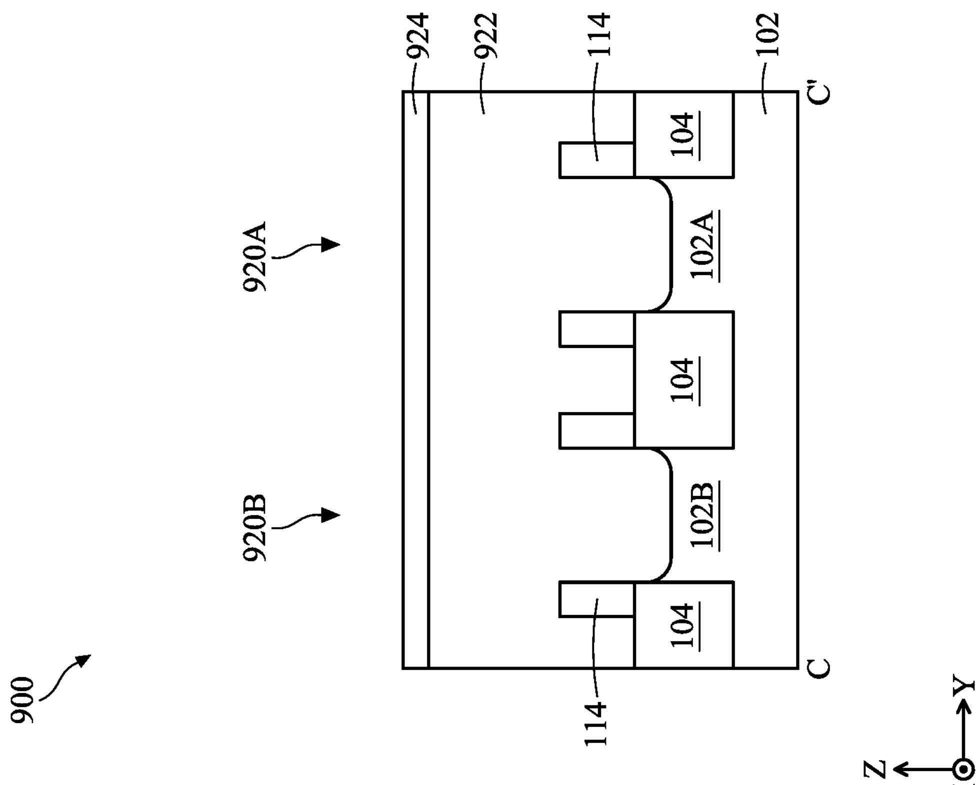

FIG. 15A

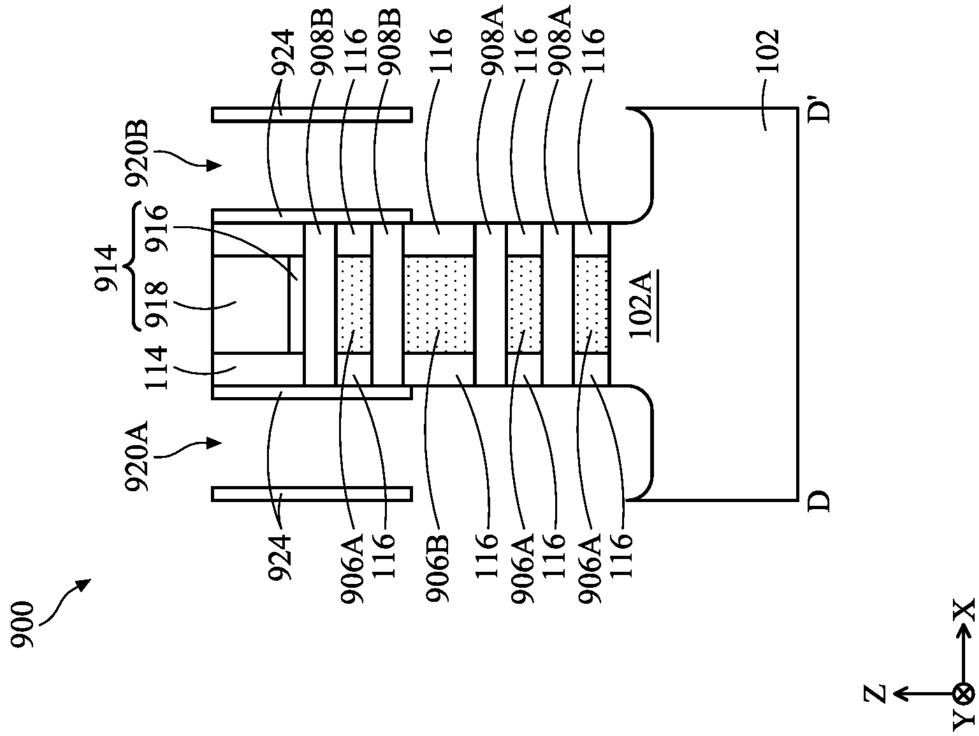

FIG. 16B

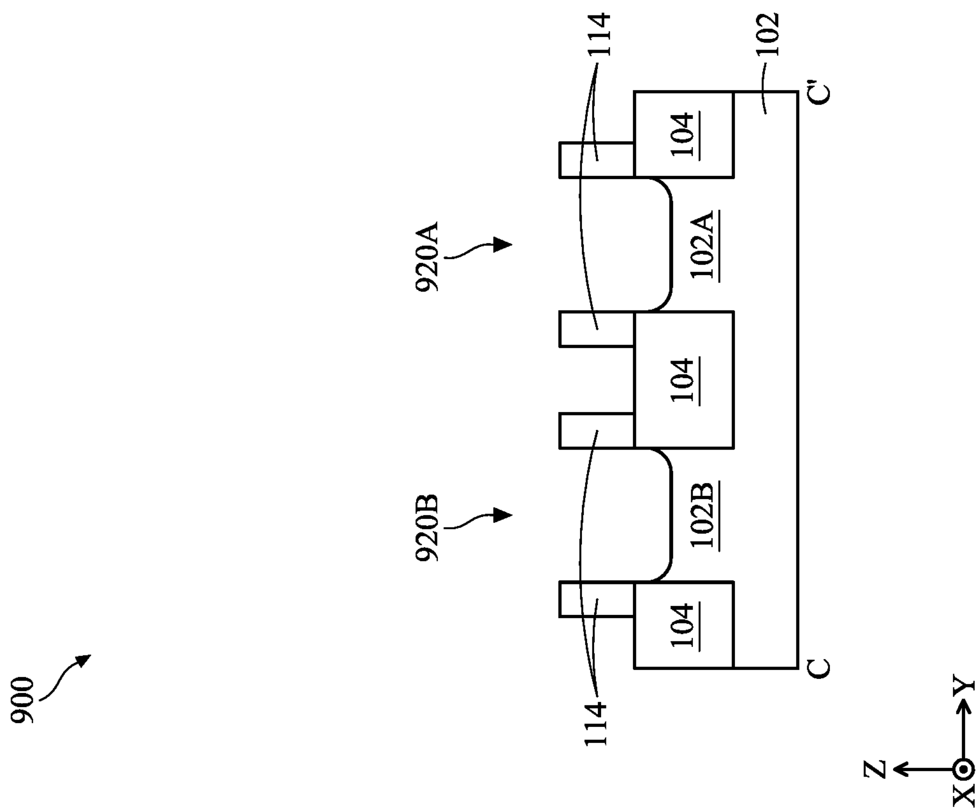

FIG. 16A

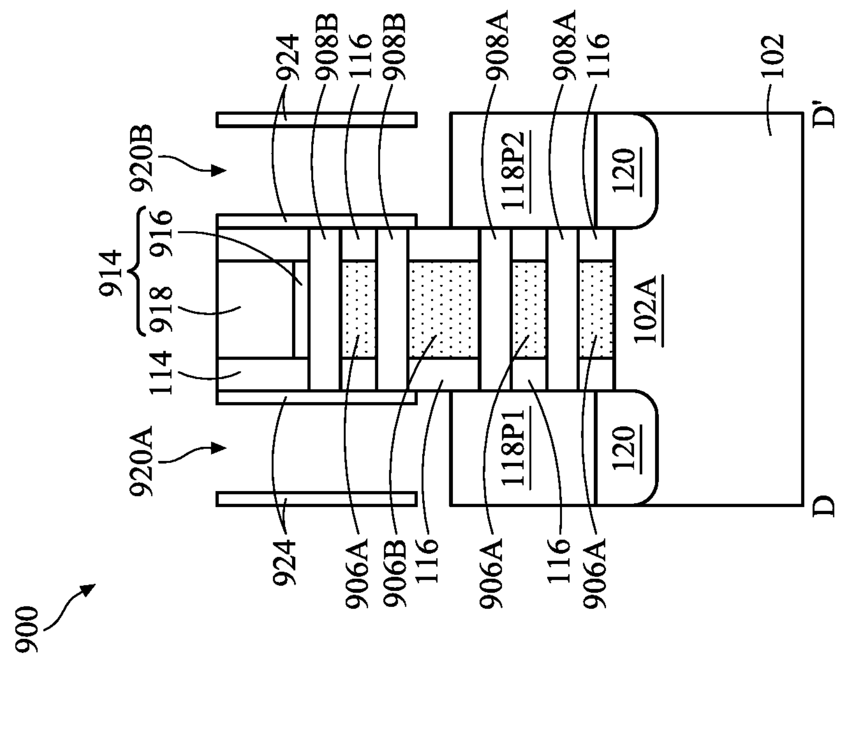

FIG. 17B

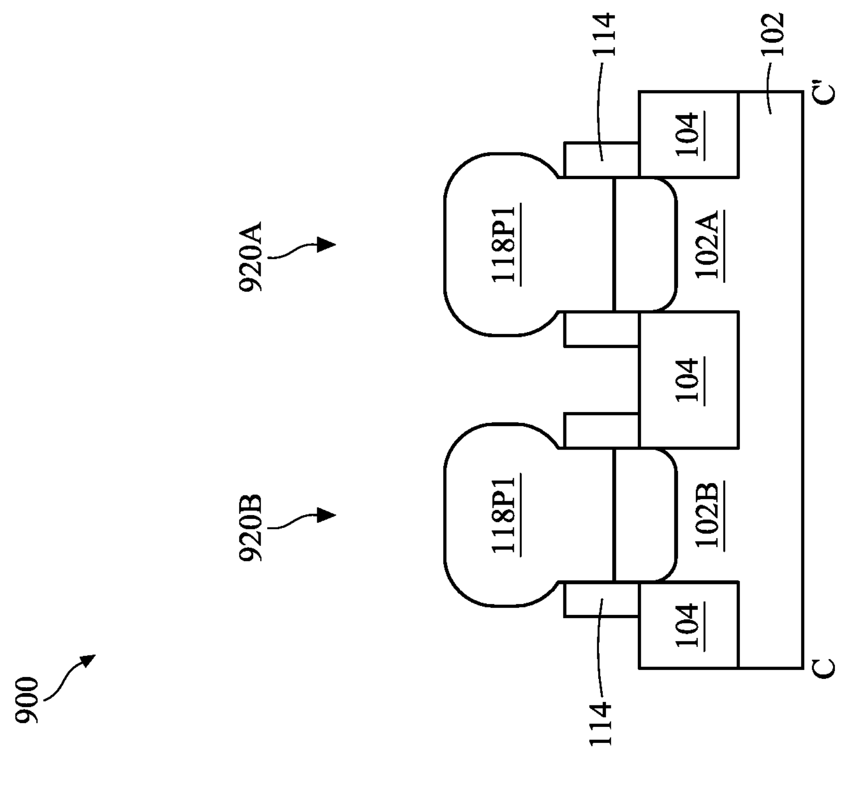

FIG. 17A

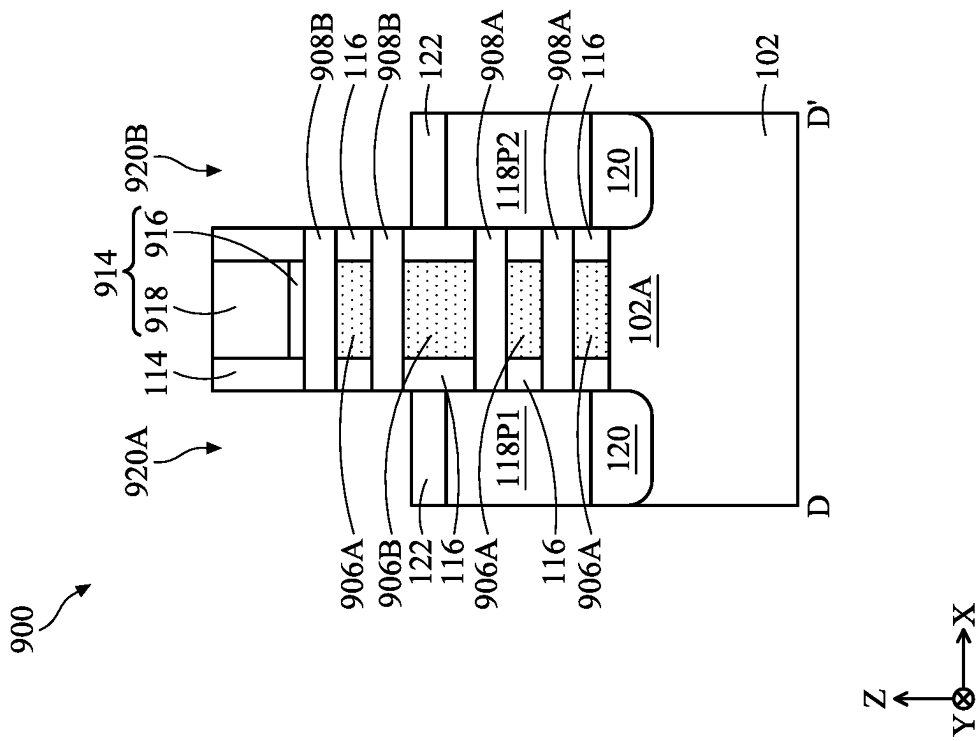

FIG. 18B

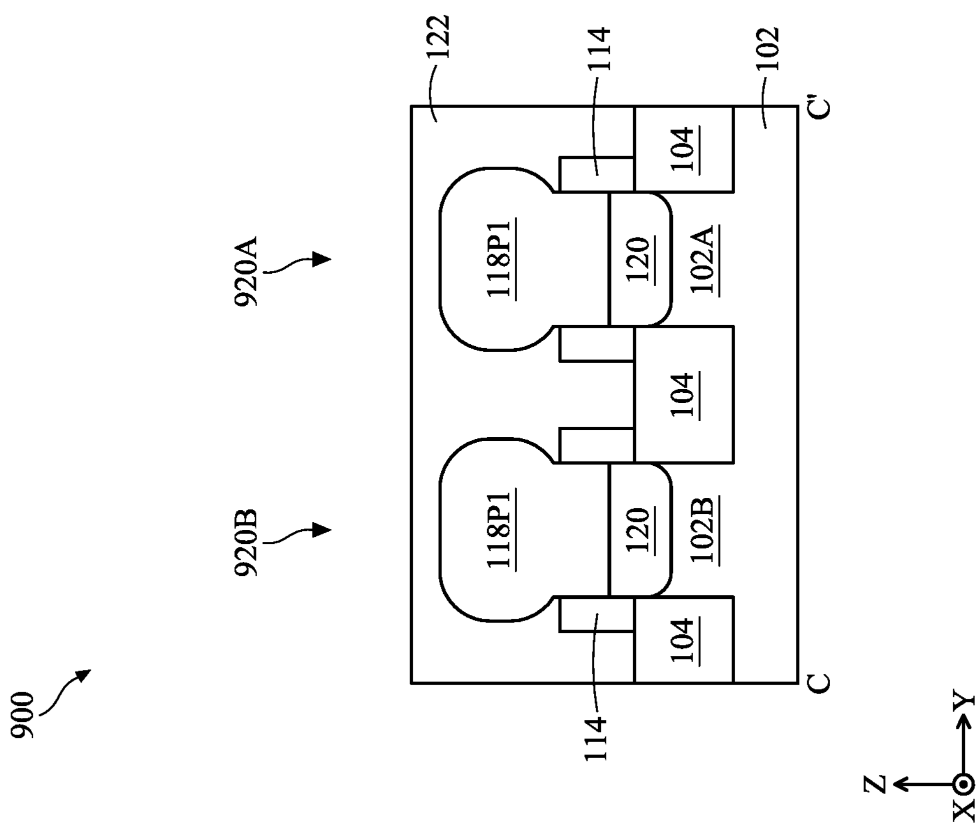

FIG. 18A

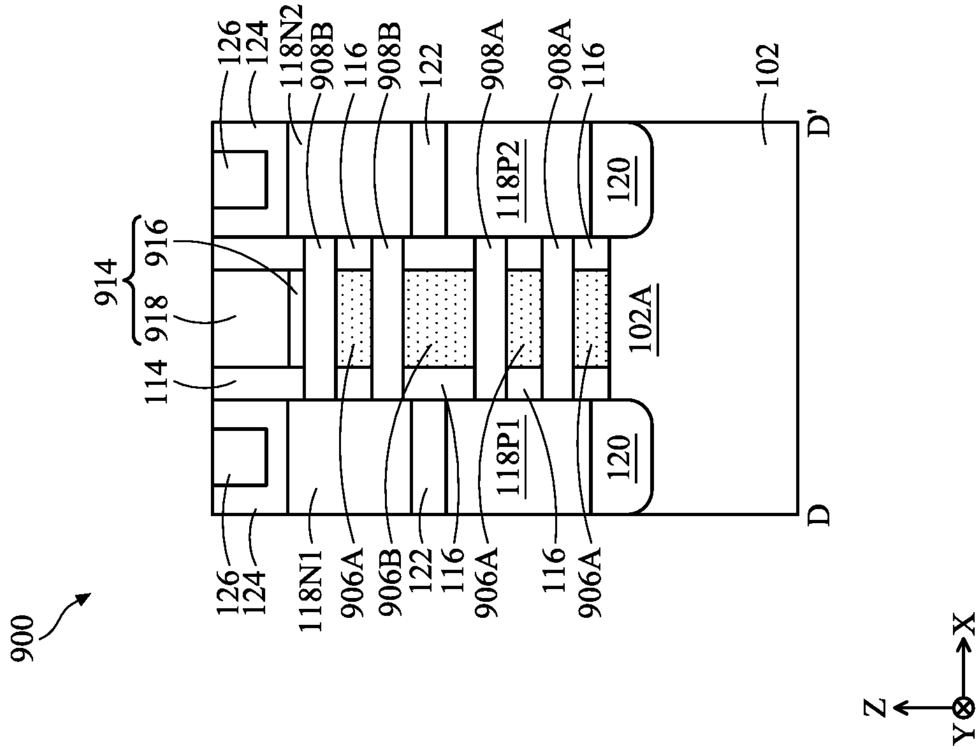

FIG. 19B

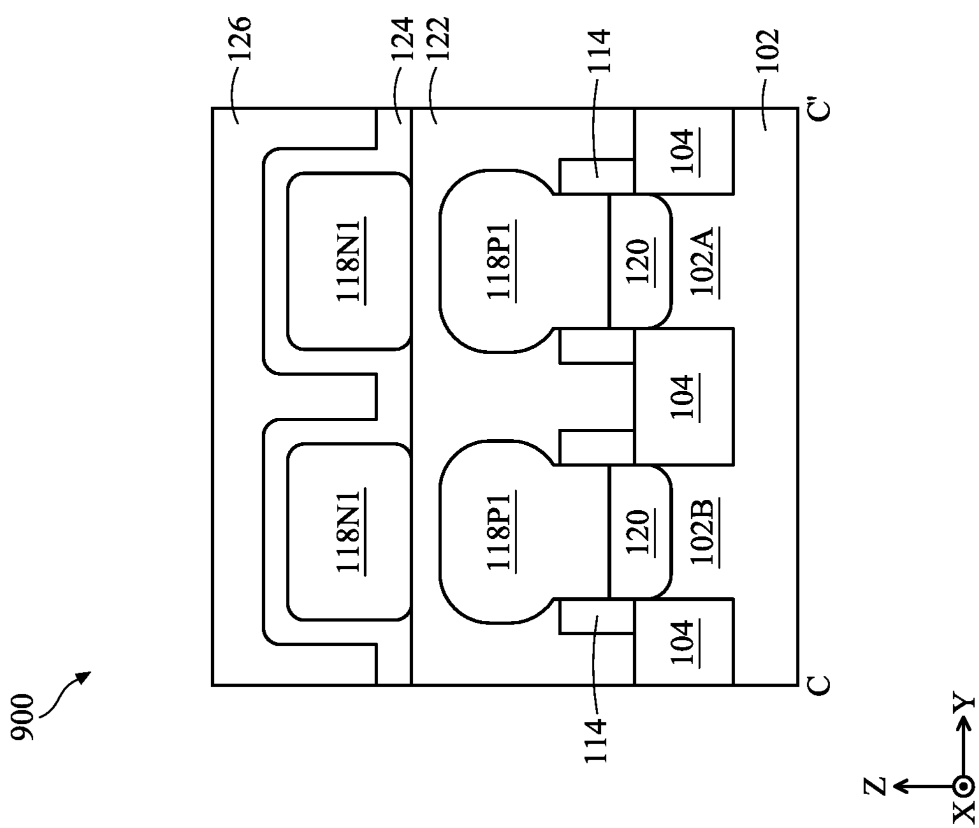

FIG. 19A

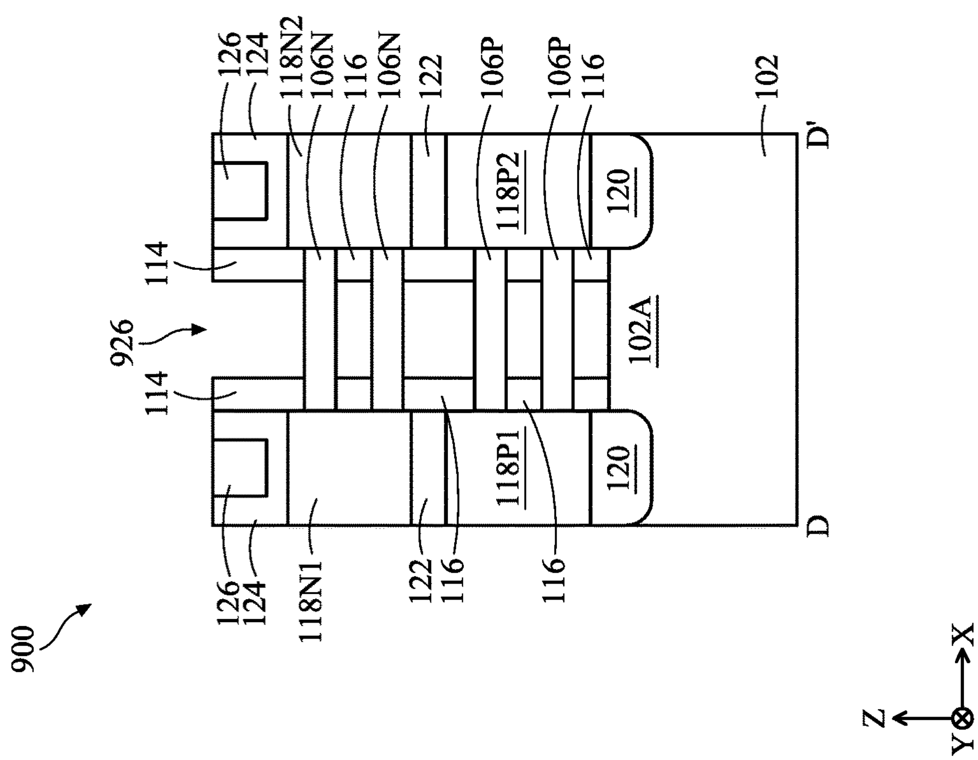

FIG. 20A

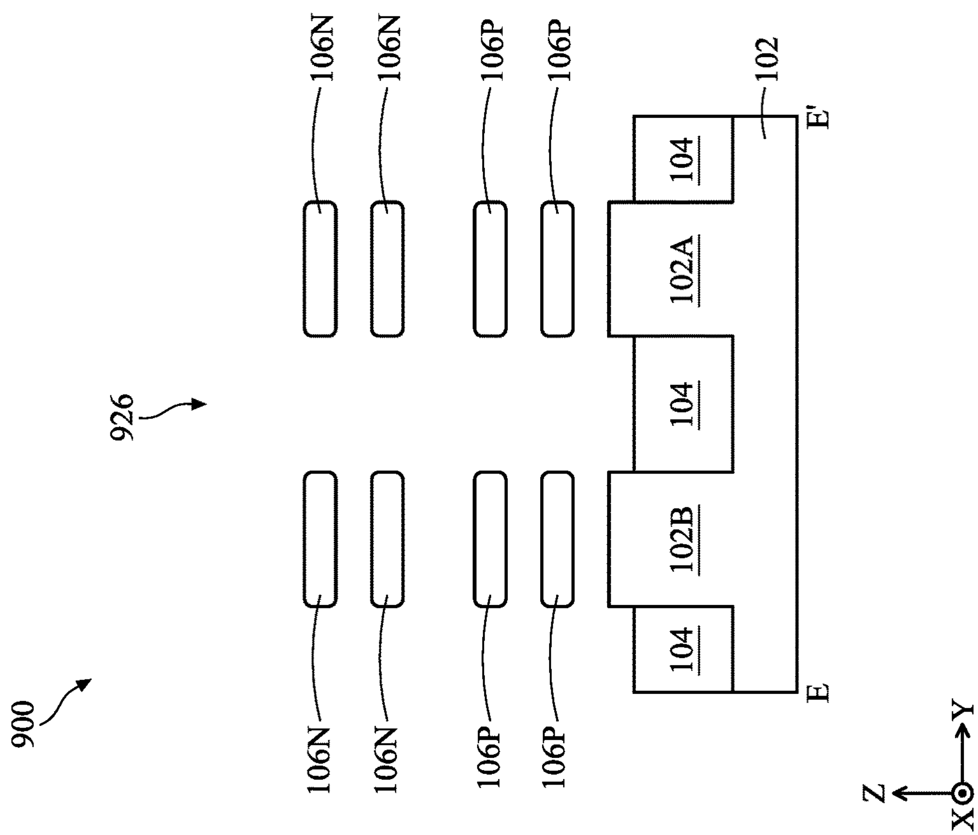

FIG. 20B

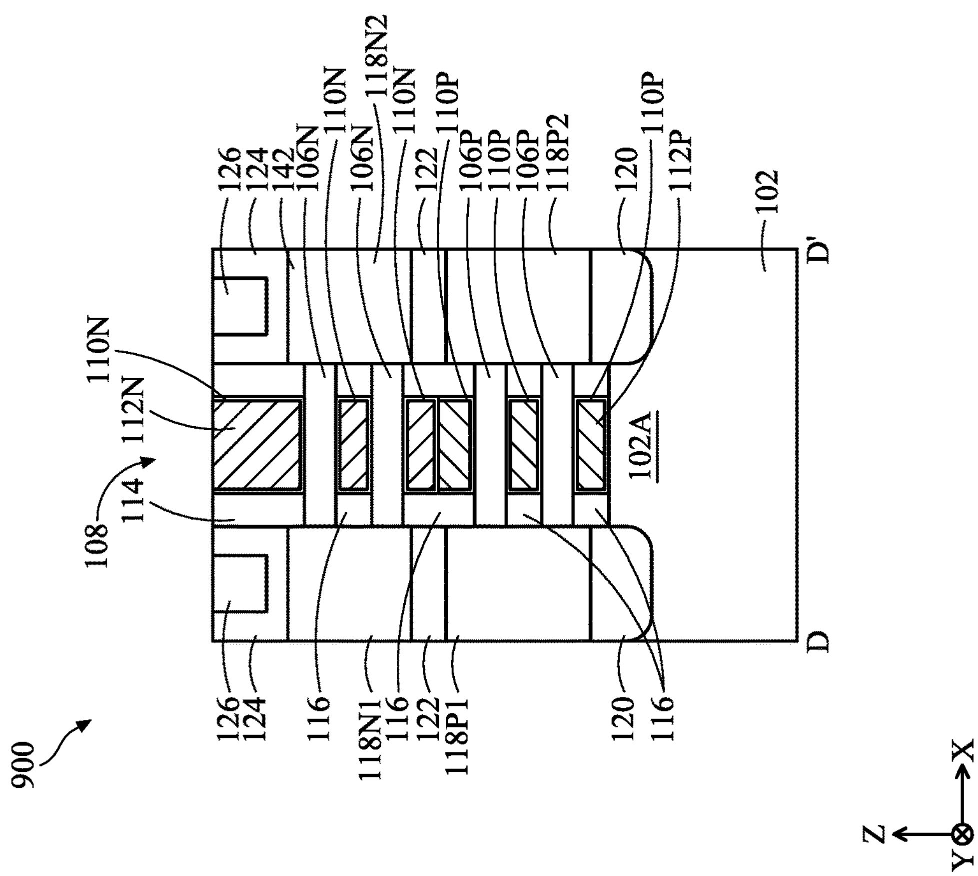

FIG. 21A

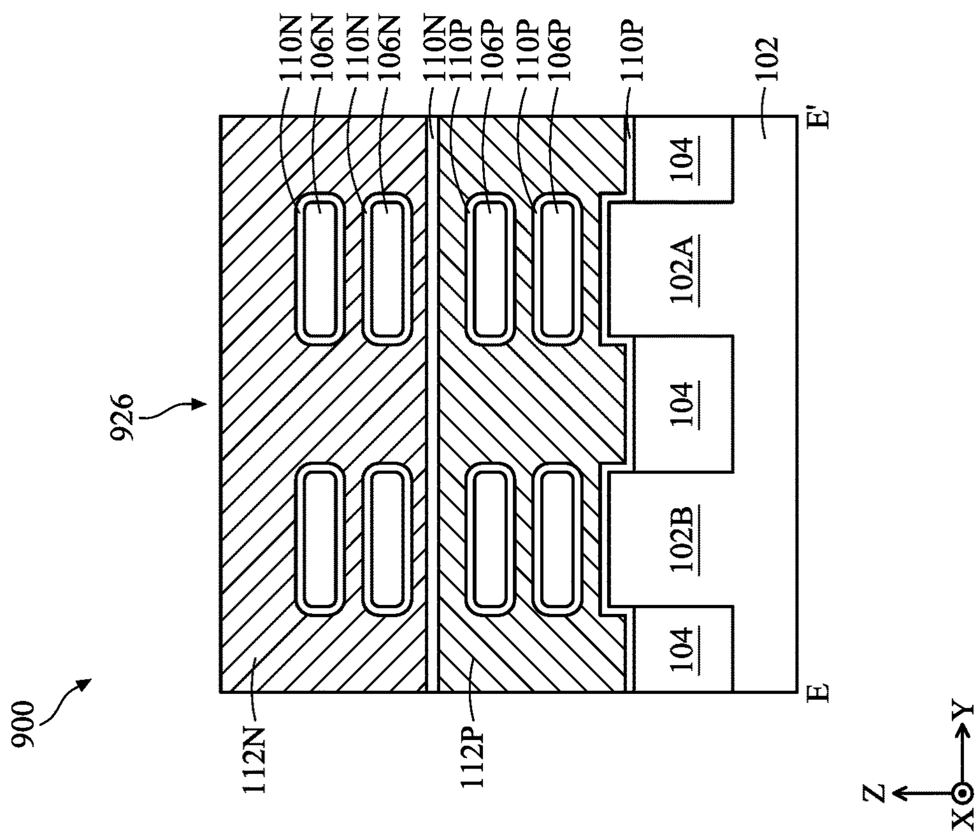

FIG. 21B

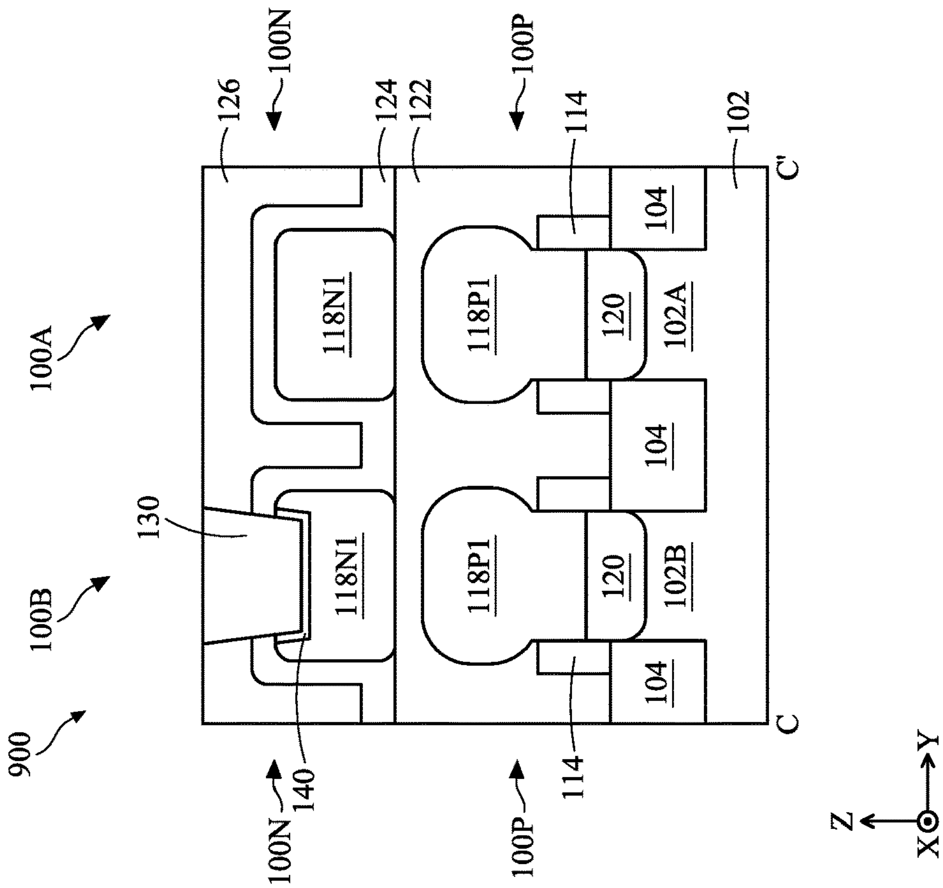

FIG. 22A

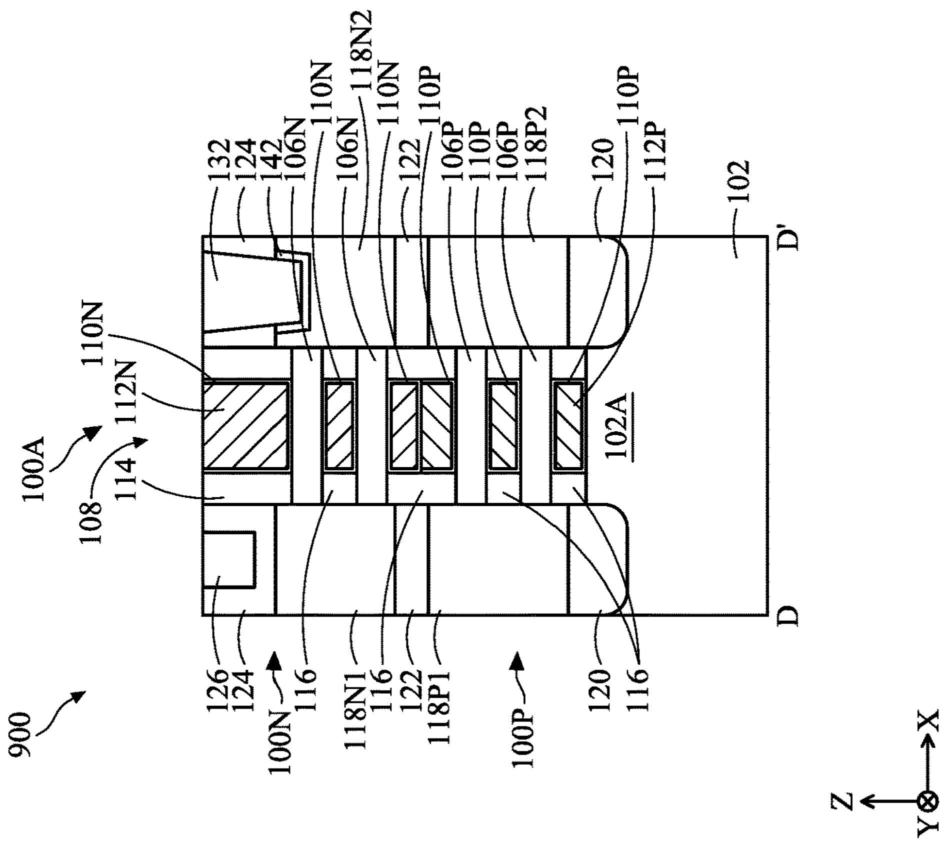

FIG. 22B

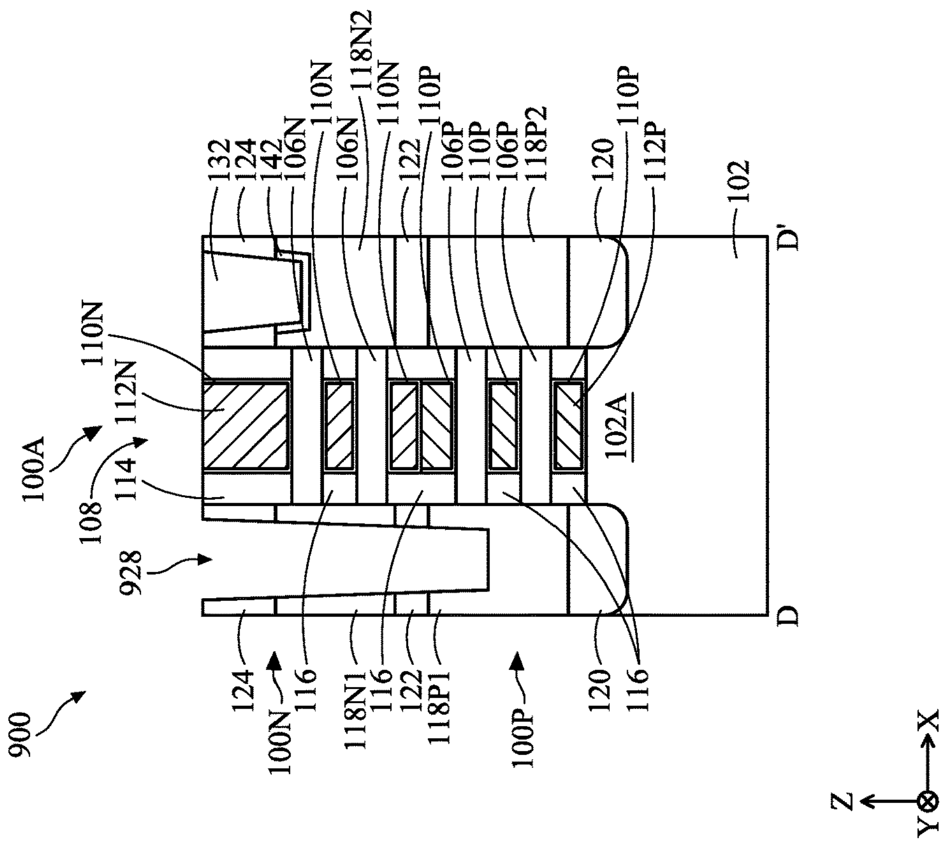

FIG. 23B

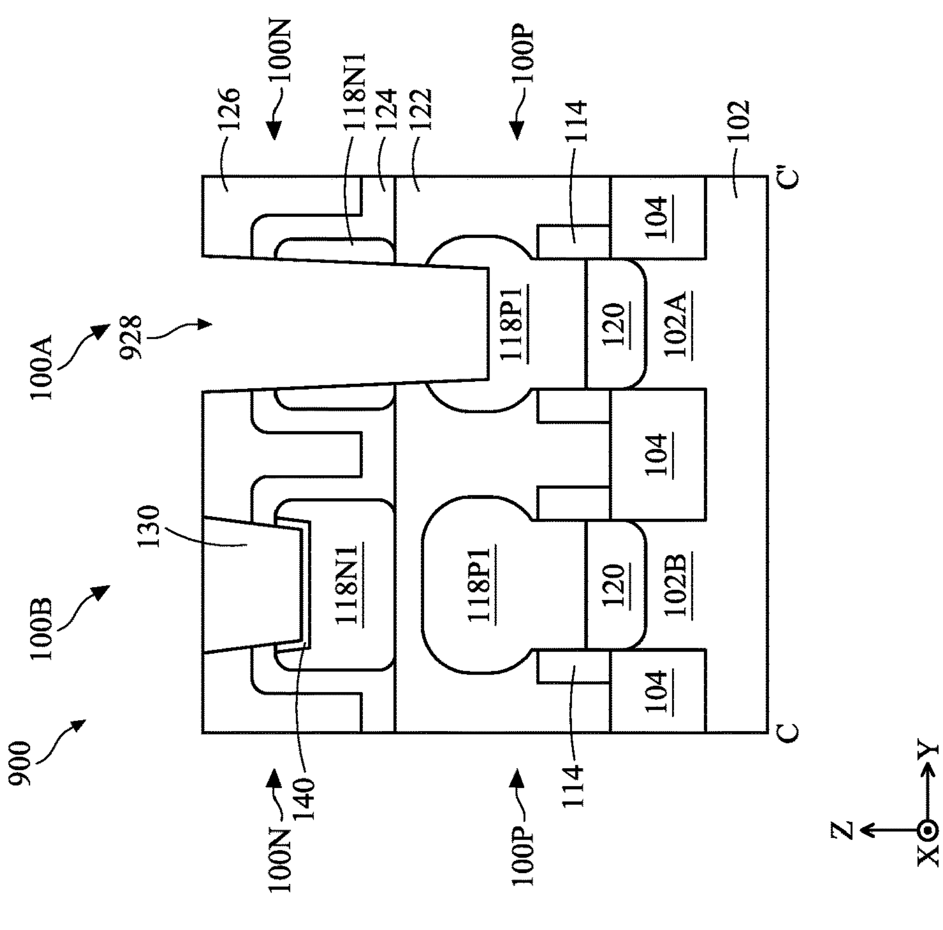

FIG. 23A

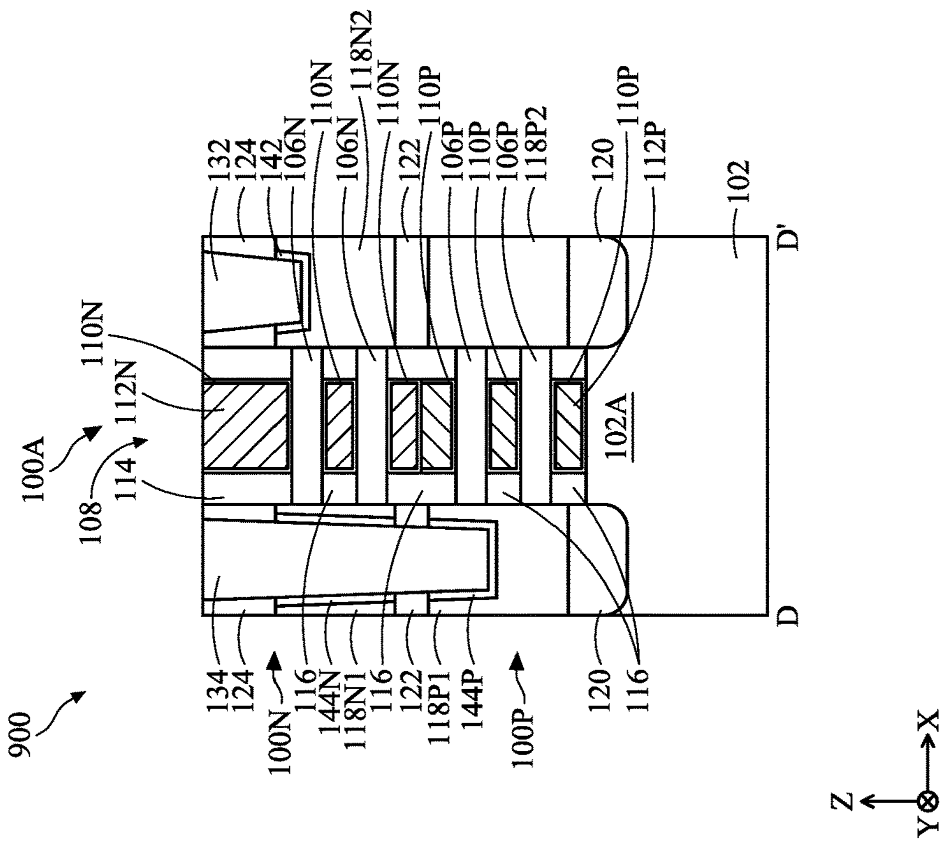

FIG. 24B

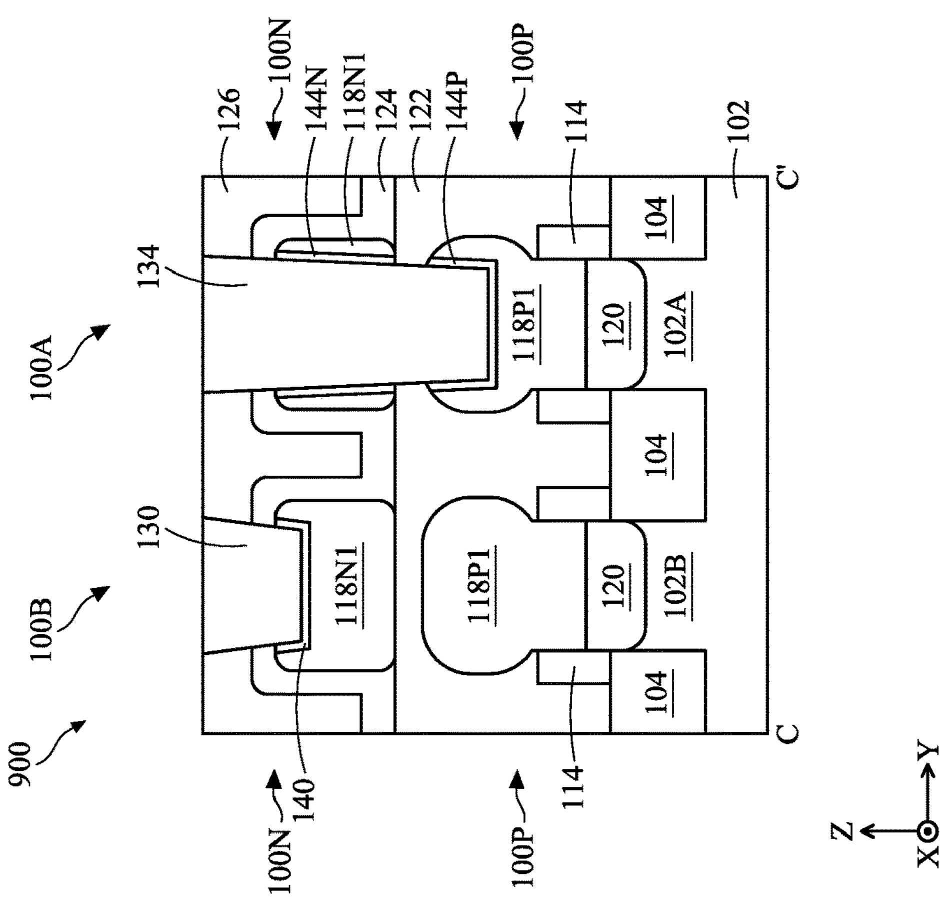

FIG. 24A

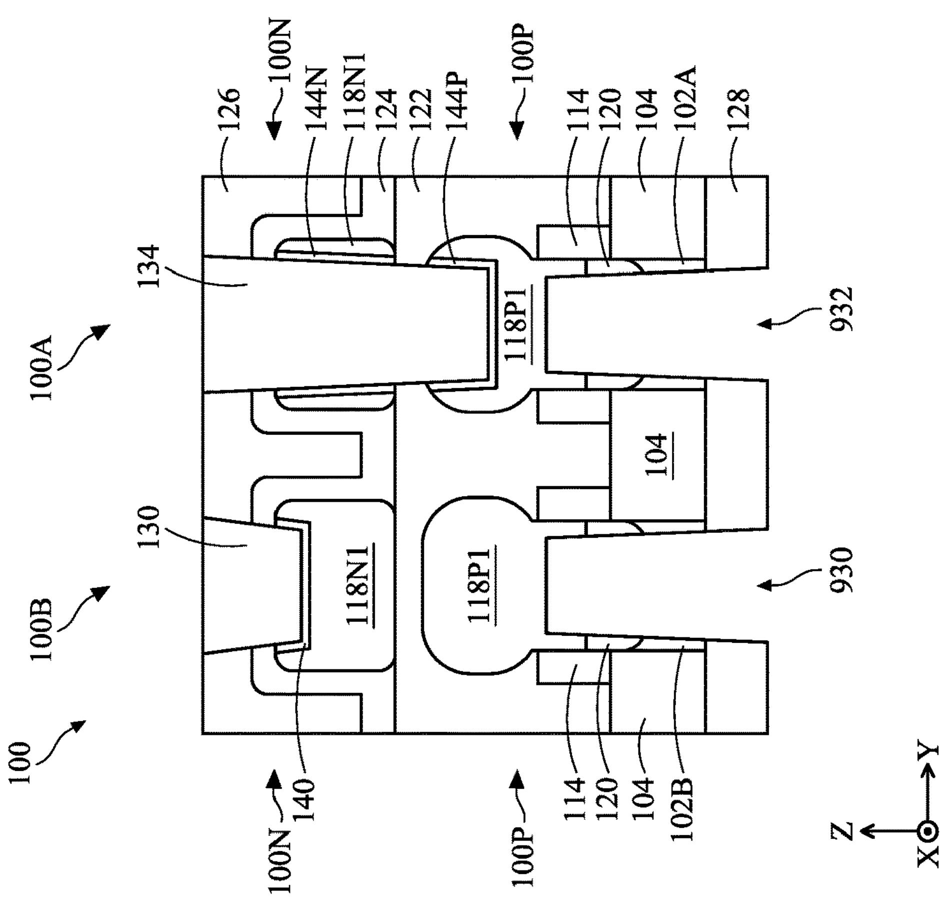

FIG. 25A

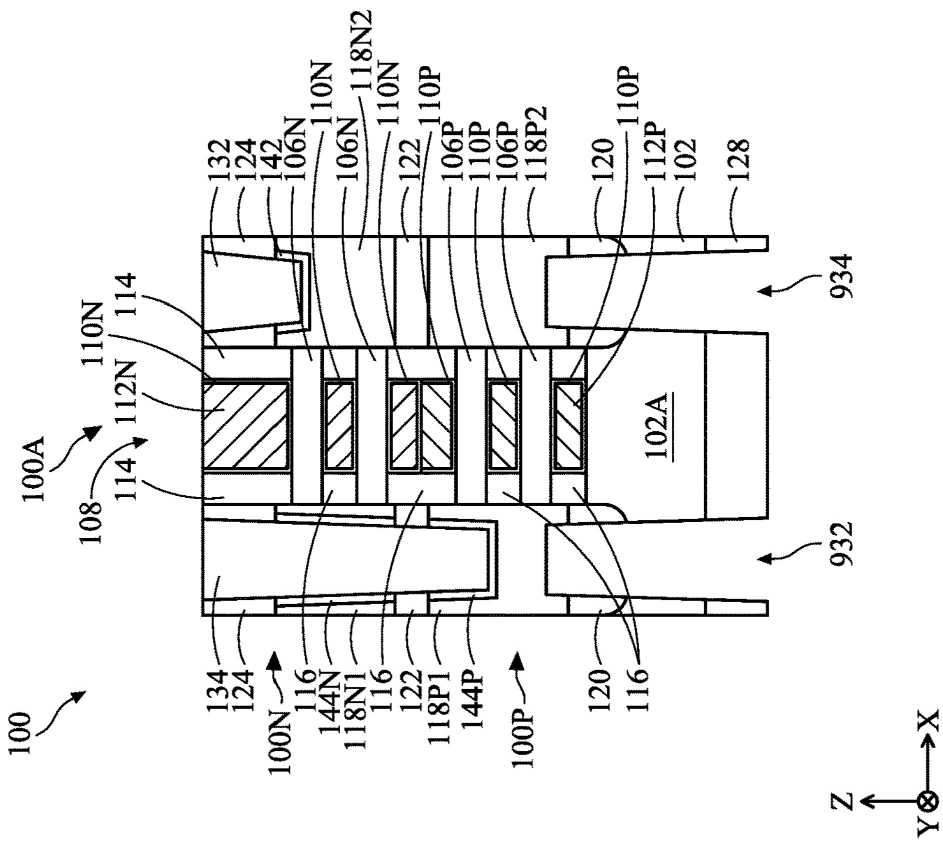

FIG. 25B

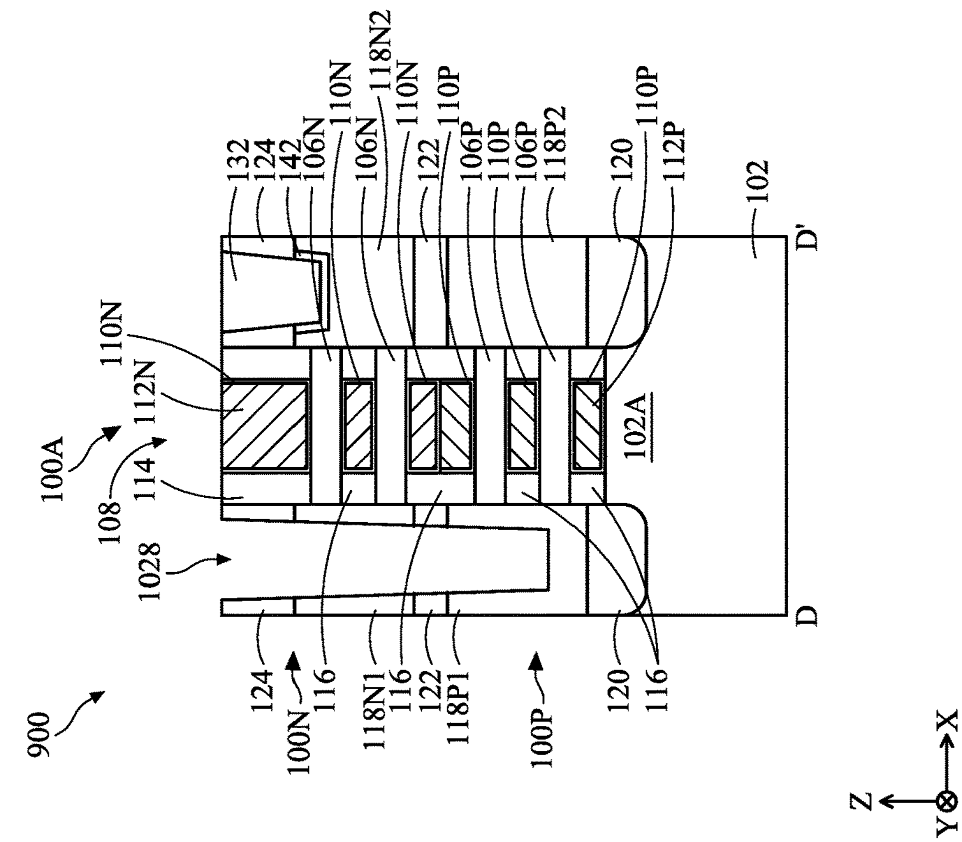

FIG. 26B

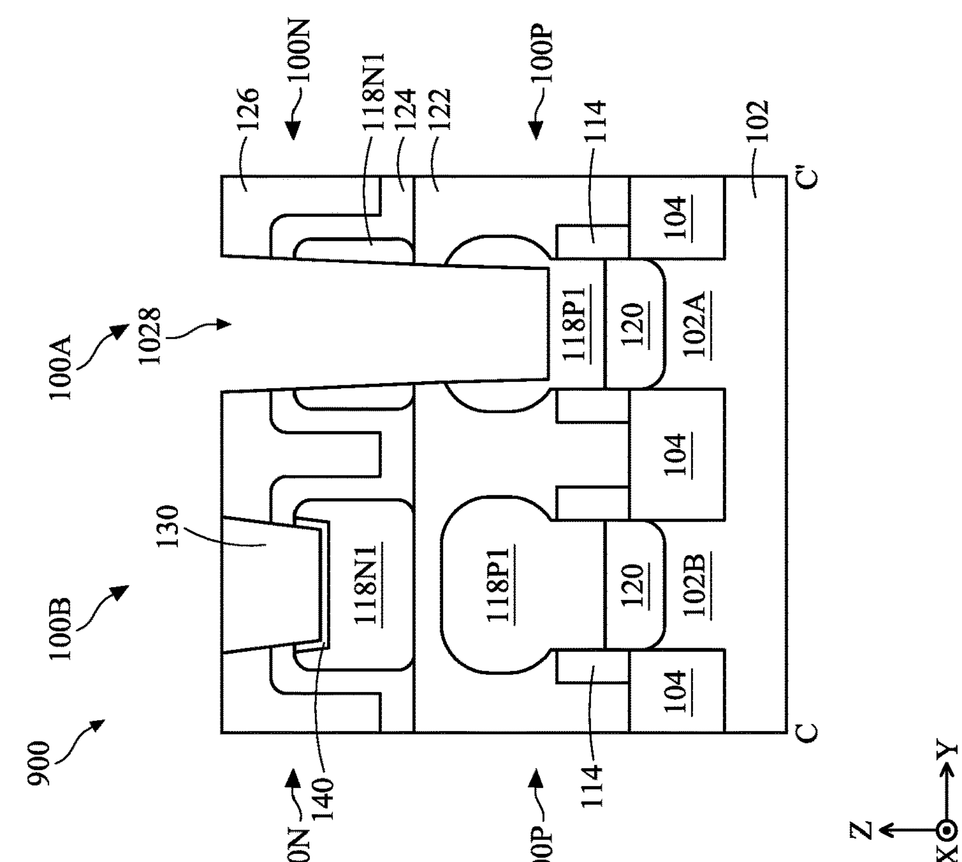

FIG. 26A

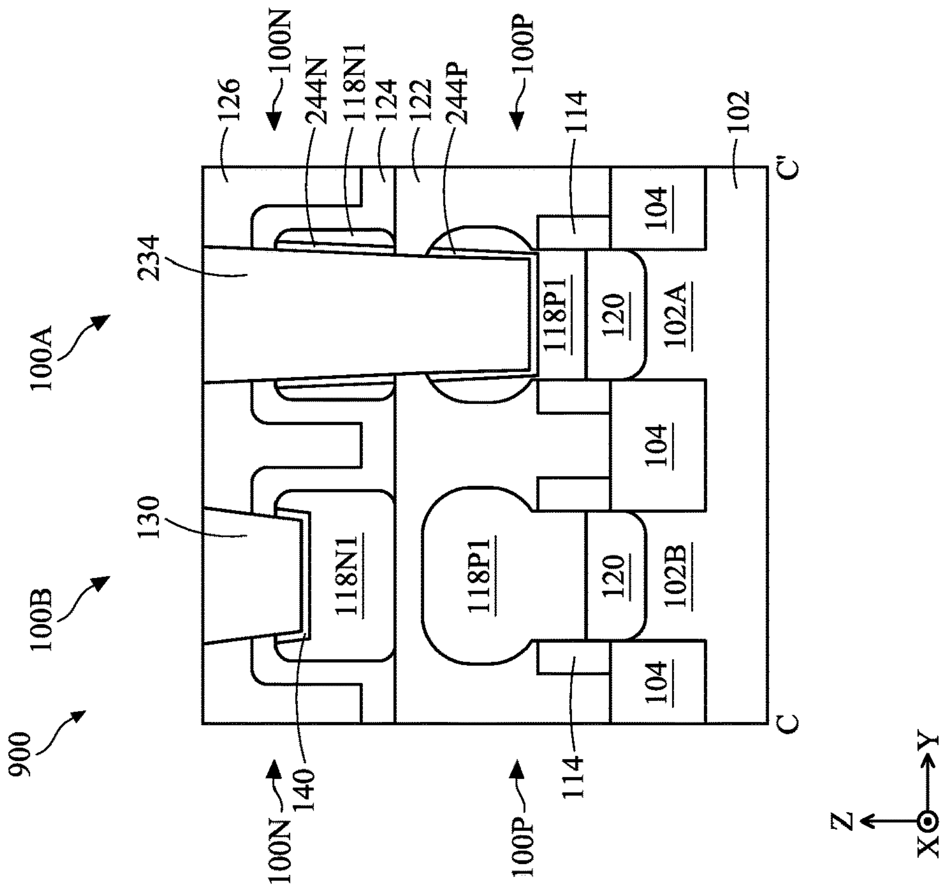

FIG. 27A

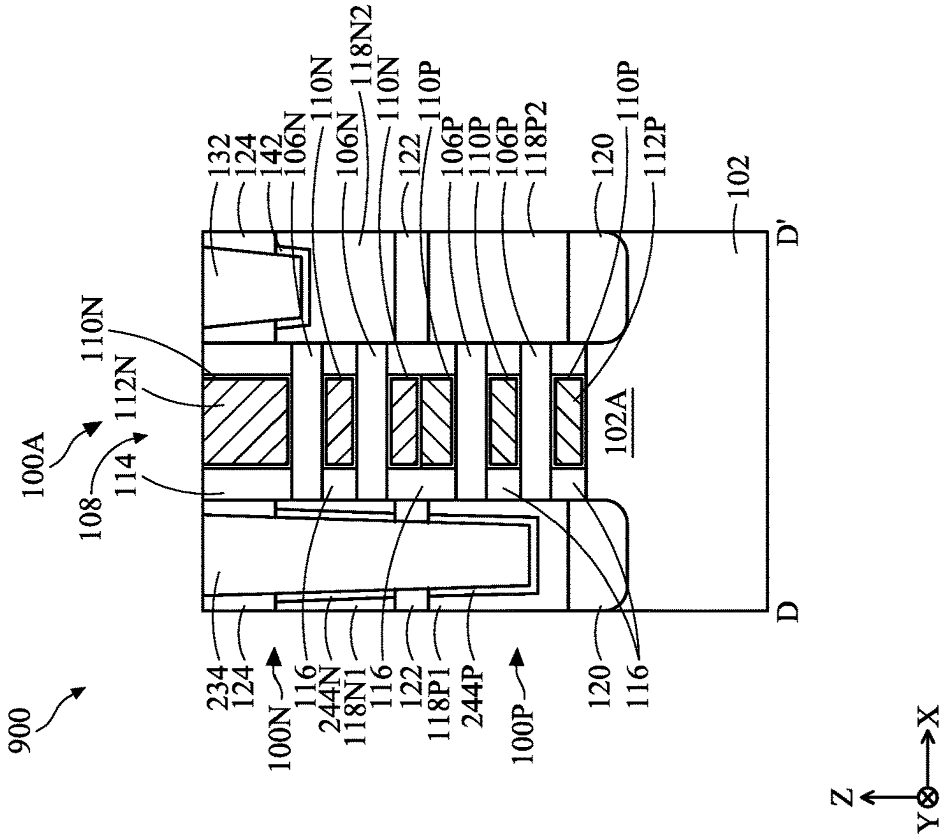

FIG. 27B

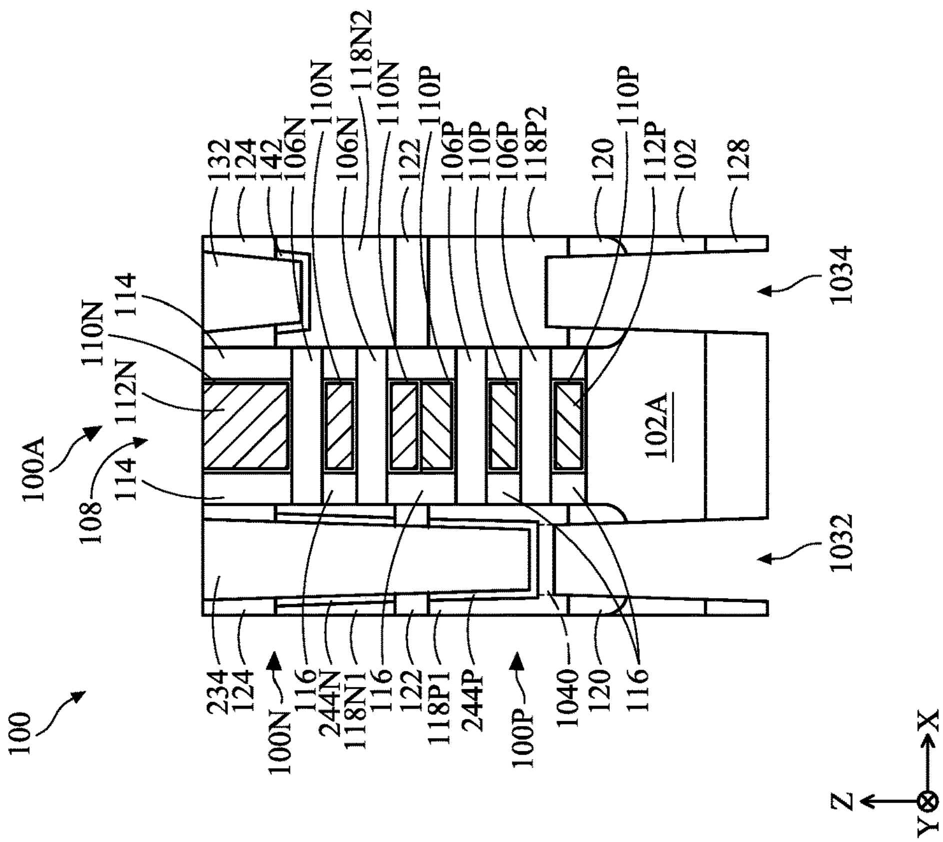

FIG. 28B

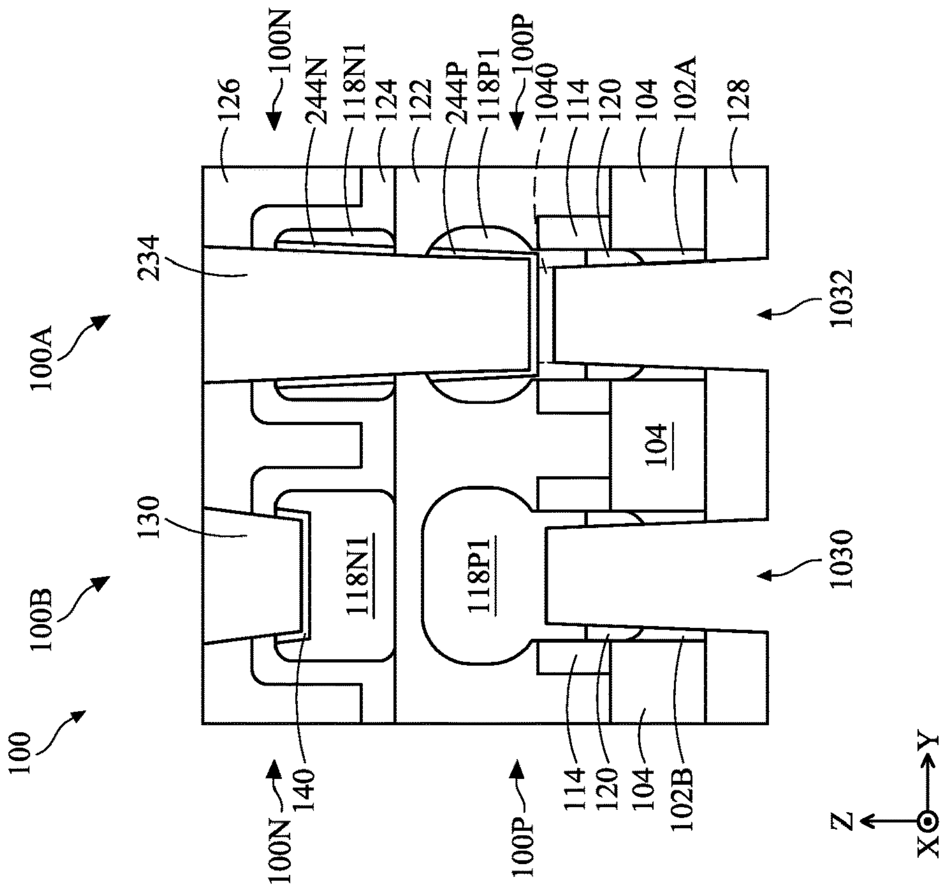

FIG. 28A

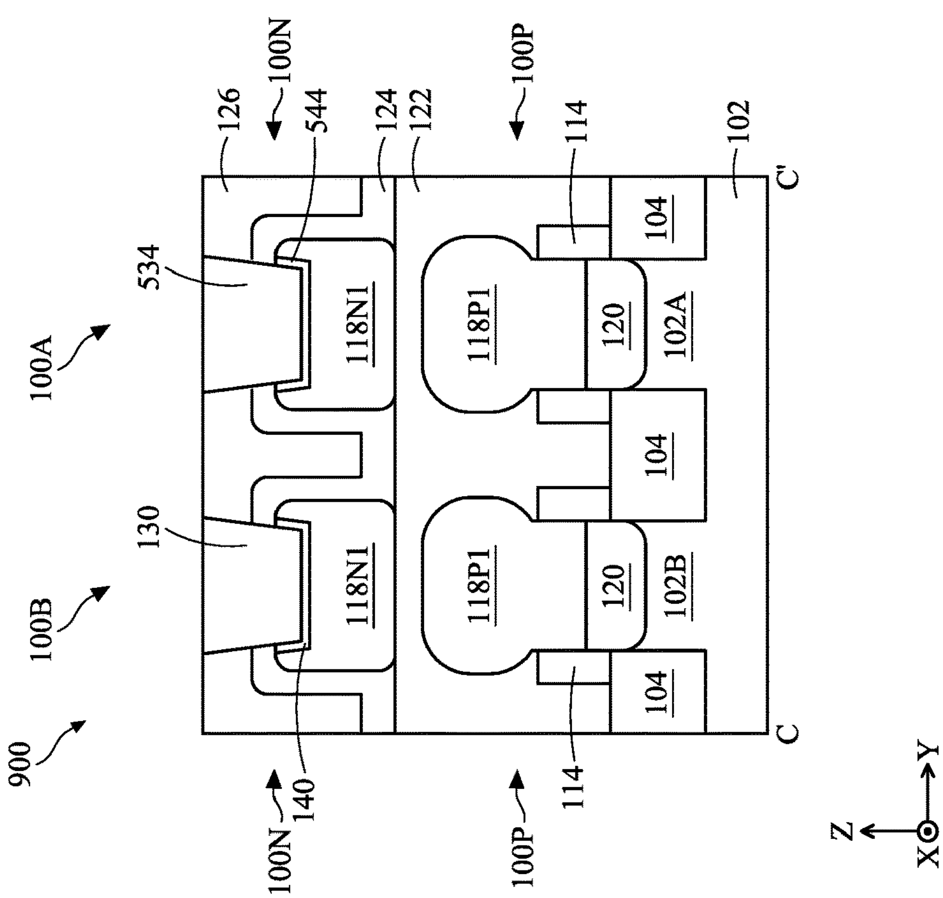

FIG. 29A

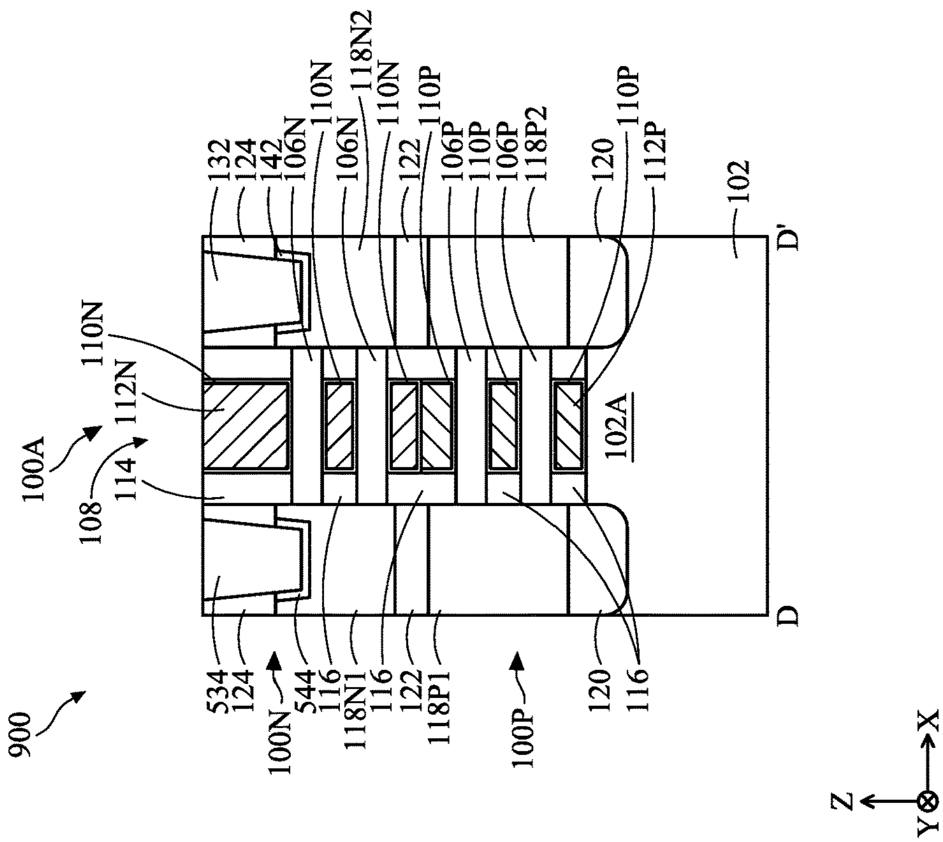

FIG. 29B

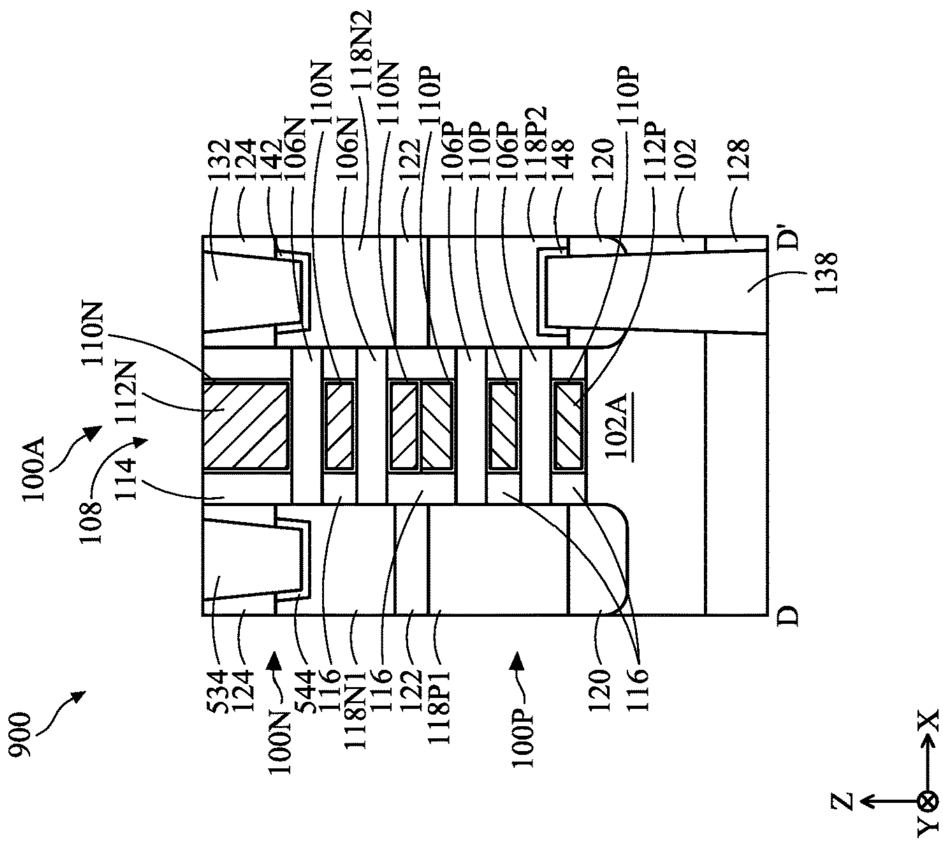

FIG. 30B

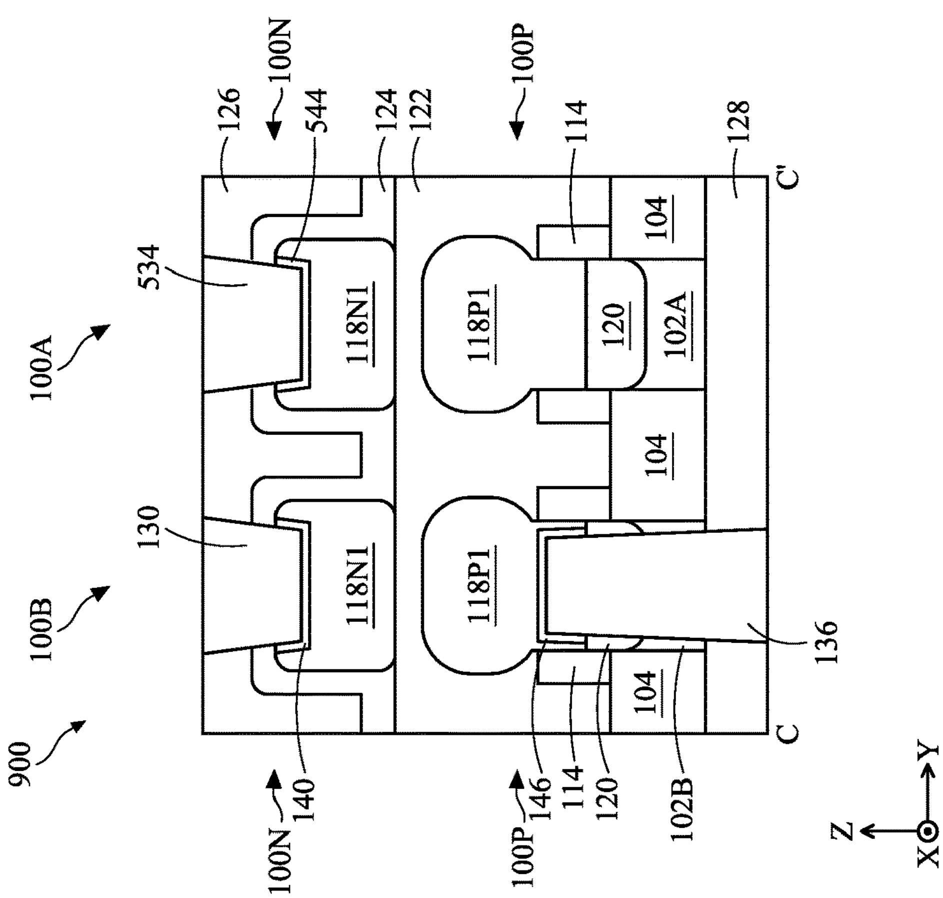

FIG. 30A

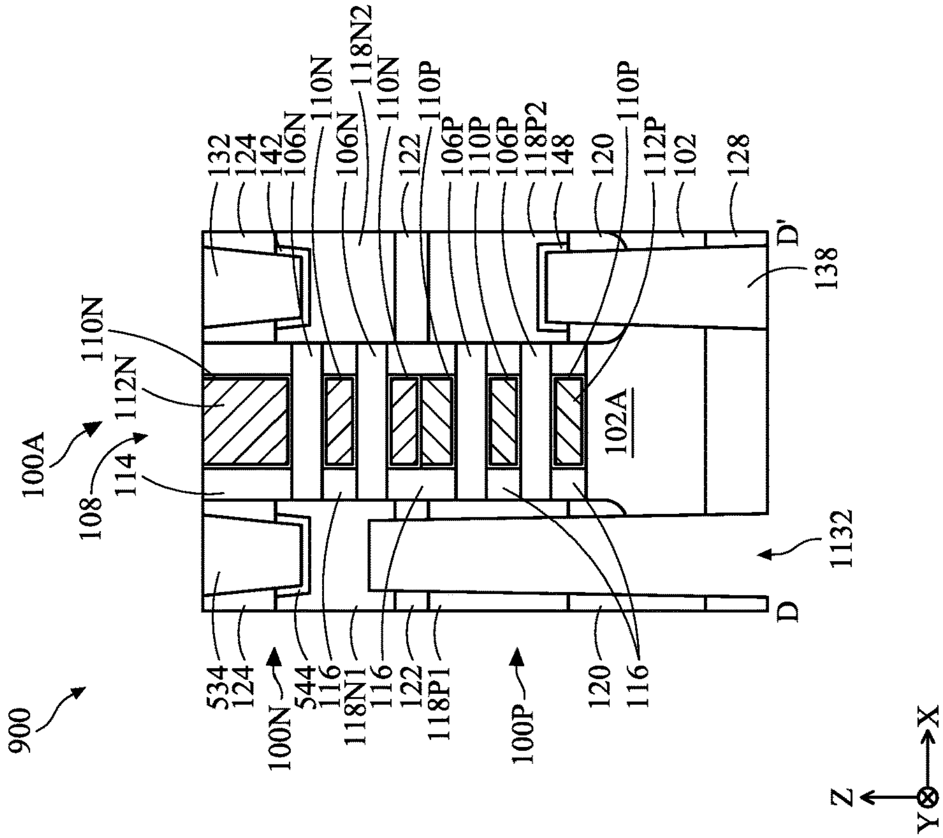

FIG. 31B

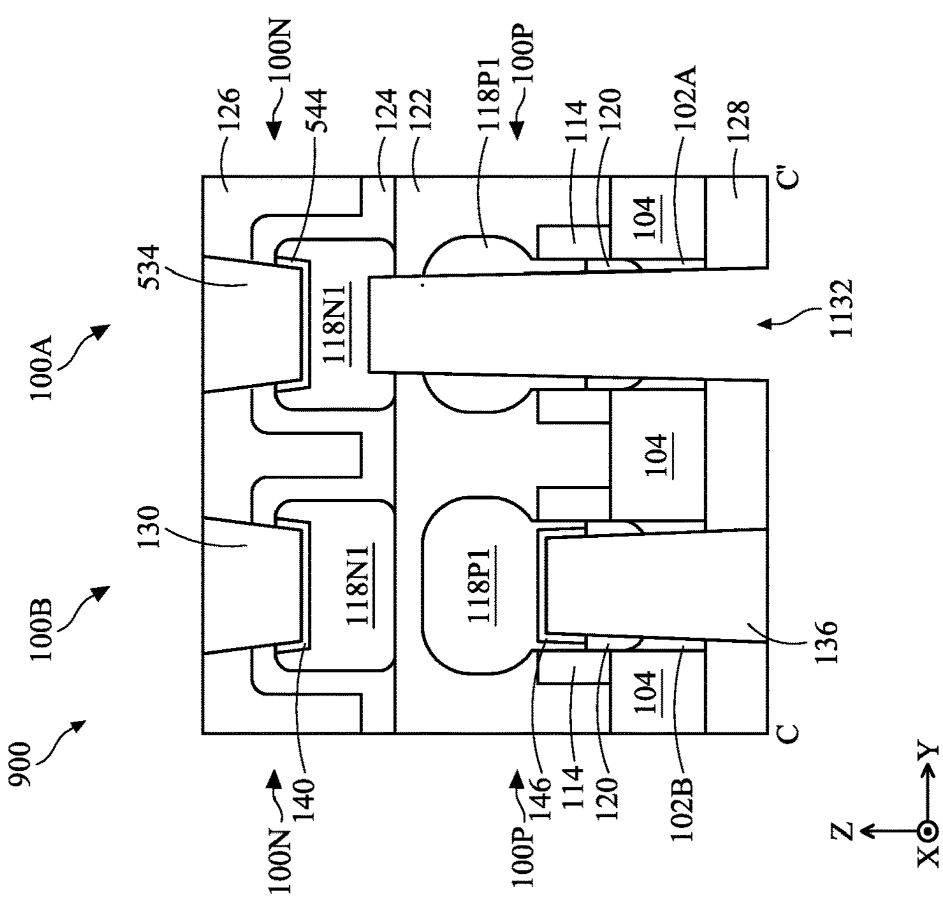

FIG. 31A

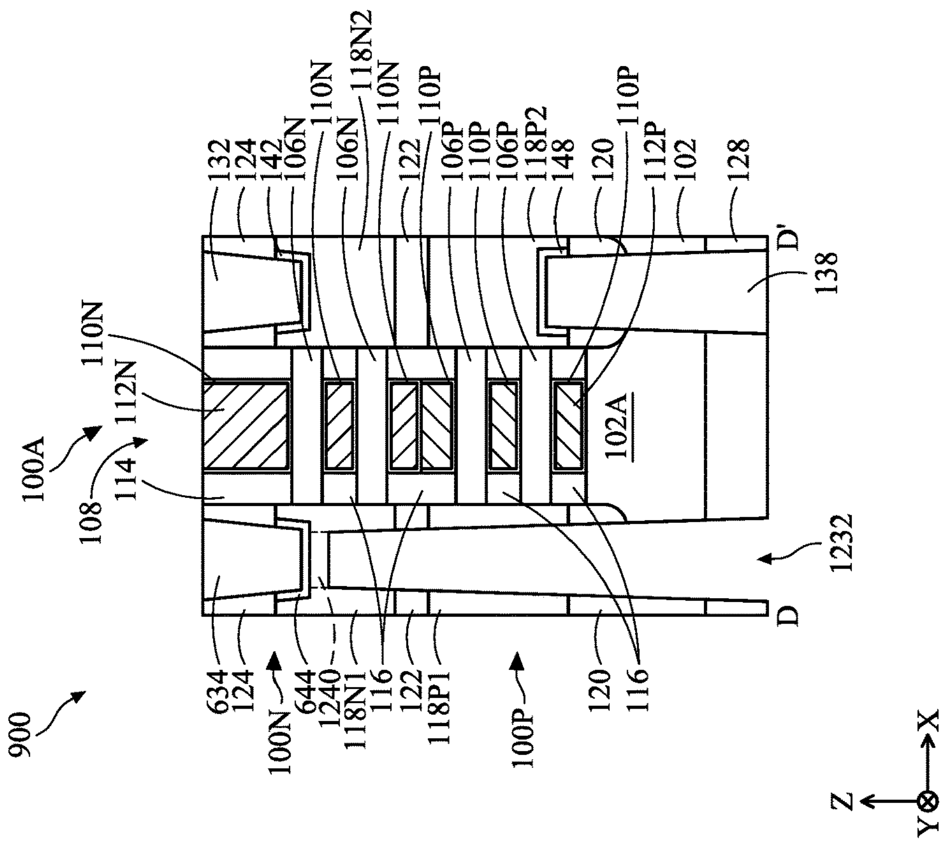

FIG. 32B

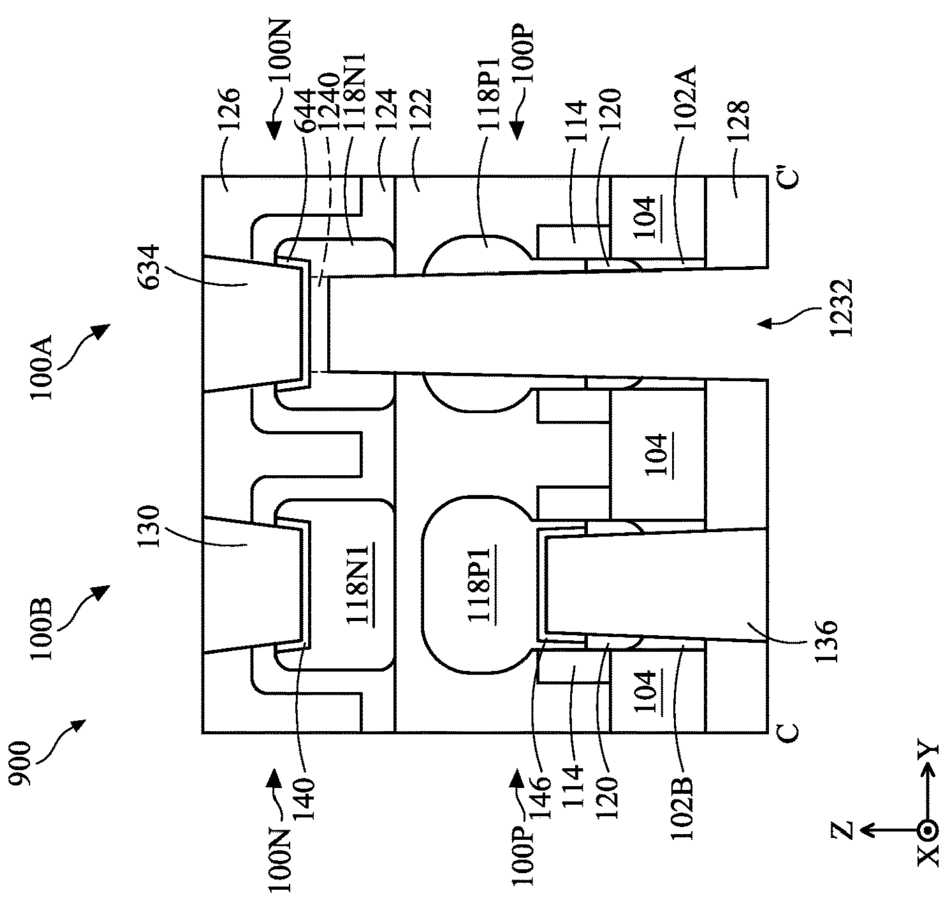

FIG. 32A

SEMICONDUCTOR STRUCTURE AND METHOD FOR FORMING THE SAME

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed.

[0002] As integrated circuit (IC) technologies progress towards smaller technology nodes, gate-all-around (GAA) devices have been introduced to improve gate control by increasing gate-channel coupling, reducing off-state current, and reducing short-channel effects (SCEs). As GAA devices continue to be developed, complementary metal-oxide-semiconductor field effect transistors (CMOSFET or CFET) has been provided due to their high noise immunity and low static power consumption. However, although existing technologies for fabricating CFETs have been generally adequate for their intended purposes, they have not been entirely satisfactory in all respects.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. **1**A is a Y-Z cross-sectional view of a semiconductor device, in accordance with some embodiments of the present disclosure.

[0005] FIG. **1**B is an X-Z cross-sectional view of the semiconductor device, in accordance with some embodiments of the present disclosure.

[0006] FIG. **2**A is a Y-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0007] FIG. **2**B is an X-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0008] FIG. **3**A is a Y-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0009] FIG. **3**B is an X-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0010] FIG. **4**A is a Y-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0011] FIG. **4**B is an X-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0012] FIG. **5**A is a Y-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0013] FIG. **5**B is an X-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0014] FIG. **6**A is a Y-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0015] FIG. **6**B is an X-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0016] FIG. **7**A is a Y-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0017] FIG. **7**B is an X-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0018] FIG. **8**A is a Y-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0019] FIG. **8**B is an X-Z cross-sectional view of the semiconductor device, in accordance with some alternative embodiments of the present disclosure.

[0020] FIG. **9** is a perspective view of a workpiece at a fabrication stage, in accordance with some embodiments of the present disclosure.

[0021] FIGS. **10**, **11**A, **12**A, **13**A, **14**A, **15**A, **16**A, **17**A, **18**A, **19**A, **22**A, **23**A, **24**A, **25**A, **26**A, **27**A, **28**A, **29**A, **30**A, **31**A, and **32**A are Y-Z cross-sectional views of the workpiece at various fabrication stages along line C-C' of FIG. **9**, in accordance with some embodiments of the present disclosure.

[0022] FIGS. **12**B, **13**B, **14**B, **15**B, **16**B, **17**B, **18**B, **19**B, **20**A, **21**A, **22**B, **23**B, **24**B, **25**B, **26**B, **27**B, **28**B, **29**B, **30**B, **31**B, and **32**B are X-Z cross-sectional views of the workpiece at various fabrication stages along line D-D' of FIG. **9**, in accordance with some embodiments of the present disclosure.

[0023] FIGS. **11**B, **20**B, and **21**B are Y-Z cross-sectional views of the workpiece at various fabrication stages along line E-E' of FIG. **9**, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0024] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0025] Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0026] The present disclosure is generally related to semiconductor devices, and more particularly to field-effect transistors (FETs), such as three-dimensional complementary field effect transistors (CFETs) with gate-all-around (GAA) structures. Generally, a CFET may include an n-type FET (NFET) and a p-type FET (PFET) disposed vertically with a plurality of vertically stacked sheets (e.g., nanosheets), wires (e.g., nanowires), or rods (e.g., nanorods) in a channel region of the CFET, thereby allowing better gate control, lowered leakage current, and improved scaling capability for various IC applications.

[0027] The GAA structures may be patterned by any suitable method. For example, the GAA structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally speaking, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over a substrate and is patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

[0028] In the existing CFET device, source/drain (S/D) features of NFET and PFET may sometimes be connected together by a frontside S/D local-interconnection (LI) contact, and the frontside S/D LI contact may be connected to signals. For example, in a static random access memory (SRAM) device, a frontside S/D LI contact may connect drains of a pull-down (PD) transistor and a pull-up (PD) transistor that form an inverter, and then connect to a pass-gate (PG) transistor and another inverter of the SRAM device (i.e., signals). However, it results in signal routing being restricted to the frontside of the device. In advanced technology node, since the geometric size is decreased, the routing restricted to the frontside of the device may not provide interconnection that is good enough. Therefore, a novel structure and fabricating method are needed to provide flexibility for the signal routing.

[0029] Embodiments of the present disclosure offer advantages over the existing art, though it is understood that other embodiments may offer different advantages, not all advantages are necessarily discussed herein, and no particular advantage is required for all embodiments. For example, embodiments discussed herein include methods and structures that have a backside S/D contact that corresponds to the frontside S/D LI contact and is formed on the backside of the device. In this way, the signals could be transmitted between the frontside S/D LI contact and the backside S/D contact. As a result, the signal routing can be arranged on the frontside of the device (e.g., by connecting to the frontside S/D LI contact), arranged on the backside of the device (e.g., by connecting to the backside S/D contact), or distributed on both the frontside and the backside (e.g., by connecting to both the frontside S/D LI contact and the backside S/D contact). Therefore, the flexibility for the signal routing is provided. Similarly, alternative embodiments discussed herein include methods and structures that have a frontside S/D contact and a corresponding backside S/D LI contact that connects S/D features of NFET and PFET together. It also provides flexibility for the signal routing. On the other hand, the embodiments provided by the present disclosure are not limited to the signal routing, they can further be applied to any device having a S/D LI contact to provide the flexibility for the routing of the interconnection structure.

[0030] The various aspects of the present disclosure will now be described in more detail with reference to the figures. For avoidance of doubts, an X-direction, a Y-direction, and a Z-direction in the figures are perpendicular to one another and are used consistently. Throughout the present disclosure, like reference numerals denote like features unless otherwise indicated.

[0031] FIG. **1**A is a Y-Z cross-sectional view of a semiconductor device **100**, in accordance with some embodiments of the present disclosure. FIG. **1**B is an X-Z cross-sectional view of the semiconductor device **100** along line B-B' of FIG. **1**A, in accordance with some embodiments of the present disclosure.

[0032] Referring to FIGS. **1**A and **1**B, the semiconductor device **100** includes two complementary field effect transistors (CFETs) **100**A and **100**B that are arranged in the Y-direction, in accordance with some embodiments. The CFETs **100**A and **100**B have similar structure and features. Furthermore, each of the CFETs **100**A and **100**B has a p-type field effect transistor (PFET) **100**P and an n-type field effect transistors (NFETs) **100**N. In each of the CFETs **100**A and **100**B, the NFET **100**N is disposed over (or vertically overlaps) the PFET **100**P in the Z-direction, as shown in FIGS. **1**A and **1**B. In other embodiments, the PFET may be disposed over the NFET in the Z-direction.

[0033] The semiconductor device **100** further includes a substrate **102**, as shown in FIGS. **1**A and **1**B. The substrate **102** includes a base portion **102**A for the CFET **100**A and a base portion **102**B for the CFET **100**B, which are protruded from the substrate **102** under the nanostructures **106**P and **106**N. Subsequent features for the CFETs **100**A and **100**B are formed over the base portions **102**A and **102**B of the substrate **102**, as described in further detail below. In some embodiments, after the resultant NFETs **100**N and PFETs **100**P of the CFETs **100**A and **100**B are formed, the substrate **102** may be thinned (or partially removed) by a suitable process (e.g., a chemical mechanical polishing (CMP) process) for forming backside interconnection.

[0034] In some embodiments, the semiconductor device **100** further includes isolation structures **104** in and/or over the substrate **102**. The isolation structures **104** are formed between the base portions **102**A and **102**B of the substrate **102**. In some embodiments, top surfaces of the isolation structures **104** are lower than top surfaces of the substrate **102** (more specifically, top surfaces of the base portions **102**A and **102**B).

[0035] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes two groups of nanostructures, such as a group of nanostructures **106**P and a group of nanostructures **106**N (which may be collectively referred to as the nanostructures **106**), in accordance with some embodiments. In some embodiments, the nanostructures **106** may also be referred to as channels, channel layers, nanosheets, or nanowires. In some embodiments, the nanostructures **106**N are disposed over (or vertically overlap) the nanostructures **106**P in the Z-direction. The nanostructures **106**P are used for the PFETs in the CFETs and the nanostructures **106**N are used for the NFETs in the CFETs. Furthermore, the nanostructures **106** are suspended over the base portions **102**A and **102**B of the substrate **102**.

[0036] In some embodiments, the nanostructures **106** are extended in the X-direction and vertically stacked (or arranged) in the Z-direction. Specifically, the nanostructures **106** are spaced apart from each other in the Z-direction. In some embodiments, the distance between the topmost nanostructure **106**P and the bottommost nanostructure **106**N is greater than the distance between two nanostructures **106**P or the distance between two nanostructures **106**N, as shown in FIG. **1**B.

[0037] In the embodiment depicted in FIG. **1**B, two nanostructures **106** are vertically stacked (or vertically arranged) from each other in the Z-direction for one transistor. For example, in CFET **100**A, the NFET **100**N has two nanostructures **106**N vertically stacked from each other in the Z-direction, and the PFET **100**P has two nanostructures **106**P vertically stacked from each other in the Z-direction. However, there may be another appropriate number of nanostructures in one transistor. For example, there may be 1, 2, 3, 4, or more than 4 nanostructures **106** in one transistor.

[0038] In some embodiments, the semiconductor device **100** further includes a gate structure **108** wrapped around the nanostructures **106**. Specifically, the gate structure **108** wraps around the nanostructures **106**P in the PFETs **100**P and the nanostructures **106**N in the NFETs **100**N. The CFETs **100**A and **100**B share the gate structure **108**. More specifically, the gate structure **108** extend in the Y-direction to wrap around the nanostructures **106** in the CFET **100**A and the CFET **100**B. In some embodiments, the gate structure **108** has a length in the X-direction (i.e., gate length) that is in a range from about 6 nm (nanometer) to about 20 nm.

[0039] The gate structure **108** has a gate dielectric layer **110**P, a gate electrode layer **112**P, a gate dielectric layer **110**N, and a gate electrode layer **112**N (the gate dielectric layers **110**P and **110**N may be collectively referred to as the gate dielectric layers **110**, and the gate electrode layers **112**P and **112**N may be collectively referred to as the gate electrode layers **112**). As shown in FIG. **1**B, the gate dielectric layer **110**P wraps around each of the nanostructures **106**P in the PFETs **100**P, and the gate dielectric layer **110**N wraps around each of the nanostructures **106**N in the NFETs **100**N. The gate electrode layer **112**P wraps around the gate dielectric layer **110**P and each of the nanostructures **106**P in the PFETs **100**P, and the gate electrode layer **112**N wraps around the gate dielectric layer **110**N and each of the nanostructures **106**N in the NFETs **100**N.

[0040] In some embodiments, the gate dielectric layer **110**P is also formed on top surfaces of the isolation structures **104** and top surfaces and sidewalls of the substrate **102** (more specifically, the top surfaces and sidewalls of the base portions **102**A and **102**B) (not shown). The gate dielectric layer **110**N is also formed on the top surface of the gate electrode layer **112**P, as shown in FIG. **1**B. In some embodiments, the gate dielectric layer **110**P is further formed on sidewalls of inner spacers **116** (discussed below), and the gate dielectric layer **110**N is further formed on sidewalls of the gate spacer **114** (discussed below) and the inner spacers **116**, as shown in FIG. **1**B. In some embodiments, the gate dielectric layer **110**P is in contact with the gate dielectric layer **110**N, as shown in FIG. **1**B.

[0041] In some embodiments, the gate structure **108** further includes an interfacial layer (not shown, such as having silicon dioxide, silicon oxynitride, or other suitable materials) between the gate dielectric layers **110** and the nanostructures **106**. As discussed above, the gate electrode layers **112**P and **112**N are formed to wrap around the gate dielectric layers **110**P and **110**N and the nanostructures **106**, as shown in FIG. **1**B.

[0042] In some embodiments, the semiconductor device **100** further includes gate spacers **114** on opposite sides of the gate structure **108**. More specifically, the gate spacers **114** are on sidewalls of the gate structures **108** and over the nanostructures **106**, as shown in FIG. **1**B. Furthermore, the gate spacers **114** extend lengthwise in the Y-direction (e.g., parallel to the gate structure **108**), and are on opposite sides (or on opposite sidewalls) of the gate structure **108** in the X-direction. The gate spacers **114** are over the nanostructures **106** and on top sidewalls of the gate structures **108**, and thus are also referred to as gate top spacers or top spacers.

[0043] In some embodiments, the semiconductor device **100** further includes inner spacers **116** on opposite sides of the gate structure **108**. More specifically, the inner spacers **116** are on the sidewalls of the gate structure **108**, and below the gate spacers **114** and the topmost nanostructure **106**N. As shown in FIG. **1**B, the inner spacers **116** are also vertically between adjacent nanostructures **106**N, vertically between adjacent nanostructures **106**N and **106**P, vertically between adjacent nanostructures **106**P, and vertically between the bottommost nanostructures **106**P and the substrate **102**, in accordance with some embodiments. In some embodiments, a thickness of the inner spacers **116** vertically between adjacent nanostructures **106**N and **106**P in the Z-direction is greater than thicknesses of other inner spacers **116**, as shown in FIG. **1**B. Furthermore, the inner spacers **116** are laterally between the source/drain features **118**N/**118**P (described below) and the gate structure **108** in the X-direction.

[0044] Still referring to FIGS. **1**A and **1**B, each of the CFETs **100**A and **100**B includes source/drain features **118P1**, **118P2** (may be collectively referred to as the source/drain features **118**P) and source/drain features **118N1**, **118N2** (may be collectively referred to as the source/drain features **118**N) over the substrate **102**, in accordance with some embodiments. More specifically, the source/drain features **118P1** and **118P2** are disposed over the substrate **102**, and the source/drain features **118N1** and **118N2** are disposed over (or vertically overlap) the source/drain features **118P1** and **118P2**, respectively. In some embodiments, the source/drain features **118**N are vertically separated from the source/drain features **118**P in the Z-direction, as shown in FIGS. **1**A and **1**B. In some embodiments, the source/drain features **118**N are disposed on opposite sides of the gate structure **108** in the X-direction to form the NFETs **100**N, as shown in FIG. **1**B. Similarly, the source/drain features **118**P are disposed on opposite sides of the gate structure **108**P in the X-direction to form the PFETs **100**P, as shown in FIG. **1**B.

[0045] The nanostructures **106** extend in the X-direction to connect one source/drain feature **118**P/**118**N to the other source/drain feature **118**P/**118**N, in accordance with some embodiments. More specifically, the source/drain features **118**P**1** and **118**P**2** are disposed on opposite sides of the nanostructures **106**P and the source/drain features **118**N**1** and **118**N**2** are disposed on opposite sides of the nanostructures **106**N in the X-direction. Therefore, the source/drain features **118**P are attached and electrically connected to the nanostructures **106**P in the X-direction, and the source/drain features **118**N are attached and electrically connected to the nanostructures **106**N in the X-direction. The source/drain features **118**P/**118**N may also be referred to as source/drain, or source/drain regions. In some embodiments, source/drain feature(s) may refer to a source or a drain, individually or collectively dependent upon the context.

[0046] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes bottom isolation layers **120** under the source/drain features **118**N and **118**P and over the substrate **102** in the Z-direction, in accordance with some embodiments. In some embodiments, the bottom isolation layers **120** are vertically between and in contact with the source/drain features **118**P and the substrate **102** in the Z-direction. In some embodiments, the top surfaces of the bottom isolation layers **120** are higher than the bottommost surface of the gate structure **108**, as shown in FIG. **1**B. In some aspects, the top surfaces of the bottom isolation layers **120** are higher than the topmost surfaces of the substrate **102** (i.e., the top surfaces of the base portions **102**A and **102**B), so as to ensure that the bottom isolation layers **120** separate the source/drain features **118**P from the substrate **102**. The bottom isolation layers **120** may be a single layer structure or a multi-layer structure.

[0047] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes an interlayer dielectric (ILD) layer **122** over the substrate **102**, the isolation structures **104**, and the source/drain features **118**P, in accordance with some embodiments. In some embodiments, the ILD layer **122** is also between and fill the spaces between the source/drain features **118**P in the Y-direction, as shown in FIG. **1**A. In these embodiments, the source/drain features **118**P are surrounded by the ILD layer **122** in the Y-direction. In some embodiments, the ILD layer **122** is vertically between and in contact with the source/drain features **118**P and the source/drain features **118**N in the Z-direction. For example, the ILD layer **122** is vertically between the source/drain feature **118**P**1** and **118**N**1**, and is vertically between the source/drain feature **118**P**2** and **118**N**2**. Thus, the source/drain features **118**P are separated and electrically isolated from the source/drain features **118**N by the ILD layer **122**.

[0048] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes contact etch stop layers (CESLs) **124** and an ILD layer **126** over the CESLs **124**, in accordance with some embodiments. The CESLs **124** are over the source/drain features **118**N and the ILD layer **122**, and the ILD layer **126** is formed to fill the space between the source/drain features **118**N. Specifically, the CESLs **124** are conformally formed on the sidewalls of the gate spacers **114**. In some embodiments, the CESLs **124** are also conformally formed on the top surfaces and the sidewalls of the source/drain features **118**N, as shown in FIG. **1**A. The ILD layer **126** is formed over and between the CESLs **124** to fill the space in the CESLs **124** and between the gate spacers **114**.

[0049] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes an ILD layer **128** under the substrate **102**, in accordance with some embodiments.

[0050] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes source/drain contacts **130** and **132**, in accordance with some embodiments. In some embodiments, the source/drain contacts **130** and **132** pass through the ILD layer **126**, the CESLs **124**, and portions of the source/drain features **118**N, so as to be in contact with and electrically connected to the source/drain features **118**N. In other words, the source/drain contacts **130** and **132** extend through the ILD layer **126** and the CESLs **124**, and partially extend into the source/drain features **118**N. In some embodiments, the source/drain contact **130** is in contact with and electrically connected to the source/drain feature **118**N**1** of the CFET **100**B, as shown in FIG. **1**A. In some embodiments, the source/drain contact **132** is in contact with and electrically connected to the source/drain feature **118**N**2** of the CFET **100**A, as shown in FIG. **1**B.

[0051] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes silicide layers **140** and **142** that are between the source/drain contacts **130** and **132** and the source/drain features **118**N, respectively, in accordance with some embodiments. In some embodiments, the silicide layer **140** is formed on the interface between the source/drain contact **130** and the source/drain feature **118**N**1** of the CFET **100**B, as shown in FIG. **1**A. That is, the silicide layer **140** is formed on sidewalls and the bottom surface of a portion of the source/drain contact **130** inside the source/drain feature **118**N**1** of the CFET **100**B. In some embodiments, the silicide layer **142** is formed on the interface between the source/drain contact **132** and the source/drain feature **118**N**2** of the CFET **100**A, as shown in FIG. **1**B. That is, the silicide layer **142** is formed on sidewalls and the bottom surface of a portion of the source/drain contact **132** inside the source/drain feature **118**N**2** of the CFET **100**A.

[0052] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes a source/drain contact **134**, in accordance with some embodiments. In some embodiments, the source/drain contact **134** passes through the ILD layer **126**, the CESLs **124**, the source/drain feature **118**N, the ILD layer **122**, and portions of the source/drain feature **118**P, so as to be in contact with and electrically connected to both of the source/drain features **118**N and **118**P. In other words, the source/drain contact **134** extends through the ILD layer **126**, the CESLs **124**, the source/drain feature **118**N, and the ILD layer **122**, and partially extend into the source/drain feature **118**P. In some embodiments, the source/drain contact **134** is in contact with and electrically connected to the source/drain feature **118**N**1** and the source/drain feature **118**P**1** of the CFET **100**A, as shown in FIGS. **1**A and **1**B. That is, the source/drain contact **134** electrically connects the source/drain feature **118**N**1** and the source/drain feature **118**P**1** of the CFET **100**A together.

[0053] In some embodiments, since the source/drain contacts **130**, **132**, and **134** are formed on the frontside of the semiconductor device **100**, the source/drain contacts **130**, **132**, and **134** may be referred to as frontside source/drain contacts. Furthermore, since the source/drain contact **134** connects the source/drain features **118**N and **118**P of NFET **100**N and PFET **100**P together, the source/drain contact **134** may be referred to as the frontside S/D local-interconnection (LI) contact.

[0054] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes silicide layers **144**N and **144**P that are between the source/drain contact **134** and the source/drain features **118**N and **118**P, respectively, in accordance with some embodiments. In some embodiments, the silicide layer **144**N is formed on the interface between the source/drain contact **134** and the source/drain feature **118**N**1** of the CFET **100**A, as shown in FIGS. **1**A and **1**B. That is, the silicide layer **144**N is formed on sidewalls of a portion of the source/drain contact **134** inside the source/drain feature **118**N**1** of the CFET **100**A. In some embodiments, the silicide layer **144**P is formed on the interface between the source/drain contact **134** and the source/drain feature **118**P**1** of the CFET **100**A, as shown in FIGS. **1**A and **1**B. That is, the silicide layer **144**P is formed on sidewalls and the bottom surface of a portion of the source/drain contact **134** inside the source/drain feature **118**P**1** of the CFET **100**A. In some embodiments, the material of the silicide layer **144**N is different from the material of the silicide layer **144**P.

[0055] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes source/drain contacts **136**, **137**, and **138**, in accordance with some embodiments. In some embodiments, the source/drain contacts **136**, **137**, and **138** pass through the ILD layer **128**, the substrate **102**, the bottom isolation layers **120**, and portions of the source/drain features **118**P, so as to be in contact with and electrically connected to the source/drain features **118**P. In other words, the source/drain contacts **136**, **137**, and **138** extend through the ILD layer **128**, the substrate **102**, and the bottom isolation layers **120**, and partially extend into the source/drain features **118**P. In some embodiments, since the source/drain contacts **136**, **137**, and **138** are formed on backside of the semiconductor device **100**, the source/drain contacts **136**, **137**, and **138** may be referred to as backside source/drain contacts or bottom source/drain contacts.

[0056] In some embodiments, the source/drain contact **136** is in contact with and electrically connected to the source/drain feature **118**P**1** of the CFET **100**B, as shown in FIG. **1**A. In some embodiments, the source/drain contact **137** is in contact with and electrically connected to the source/drain feature **118**P**1** of the CFET **100**A, as shown in FIGS. **1**A and **1**B. In some embodiments, the source/drain contact **138** is in contact with and electrically connected to the source/drain feature **118**P**2** of the CFET **100**A, as shown in FIG. **1**B.

[0057] Still referring to FIGS. **1**A and **1**B, the semiconductor device **100** further includes silicide layers **146**, **147**, and **148** that are between the source/drain contacts **136**, **137**, and **138** and the source/drain features **118**P, respectively, in accordance with some embodiments. In some embodiments, the silicide layer **146** is formed on the interface between the source/drain contact **136** and the source/drain feature **118**P**1** of the CFET **100**B, as shown in FIG. **1**A. That is, the silicide layer **146** is formed on sidewalls and the top surface of a portion of the source/drain contact **136** inside the source/drain feature **118**P**1** of the CFET **100**B.

[0058] In some embodiments, the silicide layer **147** is formed on the interface between the source/drain contact **137** and the source/drain feature **118**P**1** of the CFET **100**A, as shown in FIGS. **1**A and **1**B. That is, the silicide layer **147** is formed on sidewalls and the top surface of a portion of the source/drain contact **137** inside the source/drain feature **118**P**1** of the CFET **100**A. In some embodiments, the material of the silicide layer **147** is different from the materials of the silicide layers **144**N and **144**P. In some embodiments, the silicide layer **148** is formed on the interface between the source/drain contact **138** and the source/drain feature **118**P**2** of the CFET **100**A, as shown in FIG. **1**B. That is, the silicide layer **148** is formed on sidewalls and the top surface of a portion of the source/drain contact **138** inside the source/drain feature **118**P**2** of the CFET **100**A.

[0059] In some embodiments, the source/drain contact **134** is electrically connected to the source/drain contact **137** through the silicide layer **144**P, the source/drain feature **118**P**1** of the CFET **100**A, and the silicide layer **147**. In these embodiments, a portion of the source/drain feature **118**P**1** of the CFET **100**A is between the source/drain contact **134** (and the silicide layers **144**P formed thereon) and the source/drain contact **137** (and the silicide layers **147** formed thereon), as shown in FIGS. **1**A and **1**B. In other words, a horizontal portion of the silicide layer **144**P formed on the bottom surface of the source/drain contact **134** is separated from a horizontal portion of the silicide layer **147** formed on the top surface of the source/drain contact **137** by a portion of the source/drain feature **118**P**1** of the CFET **100**A.

[0060] In some embodiments, the source/drain contact **134** and/or the source/drain contact **137** are connected to signals. Since the source/drain contact **134** is electrically connected to the source/drain contact **137**, the signals could be transmitted between the source/drain contacts **134** and **137**. Therefore, the source/drain features **118**N**1** and **118**P**1** of the CFET **100**A connected together by the source/drain contact **134** can be connected to signals through the source/drain contact **134**, through the source/drain contact **137**, or through both of the source/drain contacts **134** and **137**. In this way, the signal routing can be arranged on the frontside of the semiconductor device **100** (by connecting to the source/drain contact **134**), arranged on the backside of the semiconductor device **100** (by connecting to the source/drain contact **137**), or distributed on the frontside and backside of the semiconductor device **100** (by connecting to both of the source/drain contacts **134** and **137**). Accordingly, the flexibility for the signal routing can be improved.

[0061] In some embodiments where the semiconductor device **100** is applied to a static random access memory (SRAM) device, the NFET **100**N and the PFET **100**P of the CFET **100**A may act as a PD transistor and a PU transistor and constitute an inverter. Furthermore, the source/drain features **118**N**1** and **118**P**1** of the CFET **100**A may act as drains of the PD transistor and the PU transistor, and are connected together by the source/drain contact **134**. In these embodiments, the signals connected to the source/drain features **118**N**1** and **118**P**1** of the CFET **100**A may come from another inverter and a PG transistor of the SRAM device.

[0062] In some embodiments, an interconnection structure (not shown) of the signal routing is arranged on the frontside of the semiconductor device **100**, and connects the source/drain contact **134** to the PG transistor and another inverter. In these embodiments, the signals coming from the PG transistor and another inverter are transmitted to the source/drain features **118**N**1** and **118**P**1** of the CFET **100**A through the source/drain contact **134**. In other embodiments, an interconnection structure (not shown) of the signal routing is arranged on the backside of the semiconductor device **100**, and connects the source/drain contact **137** to the PG transistor and another inverter. In these embodiments, the signals coming from the PG transistor and another inverter are transmitted to the source/drain features **118N1** and **118P1** of the CFET **100A** through the source/drain contacts **137** and **134**.

[0063] In certain embodiments, an interconnection structure (not shown) of the signal routing is distributed on both of the frontside and backside of the semiconductor device **100**, and connects the source/drain contacts **134** and **137** to the PG transistor and another inverter. For example, the interconnection structure of the signal routing may connect the source/drain contacts **134** and **137** to the PG transistor and another inverter, respectively. Alternatively, the interconnection structure of the signal routing may connect the source/drain contacts **134** and **137** to another inverter and the PG transistor, respectively. For example, the interconnection structure of the signal routing may first connect the source/drain contacts **134** and **137** together, and then connect them to both of the PG transistor and another inverter. In these embodiments, the signals coming from the PG transistor and another inverter are transmitted to the source/drain features **118N1** and **118P1** of the CFET **100A** through the source/drain contacts **137** and **134**.

[0064] In some embodiments, the source/drain contacts **130**, **132**, **136**, and **138** are connected to powers. For example, when the semiconductor device **100** is applied to a SRAM device, the source/drain contacts **130**, **132**, **136**, and **138** may connect to a positive power supply VDD or a negative power supply (or ground) VSS.

[0065] FIG. **2A** is a Y-Z cross-sectional view of a semiconductor device **200**, in accordance with alternative embodiments of the present disclosure. FIG. **2B** is an X-Z cross-sectional view of the semiconductor device **200** along line B-B' of FIG. **2A**, in accordance with alternative embodiments of the present disclosure. The semiconductor device **200** shown in FIGS. **2A** and **2B** may be similar to the semiconductor device **100** shown in FIGS. **1A** and **1B**, except the source/drain contacts **134**, **137** and the silicide layers **144N**, **144P**, **147** shown in FIGS. **1A** and **1B** are replaced by source/drain contacts **234**, **237** and silicide layers **244N**, **244P**, **247** shown in FIGS. **2A** and **2B**.

[0066] Referring to FIGS. **2A** and **2B**, similar to the source/drain contact **134** shown in FIGS. **1A** and **1B**, the source/drain contact **234** extends through the ILD layer **126**, the CESLs **124**, the source/drain feature **118N**, and the ILD layer **122**, and partially extend into the source/drain feature **118P**, so as to be in contact with and electrically connected to both of the source/drain features **118N** and **118P**, in accordance with some embodiments. In some embodiments, the source/drain contact **234** is in contact with the source/drain features **118N1** and **118P1** of the CFET **100A**, and electrically connects them together, as shown in FIGS. **2A** and **2B**. Similar to the source/drain contact **134**, the source/drain contact **234** may be referred to as the frontside S/D LI contact.

[0067] In some embodiments, similar to the silicide layer **144N** shown in FIGS. **1A** and **1B**, the silicide layer **244N** is formed on the interface between the source/drain contact **234** and the source/drain feature **118N1** of the CFET **100A**, as shown in FIGS. **2A** and **2B**. In some embodiments, similar to the silicide layer **144P** shown in FIGS. **1A** and **1B**, the silicide layer **244P** is formed on the interface between the source/drain contact **234** and the source/drain feature **118P1** of the CFET **100A**, as shown in FIGS. **2A** and **2B**. That is, the silicide layer **244P** is formed on sidewalls and the bottom surface of a portion of the source/drain contact **234** inside the source/drain feature **118P1** of the CFET **100A**. In some embodiments, the material of the silicide layer **244N** is different from the material of the silicide layer **244P**.

[0068] Still referring to FIGS. **2A** and **2B**, similar to the source/drain contact **137** shown in FIGS. **1A** and **1B**, the source/drain contact **237** extends through the ILD layer **128**, the substrate **102**, and the bottom isolation layer **120**, and partially extend into the source/drain feature **118P**, so as to be in contact with and electrically connected to the source/drain feature **118P**, in accordance with some embodiments. In some embodiments, the source/drain contact **237** is in contact with and electrically connected to the source/drain feature **118P1** of the CFET **100A**, as shown in FIGS. **2A** and **2B**. Similar to the source/drain contact **137**, the source/drain contact **237** may be referred to as backside source/drain contact or bottom source/drain contact.

[0069] In some embodiments, similar to the silicide layer **147** shown in FIGS. **1A** and **1B**, the silicide layer **247** is formed on the interface between the source/drain contact **237** and the source/drain feature **118P1** of the CFET **100A**, as shown in FIGS. **2A** and **2B**. That is, the silicide layer **247** is formed on sidewalls and the top surface of a portion of the source/drain contact **237** inside the source/drain feature **118P1** of the CFET **100A**. In some embodiments, the material of the silicide layer **247** is different from the materials of the silicide layers **244N** and **244P**.

[0070] In some embodiments, the source/drain contact **234** is electrically connected to the source/drain contact **237** through the silicide layer **244P** and the silicide layer **247**. In these embodiments, the silicide layer **244P** is in direct contact with the silicide layer **247** in the source/drain feature **118P1** of the CFET **100A**, as shown in FIGS. **2A** and **2B**. That is, a horizontal portion of the silicide layer **244P** formed on the bottom surface of the source/drain contact **234** is in direct contact with a horizontal portion of the silicide layer **247** formed on the top surface of the source/drain contact **237** in the source/drain feature **118P1** of the CFET **100A**.

[0071] In some embodiments, the source/drain contact **234** and/or the source/drain contact **237** are connected to signals. Similar to the source/drain contacts **134** and **137** described above, the arrangement of the source/drain contacts **234** and **237** can allow the signal routing to be arranged on the frontside, the backside, or both of the frontside and backside of the semiconductor device **200**. As a result, the flexibility for the signal routing can be improved. In some embodiments where the semiconductor device **200** is applied to a SRAM device, the source/drain contacts **234** and **237** may function as the source/drain contacts **134** and **137** in the embodiments where the semiconductor device **100** is applied to the SRAM device.

[0072] FIG. **3A** is a Y-Z cross-sectional view of a semiconductor device **300**, in accordance with alternative embodiments of the present disclosure. FIG. **3B** is an X-Z cross-sectional view of the semiconductor device **300** along line B-B' of FIG. **3A**, in accordance with alternative embodiments of the present disclosure. The semiconductor device **300** shown in FIGS. **3A** and **3B** may be similar to the semiconductor device **100** shown in FIGS. **1A** and **1B**, except the source/drain contacts **134**, **137** and the silicide layers **144N**, **144P**, **147** shown in FIGS. **1A** and **1B** are replaced by source/drain contacts **334**, **337** and silicide layers **344N**, **344P**, **347** shown in FIGS. **3A** and **3B**.

[0073] Referring to FIGS. **3A** and **3B**, similar to the source/drain contact **134** shown in FIGS. **1A** and **1B**, the source/drain contact **334** extends through the ILD layer **126**, the CESLs **124**, the source/drain feature **118**N, and the ILD layer **122**, and partially extend into the source/drain feature **118**P, so as to be in contact with and electrically connected to both of the source/drain features **118**N and **118**P, in accordance with some embodiments. In some embodiments, the source/drain contact **334** is in contact with the source/drain features **118N1** and **118P1** of the CFET **100**A, and electrically connects them together, as shown in FIGS. **3**A and **3**B. Similar to the source/drain contact **134**, the source/drain contact **334** may be referred to as the frontside S/D LI contact.

[0074] In some embodiments, similar to the silicide layer **144**N shown in FIGS. **1**A and **1**B, the silicide layer **344**N is formed on the interface between the source/drain contact **334** and the source/drain feature **118N1** of the CFET **100**A, as shown in FIGS. **3**A and **3**B. In some embodiments, similar to the silicide layer **144**P shown in FIGS. **1**A and **1**B, the silicide layer **344**P is formed on the interface between the source/drain contact **334** and the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **3**A and **3**B. That is, the silicide layer **344**P is formed on sidewalls and the bottom surface of a portion of the source/drain contact **334** inside the source/drain feature **118P1** of the CFET **100**A. In some embodiments, the material of the silicide layer **344**N is different from the material of the silicide layer **344**P.

[0075] Still referring to FIGS. **3**A and **3**B, similar to the source/drain contact **137** shown in FIGS. **1**A and **1**B, the source/drain contact **337** extends through the ILD layer **128**, the substrate **102**, and the bottom isolation layer **120**, and partially extend into the source/drain feature **118**P, so as to be in contact with and electrically connected to the source/drain feature **118**P, in accordance with some embodiments. In some embodiments, the source/drain contact **337** is in contact with and electrically connected to the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **3**A and **3**B. Similar to the source/drain contact **137**, the source/drain contact **337** may be referred to as backside source/drain contact or bottom source/drain contact.

[0076] In some embodiments, similar to the silicide layer **147** shown in FIGS. **1**A and **1**B, the silicide layer **347** is formed on the interface between the source/drain contact **337** and the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **3**A and **3**B. That is, the silicide layer **347** is formed on sidewalls of a portion of the source/drain contact **337** inside the source/drain feature **118P1** of the CFET **100**A. In some embodiments, the material of the silicide layer **347** is different from the materials of the silicide layers **344**N and **344**P.

[0077] In some embodiments, the source/drain contact **334** is electrically connected to the source/drain contact **337** through the silicide layer **344**P. In these embodiments, the silicide layer **344**P is in direct contact with the source/drain contact **337** in the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **3**A and **3**B. That is, a horizontal portion of the silicide layer **344**P formed on the bottom surface of the source/drain contact **334** is in direct contact with the top surface of the source/drain contact **337** in the source/drain feature **118P1** of the CFET **100**A.

[0078] In some embodiments, the source/drain contact **334** and/or the source/drain contact **337** are connected to signals. Similar to the source/drain contacts **134** and **137** described above, the arrangement of the source/drain contacts **334** and **337** can allow the signal routing to be arranged on the frontside, the backside, or both of the frontside and backside of the semiconductor device **300**. As a result, the flexibility for the signal routing can be improved. In some embodiments where the semiconductor device **300** is applied to a SRAM device, the source/drain contacts **334** and **337** may function as the source/drain contacts **134** and **137** in the embodiments where the semiconductor device **100** is applied to the SRAM device.

[0079] FIG. **4**A is a Y-Z cross-sectional view of a semiconductor device **400**, in accordance with alternative embodiments of the present disclosure. FIG. **4**B is an X-Z cross-sectional view of the semiconductor device **400** along line B-B' of FIG. **4**A, in accordance with alternative embodiments of the present disclosure. The semiconductor device **400** shown in FIGS. **4**A and **4**B may be similar to the semiconductor device **100** shown in FIGS. **1**A and **1**B, except the source/drain contacts **134**, **137** and the silicide layers **144**N, **144**P, **147** shown in FIGS. **1**A and **1**B are replaced by source/drain contacts **434**, **437** and silicide layers **444**N, **444**P, **447** shown in FIGS. **4**A and **4**B.

[0080] Referring to FIGS. **4**A and **4**B, similar to the source/drain contact **134** shown in FIGS. **1**A and **1**B, the source/drain contact **434** extends through the ILD layer **126**, the CESLs **124**, the source/drain feature **118**N, and the ILD layer **122**, and partially extend into the source/drain feature **118**P, so as to be in contact with and electrically connected to both of the source/drain features **118**N and **118**P, in accordance with some embodiments. In some embodiments, the source/drain contact **434** is in contact with the source/drain features **118N1** and **118P1** of the CFET **100**A, and electrically connects them together, as shown in FIGS. **4**A and **4**B. Similar to the source/drain contact **134**, the source/drain contact **434** may be referred to as the frontside S/D LI contact.

[0081] In some embodiments, similar to the silicide layer **144**N shown in FIGS. **1**A and **1**B, the silicide layer **444**N is formed on the interface between the source/drain contact **434** and the source/drain feature **118N1** of the CFET **100**A, as shown in FIGS. **4**A and **4**B. In some embodiments, similar to the silicide layer **144**P shown in FIGS. **1**A and **1**B, the silicide layer **444**P is formed on the interface between the source/drain contact **434** and the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **4**A and **4**B. That is, the silicide layer **444**P is formed on sidewalls of a portion of the source/drain contact **434** inside the source/drain feature **118P1** of the CFET **100**A. In some embodiments, the material of the silicide layer **444**N is different from the material of the silicide layer **444**P.

[0082] Still referring to FIGS. **4**A and **4**B, similar to the source/drain contact **137** shown in FIGS. **1**A and **1**B, the source/drain contact **437** extends through the ILD layer **128**, the substrate **102**, and the bottom isolation layer **120**, and partially extend into the source/drain feature **118**P, so as to be in contact with and electrically connected to the source/drain feature **118**P, in accordance with some embodiments. In some embodiments, the source/drain contact **437** is in contact with and electrically connected to the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **4**A and **4**B. Similar to the source/drain contact **137**, the source/drain contact **437** may be referred to as backside source/drain contact or bottom source/drain contact.

[0083] In some embodiments, similar to the silicide layer **147** shown in FIGS. **1**A and **1**B, the silicide layer **447** is formed on the interface between the source/drain contact

437 and the source/drain feature 118P1 of the CFET 100A, as shown in FIGS. 4A and 4B. That is, the silicide layer 447 is formed on sidewalls of a portion of the source/drain contact 437 inside the source/drain feature 118P1 of the CFET 100A. In some embodiments, since the source/drain contact 437 is formed after the formation of the source/drain contact 434 and the silicide layer 444P, a portion of the silicide layer 444P is formed on sidewalls of the top portion of the source/drain contact 437, as shown in FIGS. 4A and 4B. In some embodiments, the material of the silicide layer 447 is different from the materials of the silicide layers 444N and 444P.

[0084] In some embodiments, the source/drain contact 434 is electrically connected to the source/drain contact 437 through the direct contact. In these embodiments, the source/drain contact 434 is in direct contact with the source/drain contact 437 in the source/drain feature 118P1 of the CFET 100A, as shown in FIGS. 4A and 4B. That is, the bottom surface of the source/drain contact 434 is in direct contact with the top surface of the source/drain contact 437 in the source/drain feature 118P1 of the CFET 100A.

[0085] In some embodiments, the source/drain contact 434 and/or the source/drain contact 437 are connected to signals. Similar to the source/drain contacts 134 and 137 described above, the arrangement of the source/drain contacts 434 and 437 can allow the signal routing to be arranged on the frontside, the backside, or both of the frontside and backside of the semiconductor device 400. As a result, the flexibility for the signal routing can be improved. In some embodiments where the semiconductor device 400 is applied to a SRAM device, the source/drain contacts 434 and 437 may function as the source/drain contacts 134 and 137 in the embodiments where the semiconductor device 100 is applied to the SRAM device.

[0086] FIG. 5A is a Y-Z cross-sectional view of a semiconductor device 500, in accordance with alternative embodiments of the present disclosure. FIG. 5B is an X-Z cross-sectional view of the semiconductor device 500 along line B-B' of FIG. 5A, in accordance with alternative embodiments of the present disclosure. The semiconductor device 500 shown in FIGS. 5A and 5B may be similar to the semiconductor device 100 shown in FIGS. 1A and 1B, except the source/drain contacts 134, 137 and the silicide layers 144N, 144P, 147 shown in FIGS. 1A and 1B are replaced by source/drain contacts 534, 537 and silicide layers 544, 547P, 547N shown in FIGS. 5A and 5B.

[0087] Referring to FIGS. 5A and 5B, the source/drain contact 534 extends through the ILD layer 126 and the CESLs 124, and partially extend into the source/drain feature 118N, so as to be in contact with and electrically connected to the source/drain feature 118N, in accordance with some embodiments. In some embodiments, the source/drain contact 534 is in contact with and electrically connected to the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 5A and 5B.

[0088] In some embodiments, the silicide layer 544 is formed on the interface between the source/drain contact 534 and the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 5A and 5B. That is, the silicide layer 544 is formed on sidewalls and bottom surface of a portion of the source/drain contact 534 inside the source/drain feature 118P1 of the CFET 100A.

[0089] Still referring to FIGS. 5A and 5B, the source/drain contact 537 passes through the ILD layer 128, the substrate 102, the bottom isolation layer 120, the source/drain feature 118P, the ILD layer 122, and portions of the source/drain feature 118N, so as to be in contact with and electrically connected to both of the source/drain features 118N and 118P. In other words, the source/drain contact 537 extends through the ILD layer 128, the substrate 102, the bottom isolation layer 120, the source/drain feature 118P, and the ILD layer 122, and partially extend into the source/drain feature 118N. In some embodiments, the source/drain contact 537 is in contact with and electrically connected to the source/drain features 118N1 and 118P1 of the CFET 100A, as shown in FIGS. 5A and 5B. That is, the source/drain contact 537 electrically connects the source/drain features 118N1 and 118P1 of the CFET 100A together. Similar to the source/drain contact 137, the source/drain contact 537 may be referred to as backside source/drain contact or bottom source/drain contact. Furthermore, since the source/drain contact 537 connects the source/drain features 118N and 118P of NFET 100N and PFET 100P together, the source/drain contact 537 may be referred to as the backside S/D LI contact.

[0090] In some embodiments, the silicide layers 547N and 547P are formed on the interfaces between the source/drain contact 537 and the source/drain features 118N and 118P, respectively, in accordance with some embodiments. In some embodiments, the silicide layer 547N is formed on the interface between the source/drain contact 537 and the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 5A and 5B. That is, the silicide layer 547N is formed on sidewalls and the top surface of a portion of the source/drain contact 537 inside the source/drain feature 118N1 of the CFET 100A. In some embodiments, the silicide layer 547P is formed on the interface between the source/drain contact 537 and the source/drain feature 118P1 of the CFET 100A, as shown in FIGS. 5A and 5B. That is, the silicide layer 547P is formed on sidewalls of a portion of the source/drain contact 537 inside the source/drain feature 118P1 of the CFET 100A. In some embodiments, the material of the silicide layer 547N is different from the material of the silicide layer 547P. In further embodiments, the material of the silicide layer 544 is different from the materials of the silicide layers 547N and 547P.

[0091] In some embodiments, the source/drain contact 534 is electrically connected to the source/drain contact 537 through the silicide layer 544, the source/drain feature 118N1 of the CFET 100A, and the silicide layer 547N. In these embodiments, a portion of the source/drain feature 118N1 of the CFET 100A is between the source/drain contact 534 (and the silicide layers 544 formed thereon) and the source/drain contact 537 (and the silicide layers 547N formed thereon), as shown in FIGS. 5A and 5B. In other words, a horizontal portion of the silicide layer 544 formed on the bottom surface of the source/drain contact 534 is separated from a horizontal portion of the silicide layer 547N formed on the top surface of the source/drain contact 537 by a portion of the source/drain feature 118N1 of the CFET 100A.

[0092] In some embodiments, the source/drain contact 534 and/or the source/drain contact 537 are connected to signals. Since the source/drain contact 534 is electrically connected to the source/drain contact 537, the signals could be transmitted between the source/drain contacts 534 and 537. Therefore, the source/drain features 118N1 and 118P1 of the CFET 100A connected together by the source/drain contact

537 can be connected to signals through the source/drain contact **534**, through the source/drain contact **537**, or through both of the source/drain contacts **534** and **537**. In this way, the signal routing can be arranged on the frontside of the semiconductor device **500** (by connecting to the source/drain contact **534**), arranged on the backside of the semiconductor device **500** (by connecting to the source/drain contact **537**), or distributed on the frontside and backside of the semiconductor device **500** (by connecting to both of the source/drain contacts **534** and **537**). Accordingly, the flexibility for the signal routing can be improved. In some embodiments where the semiconductor device **500** is applied to a SRAM device, the source/drain contacts **534** and **537** may function as the source/drain contacts **134** and **137** in the embodiments where the semiconductor device **100** is applied to the SRAM device.

[0093] FIG. **6**A is a Y-Z cross-sectional view of a semiconductor device **600**, in accordance with alternative embodiments of the present disclosure. FIG. **6**B is an X-Z cross-sectional view of the semiconductor device **600** along line B-B' of FIG. **6**A, in accordance with alternative embodiments of the present disclosure. The semiconductor device **600** shown in FIGS. **6**A and **6**B may be similar to the semiconductor device **500** shown in FIGS. **5**A and **5**B, except the source/drain contacts **534**, **537** and the silicide layers **544**, **547**N, **547**P shown in FIGS. **5**A and **5**B are replaced by source/drain contacts **634**, **637** and silicide layers **644**, **647**N, **647**P shown in FIGS. **6**A and **6**B.

[0094] Referring to FIGS. **6**A and **6**B, similar to the source/drain contact **534** shown in FIGS. **5**A and **5**B, the source/drain contact **634** extends through the ILD layer **126** and the CESLs **124**, and partially extend into the source/drain feature **118**N, so as to be in contact with and electrically connected to the source/drain feature **118**N, in accordance with some embodiments. In some embodiments, the source/drain contact **634** is in contact with the source/drain feature **118N1** of the CFET **100**A, as shown in FIGS. **6**A and **6**B.

[0095] In some embodiments, similar to the silicide layer **544** shown in FIGS. **5**A and **5**B, the silicide layer **644** is formed on the interface between the source/drain contact **634** and the source/drain feature **118N1** of the CFET **100**A, as shown in FIGS. **6**A and **6**B. That is, the silicide layer **644** is formed on sidewalls and the bottom surface of a portion of the source/drain contact **634** inside the source/drain feature **118N1** of the CFET **100**A.

[0096] Still referring to FIGS. **6**A and **6**B, similar to the source/drain contact **537** shown in FIGS. **5**A and **5**B, the source/drain contact **637** extends through the ILD layer **128**, the substrate **102**, the bottom isolation layer **120**, the source/drain feature **118**P, and the ILD layer **122**, and partially extend into the source/drain feature **118**N, so as to be in contact with and electrically connected to both of the source/drain features **118**N and **118**P, in accordance with some embodiments. In some embodiments, the source/drain contact **637** is in contact with the source/drain features **118N1** and **118**P1 of the CFET **100**A, and electrically connects them together, as shown in FIGS. **6**A and **6**B. Similar to the source/drain contact **537**, the source/drain contact **637** may be referred to as backside source/drain contact, bottom source/drain contact, or the backside S/D LI contact.

[0097] In some embodiments, similar to the silicide layer **547**N shown in FIGS. **5**A and **5**B, the silicide layer **647**N is formed on the interface between the source/drain contact **637** and the source/drain feature **118N1** of the CFET **100**A, as shown in FIGS. **6**A and **6**B. In some embodiments, similar to the silicide layer **547**P shown in FIGS. **5**A and **5**B, the silicide layer **647**P is formed on the interface between the source/drain contact **637** and the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **6**A and **6**B. That is, the silicide layer **647**N is formed on sidewalls and the top surface of a portion of the source/drain contact **637** inside the source/drain feature **118N1** of the CFET **100**A. In some embodiments, the material of the silicide layer **647**N is different from the material of the silicide layer **647**P. In further embodiments, the material of the silicide layer **644** is different from the materials of the silicide layers **647**N and **647**P.

[0098] In some embodiments, the source/drain contact **634** is electrically connected to the source/drain contact **637** through the silicide layer **644** and the silicide layer **647**N. In these embodiments, the silicide layer **644** is in direct contact with the silicide layer **647**N in the source/drain feature **118N1** of the CFET **100**A, as shown in FIGS. **6**A and **6**B. That is, a horizontal portion of the silicide layer **644** formed on the bottom surface of the source/drain contact **634** is in direct contact with a horizontal portion of the silicide layer **647**N formed on the top surface of the source/drain contact **637** in the source/drain feature **118N1** of the CFET **100**A.

[0099] In some embodiments, the source/drain contact **634** and/or the source/drain contact **637** are connected to signals. Similar to the source/drain contacts **534** and **537** described above, the arrangement of the source/drain contacts **634** and **637** can allow the signal routing to be arranged on the frontside, the backside, or both of the frontside and backside of the semiconductor device **600**. As a result, the flexibility for the signal routing can be improved. In some embodiments where the semiconductor device **600** is applied to a SRAM device, the source/drain contacts **634** and **637** may function as the source/drain contacts **534** and **537** in the embodiments where the semiconductor device **500** is applied to the SRAM device.

[0100] FIG. **7**A is a Y-Z cross-sectional view of a semiconductor device **700**, in accordance with alternative embodiments of the present disclosure. FIG. **7**B is an X-Z cross-sectional view of the semiconductor device **700** along line B-B' of FIG. **7**A, in accordance with alternative embodiments of the present disclosure. The semiconductor device **700** shown in FIGS. **7**A and **7**B may be similar to the semiconductor device **500** shown in FIGS. **5**A and **5**B, except the source/drain contacts **534**, **537** and the silicide layers **544**, **547**N, **547**P shown in FIGS. **5**A and **5**B are replaced by source/drain contacts **734**, **737** and silicide layers **744**, **747**N, **747**P shown in FIGS. **7**A and **7**B.

[0101] Referring to FIGS. **7**A and **7**B, similar to the source/drain contact **534** shown in FIGS. **5**A and **5**B, the source/drain contact **734** extends through the ILD layer **126** and the CESLs **124**, and partially extend into the source/drain feature **118**N, so as to be in contact with and electrically connected to the source/drain feature **118**N, in accordance with some embodiments. In some embodiments, the source/drain contact **734** is in contact with the source/drain feature **118N1** of the CFET **100**A, as shown in FIGS. **7**A and **7**B.

[0102] In some embodiments, similar to the silicide layer **544** shown in FIGS. **5**A and **5**B, the silicide layer **744** is formed on the interface between the source/drain contact

734 and the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 7A and 7B. That is, the silicide layer 744 is formed on sidewalls and the bottom surface of a portion of the source/drain contact 734 inside the source/drain feature 118N1 of the CFET 100A.

[0103] Still referring to FIGS. 7A and 7B, similar to the source/drain contact 537 shown in FIGS. 5A and 5B, the source/drain contact 737 extends through the ILD layer 128, the substrate 102, the bottom isolation layer 120, the source/drain feature 118P, and the ILD layer 122, and partially extend into the source/drain feature 118N, so as to be in contact with and electrically connected to both of the source/drain features 118N and 118P, in accordance with some embodiments. In some embodiments, the source/drain contact 737 is in contact with the source/drain features 118N1 and 118P1 of the CFET 100A, and electrically connects them together, as shown in FIGS. 7A and 7B. Similar to the source/drain contact 537, the source/drain contact 737 may be referred to as backside source/drain contact, bottom source/drain contact, or the backside S/D LI contact.

[0104] In some embodiments, similar to the silicide layer 547N shown in FIGS. 5A and 5B, the silicide layer 747N is formed on the interface between the source/drain contact 737 and the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 7A and 7B. In some embodiments, similar to the silicide layer 547P shown in FIGS. 5A and 5B, the silicide layer 747P is formed on the interface between the source/drain contact 737 and the source/drain feature 118P1 of the CFET 100A, as shown in FIGS. 7A and 7B. That is, the silicide layer 747N is formed on sidewalls of a portion of the source/drain contact 737 inside the source/drain feature 118N1 of the CFET 100A. In some embodiments, the material of the silicide layer 747N is different from the material of the silicide layer 747P. In further embodiments, the material of the silicide layer 744 is different from the materials of the silicide layers 747N and 747P.

[0105] In some embodiments, the source/drain contact 734 is electrically connected to the source/drain contact 737 through the silicide layer 744. In these embodiments, the silicide layer 744 is in direct contact with the source/drain contact 737 in the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 7A and 7B. That is, a horizontal portion of the silicide layer 744 formed on the bottom surface of the source/drain contact 734 is in direct contact with the top surface of the source/drain contact 737 in the source/drain feature 118N1 of the CFET 100A.

[0106] In some embodiments, the source/drain contact 734 and/or the source/drain contact 737 are connected to signals. Similar to the source/drain contacts 534 and 537 described above, the arrangement of the source/drain contacts 734 and 737 can allow the signal routing to be arranged on the frontside, the backside, or both of the frontside and backside of the semiconductor device 700. As a result, the flexibility for the signal routing can be improved. In some embodiments where the semiconductor device 700 is applied to a SRAM device, the source/drain contacts 734 and 737 may function as the source/drain contacts 534 and 537 in the embodiments where the semiconductor device 500 is applied to the SRAM device.

[0107] FIG. 8A is a Y-Z cross-sectional view of a semiconductor device 800, in accordance with alternative embodiments of the present disclosure. FIG. 8B is an X-Z cross-sectional view of the semiconductor device 800 along line B-B' of FIG. 8A, in accordance with alternative embodiments of the present disclosure. The semiconductor device 800 shown in FIGS. 8A and 8B may be similar to the semiconductor device 500 shown in FIGS. 5A and 5B, except the source/drain contacts 534, 537 and the silicide layers 544, 547N, 547P shown in FIGS. 5A and 5B are replaced by source/drain contacts 834, 837 and silicide layers 844, 847N, 847P shown in FIGS. 8A and 8B.

[0108] Referring to FIGS. 8A and 8B, similar to the source/drain contact 534 shown in FIGS. 5A and 5B, the source/drain contact 834 extends through the ILD layer 126 and the CESLs 124, and partially extend into the source/drain feature 118N, so as to be in contact with and electrically connected to the source/drain feature 118N, in accordance with some embodiments. In some embodiments, the source/drain contact 834 is in contact with the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 8A and 8B.

[0109] In some embodiments, similar to the silicide layer 544 shown in FIGS. 5A and 5B, the silicide layer 844 is formed on the interface between the source/drain contact 834 and the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 8A and 8B. That is, the silicide layer 844 is formed on sidewalls of a portion of the source/drain contact 834 inside the source/drain feature 118N1 of the CFET 100A. In some embodiments, the silicide layer 844 is also formed on a portion the bottom surface of the source/drain contact 834, as shown in FIGS. 8A and 8B.

[0110] Still referring to FIGS. 8A and 8B, similar to the source/drain contact 537 shown in FIGS. 5A and 5B, the source/drain contact 837 extends through the ILD layer 128, the substrate 102, the bottom isolation layer 120, the source/drain feature 118P, and the ILD layer 122, and partially extend into the source/drain feature 118N, so as to be in contact with and electrically connected to both of the source/drain features 118N and 118P, in accordance with some embodiments. In some embodiments, the source/drain contact 837 is in contact with the source/drain features 118N1 and 118P1 of the CFET 100A, and electrically connects them together, as shown in FIGS. 8A and 8B. Similar to the source/drain contact 537, the source/drain contact 837 may be referred to as backside source/drain contact, bottom source/drain contact, or the backside S/D LI contact.

[0111] In some embodiments, similar to the silicide layer 547N shown in FIGS. 5A and 5B, the silicide layer 847N is formed on the interface between the source/drain contact 837 and the source/drain feature 118N1 of the CFET 100A, as shown in FIGS. 8A and 8B. In some embodiments, similar to the silicide layer 847P shown in FIGS. 5A and 5B, the silicide layer 847P is formed on the interface between the source/drain contact 837 and the source/drain feature 118P1 of the CFET 100A, as shown in FIGS. 8A and 8B. That is, the silicide layer 847N is formed on sidewalls of a portion of the source/drain contact 837 inside the source/drain feature 118N1 of the CFET 100A. In some embodiments, the material of the silicide layer 847N is different from the material of the silicide layer 847P. In further embodiments, the material of the silicide layer 844 is different from the materials of the silicide layers 847N and 847P. In some embodiments, since the source/drain contact 837 is formed after the formation of the source/drain contact 834 and the silicide layer **844**, the top portion of the source/drain contact **837** is surrounded by the silicide layer **844**, as shown in FIGS. **8**A and **8**B.

[0112] In some embodiments, the source/drain contact **834** is electrically connected to the source/drain contact **837** through direct contact. In these embodiments, the source/drain contact **834** is in direct contact with the source/drain contact **837** in the source/drain feature **118N1** of the CFET **100**A, as shown in FIGS. **8**A and **8**B. That is, the bottom surface of the source/drain contact **834** is in direct contact with the top surface of the source/drain contact **837** in the source/drain feature **118N1** of the CFET **100**A.

[0113] In some embodiments, the source/drain contact **834** and/or the source/drain contact **837** are connected to signals. Similar to the source/drain contacts **534** and **537** described above, the arrangement of the source/drain contacts **834** and **837** can allow the signal routing to be arranged on the frontside, the backside, or both of the frontside and backside of the semiconductor device **800**. As a result, the flexibility for the signal routing can be improved. In some embodiments where the semiconductor device **800** is applied to a SRAM device, the source/drain contacts **834** and **837** may function as the source/drain contacts **534** and **537** in the embodiments where the semiconductor device **500** is applied to the SRAM device.

[0114] The formation of the semiconductor device (e.g., the semiconductor devices **100-800**) are described in detail in below. The formation of the semiconductor device starts from a workpiece **900**. FIG. **9** is a perspective view of a workpiece **900** at a fabrication stage, in accordance with some embodiments of the present disclosure. FIGS. **10**, **11**A, **12**A, **13**A, **14**A, **15**A, **16**A, **17**A, **18**A, **19**A, **22**A, **23**A, **24**A, **25**A, **26**A, **27**A, **28**A, **29**A, **30**A, **31**A, and **32**A are Y-Z cross-sectional views of the workpiece **900** at various fabrication stages along line C-C' of FIG. **9**, in accordance with some embodiments of the present disclosure. FIGS. **12**B, **13**B, **14**B, **15**B, **16**B, **17**B, **18**B, **19**B, **20**A, **21**A, **22**B, **23**B, **24**B, **25**B, **26**B, **27**B, **28**B, **29**B, **30**B, **31**B, and **32**B are X-Z cross-sectional views of the workpiece **900** at various fabrication stages along line D-D' of FIG. **9**, in accordance with some embodiments of the present disclosure. FIGS. **11**B, **20**B, and **21**B are Y-Z cross-sectional views of the workpiece **900** at various fabrication stages along line E-E' of FIG. **9**, in accordance with some embodiments of the present disclosure.

[0115] Referring to FIG. **9**, the workpiece **900** is provided. The workpiece **900** includes a substrate **102** and a stack **904** over the substrate **102**. In some embodiments, the substrate **102** contains a semiconductor material, such as bulk silicon (Si). In some other embodiments, the substrate **102** may include other semiconductors such as Ge, SiGe, or a III-V semiconductor material. Example III-V semiconductor materials may include gallium arsenide (GaAs), indium phosphide (InP), gallium phosphide (GaP), gallium nitride (GaN), gallium arsenide phosphide (GaAsP), aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium phosphide (GaInP), and indium gallium arsenide (InGaAs). The substrate **102** may also include an insulating layer, such as a silicon oxide layer, to have a silicon-on-insulator (SOI) structure or a germanium-on-insulator (GOI) structure.

[0116] In some embodiments, the substrate **102** may include one or more well regions, such as n-type well regions doped with an n-type dopant (e.g., phosphorus (P) or arsenic (As)) or p-type well regions doped with a p-type dopant (e.g., boron (B)), for forming different types of devices. The n-type well regions and the p-type well regions may be formed by using ion implantation or thermal diffusion.

[0117] In some embodiments, the stack **904** includes semiconductor layers **906** (including semiconductor layers **906**A and a semiconductor layer **906**B) and **908** (including semiconductor layers **908**A and **908**B), and the semiconductor layers **906** and **908** are stacked in an alternating manner in the Z-direction. In some embodiments, the semiconductor layer **906**B is form vertically between a group of semiconductor layers **908**A and a group of semiconductor layers **908**B. In some embodiments, a thickness of the semiconductor layer **906**B is greater than the thickness of the semiconductor layers **906**A, as shown in FIG. **9**.

[0118] In some embodiments, the thickness of the semiconductor layers **908**A is greater than the semiconductor layers **908**B by a difference in a range from about 0.5 nm to about 5 nm. In other embodiments, the thicknesses of the semiconductor layers **908**A and **908**B are the same. In yet some embodiments, the thickness of the topmost semiconductor layer **908**B is greater than the thickness of the other semiconductor layers **908**B. In some embodiments, the thickness of the bottommost semiconductor layer **908**A is greater than the thickness of the other semiconductor layers **908**A.

[0119] The semiconductor layers **908** may include a semiconductor material, such as silicon, germanium, silicon carbide, silicon phosphide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide, SiGe, SiPC, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP. In some embodiments, the semiconductor layers **908**B include silicon for n-type transistors (i.e., NFETs **100**N of the CFETs **100**A and **100**B). In some embodiments, the semiconductor layers **908**A include silicon germanium for p-type transistors (i.e., PFETs **100**P of the CFETs **100**A and **100**B). In some embodiments, the semiconductor layers **908** are all made of silicon, and the type of the transistors depend on a work function metal layer wrapped around the nanostructures **106** (which are formed from the semiconductor layers **908**).

[0120] The semiconductor layers **906** and the semiconductor layers **908** may have different semiconductor compositions. In some embodiments, semiconductor layers **906** are formed of SiGe and the semiconductor layers **908** are formed of Si. In these embodiments, the additional germanium content in the semiconductor layers **906** allow selective removal or recess of the semiconductor layers **906** without substantial damages to the semiconductor layers **908**, so that the semiconductor layers **906** are also referred to as sacrificial layers.

[0121] In some embodiments, the semiconductor layers **906** and **908** are epitaxially grown over or on the substrate **102** using an epitaxial growth such as vapor-phase epitaxy (VPE), metal-organic CVD (MOCVD), molecular beam epitaxy (MBE), although other deposition processes, such as chemical vapor deposition (CVD), low pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), atomic layer deposition (ALD), ultrahigh vacuum CVD (UHVCVD), remote plasma CVD (RPCVD), a combination thereof, or the like, may also be utilized. The semiconductor layers **906** and **908** are grown alternatingly, one-after-another, to form the stack **904**.

[0122] The two semiconductor layers **908**A are used for the PFETs **100**P of the CFETs **100**A and **100**B and the two semiconductor layers **908**B are used for the NFETs **100**N of the CFETs **100**A and **100**B. It should be noted that four layers of the semiconductor layers **906** and four layers of the semiconductor layers **908** are alternately and vertically arranged (or stacked) as shown in FIG. **9**, which are for illustrative purposes only and not intended to be limiting beyond what is specifically recited in the claims. The number of the semiconductor layers depends on the desired number of channel members for the semiconductor device.

[0123] Referring to FIG. **10**, the substrate **102** and the stack **904** are then patterned to form fin structures **912**A and **912**B (may be collectively referred to as fin structures **912**) over the substrate **102**. For patterning purposes, the stack **904** may further include a hard mask layer **910** over the topmost semiconductor layer. The hard mask layer **910** may be a single layer structure or a multi-layer structure. In some embodiments, the hard mask layer **910** is a single layer structure and includes a silicon germanium layer. In some embodiments, the hard mask layer **910** is a multi-layer structure and includes a silicon nitride layer and a silicon oxide layer over the silicon nitride layer. In other embodiments, the hard mask layer **910** is a multi-layer structure and includes a silicon germanium layer and a silicon layer over the silicon germanium layer.

[0124] As shown in FIG. **10**, each of the fin structures **912** includes a base fin (i.e., the base portions **102**A and **102**B of the substrate **102** discussed above) formed from the substrate **102** and a stack portion formed from the stack **904** over the base fin, in accordance with some embodiments. The stack portion of each of the fin structures **912** includes the semiconductor layers **906** and **908** that are alternately stacked in the Z-direction. In some embodiments, the fin structures **912**A and **912**B extend in the X-direction, and are arranged in the Y-direction.

[0125] The fin structures **912** may be patterned using suitable processes including double-patterning or multi-patterning processes. For example, in some embodiments, a material layer is formed over the stack **904** and patterned using a photolithography process. Spacers are formed alongside the patterned material layer using a self-aligned process. The material layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fin structures **912** by etching the stack **904** and the substrate **102**. The etching process may include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes.

[0126] Still referring to FIG. **10**, isolation structures **104** are formed, in accordance with some embodiments. After the fin structures **912** are formed, the isolation structures **104** are formed over the substrate **102**. In some embodiments, the isolation structures **104** extend in the X-direction (not shown) and are arranged with the fin structures **912** in the Y-direction. In other words, the isolation structures **104** are formed on opposite sides of the fin structures **912** in the Y-direction. In some aspects, the isolation structures **104** are formed around the fin structures **912**.

[0127] The isolation structures **104** may include different structures, such as shallow trench isolation (STI) structures, deep trench isolation (DTI) structures, and/or local oxidation of silicon (LOCOS) structures. In some embodiments, STI structures include a multi-layer structure that fills the trenches, such as a silicon nitride comprising layer disposed over a thermal oxide comprising liner layer. In other embodiments, STI structures include a dielectric layer disposed over a doped liner layer (including, for example, boron silicate glass (BSG) or phosphosilicate glass (PSG)). In certain embodiments, STI structures include a bulk dielectric layer disposed over a liner dielectric layer, wherein the bulk dielectric layer and the liner dielectric layer include materials depending on design requirements.

[0128] In some embodiments, a dielectric material for the isolation structures **104** is first deposited over the workpiece **900**. In some embodiments, the dielectric material may include silicon oxide, silicon nitride, silicon oxynitride, FSG, low-k dielectrics, other suitable materials (e.g., including silicon, oxygen, nitrogen, carbon, or other suitable isolation constituent), or combinations thereof. In various embodiments, the dielectric material may be deposited using a deposition process, such as CVD, subatmospheric CVD (SACVD), FCVD, ALD, spin-on coating, or other suitable process. The deposited dielectric material is then thinned and planarized, for example by a CMP process. The planarized dielectric material is further recessed by a dry etching process, a wet etching process, and/or a combination thereof to form the isolation structures **104**. In some embodiments, the stack portions of the fin structures **912** rise above the isolation structures **104** while the base portions **102**A and **102**B are surrounded by the isolation structures **104**, as shown in FIG. **10**. In other words, the top surfaces of the substrate **102** is higher than top surfaces of the isolation structures **104**.

[0129] Referring to FIGS. **11**A and **11**B, the hard mask layer **910** is removed and a dummy gate structure **914** is formed over the fin structures **912** and the isolation structures **104**, in accordance with some embodiments. The dummy gate structure **914** may be configured to extend along the Y-direction and wrap around top surfaces and side surfaces of the fin structures **912**. In some embodiments, to form the dummy gate structure **914**, a dummy gate dielectric material for a dummy gate dielectric layer **916** is first formed over fin structures **912** and over the isolation structures **104**. In some embodiments, the dummy gate dielectric material may include, for example, a dielectric material such as a nitride (e.g., silicon nitride, silicon oxynitride), a carbide (e.g., silicon carbide), an oxide (e.g., silicon oxide), or other suitable material.

[0130] Then, in some embodiments, a dummy gate electrode material for a dummy gate electrode **918** is formed over the dummy gate dielectric material. The dummy gate electrode material may include a conductive material selected from a group comprising of polysilicon, W, Al, Cu, AlCu, Ti, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, Ta, TaN, Co, Ni, and/or combinations thereof. The dummy gate electrode material and/or the dummy gate dielectric material may be formed by way of a thermal oxidation process and/or a deposition process (e.g., PVD, CVD, PECVD, and ALD).

[0131] After the formation of the dummy gate electrode material and the dummy gate dielectric material, photolithography and etching processes may be performed to remove portions of the dummy gate electrode material and the dummy gate dielectric material, thereby forming the dummy gate structure **914** having the dummy gate electrode **918** and the dummy gate dielectric layer **916**. The dummy gate structure **914** may undergo a gate replacement process through subsequent processing to form metal gates, such as a high-k metal gate, as discussed in greater detail below.

[0132] Referring to FIGS. **12**A and **12**B, after forming of the dummy gate structure **914**, the gate spacers **114** are formed on sidewalls of the dummy gate structure **914**, and over the top surfaces and on the sidewalls of the fin structures **912**, in accordance with some embodiments. In some embodiments, the gate spacers **114** are formed on opposite sidewalls of the fin structures **912**, as shown in FIG. **12**A, and formed on opposite sidewalls of the dummy gate structures **914**, as shown in FIG. **12**B.

[0133] The gate spacers **114** may include multiple dielectric materials and be selected from a group consisting of silicon nitride (Si_3N_4), silicon oxide (SiO_2), silicon carbide (SiC), silicon oxycarbide (SiOC), silicon oxynitride (SiON), silicon oxycarbon nitride (SiOCN), carbon doped oxide, nitrogen doped oxide, porous oxide, air gap, or a combination thereof. In some embodiments, the gate spacers **114** may include a single layer or a multi-layer structure.

[0134] In some embodiments, the gate spacers **114** may be formed by conformally depositing a spacer layer (containing the dielectric material) over the isolation structures **104**, the fin structures **912**, and the dummy gate structure **914**, followed by an anisotropic etching process to remove top portions of the spacer layer from the top surfaces of the isolation structures **104**, the fin structures **912**, and the dummy gate structure **914**. After the anisotropic etching process, portions of the spacer layer on the sidewall surfaces of the fin structures **912** and the dummy gate structure **914** substantially remain and become the gate spacers **114**. In some embodiments, the anisotropic etching process is a dry (e.g., plasma) etching process. Additionally or alternatively, the formation of the gate spacers **114** may also involve chemical oxidation, thermal oxidation, CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, and/or other suitable methods.

[0135] Referring to FIGS. **13**A and **13**B, the fin structures **912** are recessed to form source/drain trenches **920** in the fin structures **912**, in accordance with some embodiments. In some embodiments, in each of the fin structures **912**A and **912**B, the source/drain trenches **920** include a source/drain trench **920**A and a source/drain trench **920**B that are on opposite sides of the dummy gate structure **914** in the X-direction. The source/drain trenches **920** may be formed by performing one or more etching processes to remove portions of the semiconductor layers **906** and **908** and the substrate **102** that do not vertically overlap or be covered by the dummy gate structure **914** and the gate spacers **114**. In some embodiments, a single etchant may be used to remove the semiconductor layers **906** and **908** and the substrate **102**. In other embodiments, multiple etchants may be used to perform the etching process. In some embodiments, portions of the gate spacers **114** on opposite sidewalls of the fin structures **912** in the Y-direction are removed, as shown in FIG. **13**A. In these embodiments, the height of the gate spacers **114** on the opposite sidewalls of the fin structures **912** in the Y-direction are reduced.

[0136] Referring to FIGS. **14**A and **14**B, the inner spacers **116** are formed between the semiconductor layers **908** as well as between the semiconductor layer **908** and the substrate **102**, in accordance with some embodiments. In some embodiments, the semiconductor layers **906** (including the semiconductor layers **906**A and **906**B) exposed in the source/drain trenches **920** are partially recessed through a selective etching process, and the semiconductor layers **908** are not etched. More specifically, the selective etching process is performed that selectively etches the side portions of the semiconductor layers **906** below the gate spacers **114** through the source/drain trenches **920**, with minimal etching (or substantially no etching) of the semiconductor layers **908** and the substrate **102**. After the selective etching process, inner spacer recesses are vertically formed between the semiconductor layers **908** as well as between the semiconductor layers **908** and the substrate **102**, below the gate spacers **114**. The selective etching process is configured to laterally etch (e.g., along the X-direction) the semiconductor layers **906** below the gate spacers **114**. The selective etching process may be a dry etching process, a wet etching process, other suitable etching process, or a combination thereof.

[0137] Next, in some embodiments, a spacer layer is conformally formed into the source/drain trenches **920** and the inner spacer recesses. More specifically, a deposition process is performed to form the spacer layer into the source/drain trenches **920** and the inner spacer recesses, such as CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, other suitable methods, or combinations thereof. The spacer layer partially (and, in some embodiments, completely) fills the source/drain trenches **920** and fully fills the inner spacer recesses. The deposition process is configured to ensure that the spacer layer fills the inner spacer recesses between the semiconductor layers **908** as well as between the semiconductor layer **908** and the substrate **102** under the gate spacers **114**. Furthermore, the spacer layer is also conformally formed on the gate spacers **114** and the isolation structures **104**.

[0138] The spacer layer may include a material that is different than the material of the semiconductor layers **908** and the material of the gate spacers **114** to achieve desired etching selectivity during the etching process. In some embodiments, the spacer layer include a dielectric material that includes silicon, oxygen, carbon, nitrogen, other suitable material, or combinations thereof (e.g., silicon oxide (SiO_2), SiON, SiOC, SiCN, SiOCN). In some embodiments, the spacer layer include a low-k dielectric material, such as those described herein. In some embodiments, the spacer layer includes a dielectric material having higher k value (dielectric constant) than the gate spacers **114**. In other embodiments, the spacer layer includes a dielectric material having lower k value than the gate spacers **114**.

[0139] Then, in some embodiments, the inner spacers **116** are formed to fill the inner spacer recesses between the semiconductor layers **908** as well as between the semiconductor layer **908** and the substrate **102**. More specifically, an etching process is performed to selectively etch the spacer layer to form the inner spacers **116** with minimal etching (or substantially no etching) of the semiconductor layers **908**, the substrate **102**, the dummy gate structure **914**, and the gate spacers **114**. The etching process may be an anisotropic etching process, such that portions of the spacer layer that do not vertically overlap or be covered by the dummy gate structure **914** and the gate spacers **114** are removed. The spacer layer on the gate spacers **114** and the isolation structures **104** are removed.

[0140] In some embodiments, sidewalls of the inner spacers **116** are aligned to the sidewalls of the gate spacers **114** and the semiconductor layers **908**. Therefore, the inner spacers **116** are formed on opposite sides of the dummy gate structure **914**. In some embodiments, the inner spacers **116** are also vertically between the semiconductor layers **908** as well as between the semiconductor layer **908** and the substrate **102**. In some embodiments, the thickness of the gate spacers **114** and the thickness of the inner spacers **116** are the same in the X-direction. In other embodiments, the thickness of the gate spacers **114** is greater than the thickness of the inner spacers **116** in the X-direction for capacitance reduction between source/drain contacts (e.g., the source/drain contacts **118**N/**118**P) and the gate structure (e.g., the gate structure **108**).

[0141] Referring to FIGS. **15**A and **15**B, polymer layers **922** and cover spacers **924** are formed in the source/drain trenches **920**, in accordance with some embodiments. More specifically, the polymer layers **922** are first formed in lower parts of the source/drain trenches **920** to cover the top surfaces of the substrate **102** and the sidewalls of the semiconductor layers **908**A (which are used for the PFET of the CFET, such as the PFETs **100**P of the CFETs **100**A and **100**B discussed above) and the inner spacers **116** (which are between the semiconductor layers **908**A) exposed in the source/drain trenches **920**. In some embodiments, top surfaces of the polymer layers **922** are lower than the semiconductor layers **908**B. In some embodiments, the polymer layers **922** is also formed on the gate spacers **114** and the isolation structures **104**, as shown in FIG. **15**A.

[0142] After forming the polymer layers **922**, the cover spacers **924** are conformally formed over the polymer layers **922** and on the sidewalls of the semiconductor layers **908**B (which are used for the NFET of the CFET, such as the NFETs **100**N of the CFETs **100**A and **100**B discussed above), the gate spacers **114**, and the inner spacers **116** (which are between the semiconductor layers **908**B). The polymer layers **922** may be formed of fluorine-containing polymer and its molecular structure includes silicon (Si), carbon (C), nitrogen (N), or fluorine (F). In some embodiments, the polymer layers **922** include fluorinated silicone or fluorinated polysilane. In some embodiments, the polymer layers **922** are spin-on-carbon layers. The polymer layers **922** may be deposited using CVD, flowable CVD (FCVD), or spin-on coating. The cover spacers **924** may include aluminum oxide (Al_2O_3).

[0143] Referring to FIGS. **16**A and **16**B, the polymer layers **922** and horizontal portions of the cover spacers **924** are removed, in accordance with some embodiments. More specifically, an anisotropic etching process is performed to remove the horizontal portions of the cover spacers **924** to exposed top surfaces of the polymer layers **922**, and then a selective etching process is performed to remove the polymer layers **922**. In some embodiments, vertical portions of the cover spacers **924** are partially removed or trimmed, but the vertical portions of the cover spacers **924** still cover the sidewalls of the gate spacers **114** and the semiconductor layers **908**B, as shown in FIG. **16**B. The selective etching process is performed that selectively etches the polymer layers **922** below the cover spacers **924** through the source/drain trenches **920**, with minimal etching (or substantially no etching) of the semiconductor layers **908**A, the substrate **102**, and the inner spacers **116**. The selective etching process is a dry etching process, a wet etching process, other suitable etching process, or combinations thereof.

[0144] Referring to FIGS. **17**A and **17**B, after removing the polymer layers **922**, the bottom isolation layers **120** and the source/drain features **118**P are formed in the lower parts of the source/drain trenches **920** and below the cover spacers **924**, in accordance with some embodiments. In some embodiments, the bottom isolation layers **120** are formed over the substrate **102** exposed in the source/drain trenches **920**, and the source/drain features **118**P are formed over the bottom isolation layers **120**. In these embodiments, the bottom isolation layers **120** are vertically between and in contact with the source/drain features **118**P and the substrate **102** in the Z-direction, and on opposite sides of the dummy gate structure **914** in the X-direction.

[0145] In some embodiments, the top surfaces of the bottom isolation layers **120** are higher than the topmost surfaces of the substrate **102** (i.e., the top surfaces of the base portions **102**A and **102**B) to ensure that the bottom isolation layers **120** separate the source/drain features **118**P from the substrate **102**. In some embodiments, the dielectric material of the bottom isolation layers **120** may include silicon nitride (Si_3N_4), silicon oxide (SiO_2), SiC, SiOC, SION, SiCN, SiOCN, high-k dielectrics, other suitable materials, or combinations thereof. In some embodiments, the bottom isolation layers **120** may be deposited by CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, other suitable methods, or combinations thereof.

[0146] In some embodiments, the source/drain features **118**P are formed on opposite sides of the dummy gate structure **914** in the X-direction. In some embodiments, source/drain features **118**P**1** and **118**P**2** are formed on opposite sides of the dummy gate structure **914**, and in the source/drain trenches **920**A and **920**B respectively, as shown in FIG. **17**B. In some embodiments, the source/drain features **118**P are connected to and in contact with the semiconductor layers **908**A. In other words, the source/drain features **118**P are attached to opposite sides of the first group of semiconductor layers **908** (i.e., the semiconductor layers **908**A). In some embodiments, the semiconductor layers **908**A connect one source/drain feature **118**P to another source/drain feature **118**P. In some embodiments, the source/drain features **118**P may have top surfaces that extend higher than top surfaces of the topmost semiconductor layers **908**A (e.g., in the Z-direction). In other embodiments, the top surfaces of the source/drain features **118**P substantially level with the top surfaces of the topmost semiconductor layers **908**A. In some embodiments, the top surfaces of the source/drain features **118**P are lower than the bottom surfaces of the cover spacers **924** and the semiconductor layers **908**B.

[0147] In some embodiments, the source/drain features **118**P may be formed by using epitaxial growth process such as VPE, MOCVD, MBE, although other deposition processes, such as CVD, LPCVD, PECVD, ALD, UHVCVD, RPCVD, a combination thereof, or the like, may also be utilized. The source/drain features **118**P are grown from the semiconductor layers **908**A rather than the semiconductor layers **908**B, the bottom isolation layers **120**, and the substrate **102**, it is because that the cover spacers **924** cover the sidewalls of the semiconductor layers **908**B, and the bottom isolation layers **120** cover the surface of the substrate **102**.

[0148] In some embodiments, the source/drain features **118**P may include epitaxially-grown material selected from a group consisting of boron-doped SiGe, boron-doped SiGeC, boron-doped Ge, boron-doped Si, boron and carbon doped SiGe, or a combination thereof. In some embodiments, the epitaxially-grown material of the source/drain features **118**P may be doped with p-type dopants (e.g., boron (B), indium (In), other p-type dopant, or a combination thereof) and have a doping concentration in a range from about $1\times10^{19}/cm^3$ to $6\times10^{20}/cm^3$. In some embodiments, the source/drain features **118**P used for the PFETs of the CFETs (e.g., the PFETs **100**P of the CFETs **100**A and **100**B shown in FIGS. **1**A to **8**B) may be referred to as p-type source/drain features. The source/drain features **118**P may be doped in-situ or ex-situ. One or more annealing processes may be performed to activate the dopants in the source/drain features **118**P. The annealing processes may include rapid thermal annealing (RTA) and/or laser annealing processes.

[0149] Referring to FIGS. **18**A and **18**B, the cover spacers **924** are removed through a selective etching process, and then the ILD layer **122** is formed in the source/drain trenches **920** and over the substrate **102**, the isolation structures **104**, and the source/drain features **118**P, in accordance with some embodiments. In some embodiments, the selective etching process is performed that selectively etches the cover spacers **924** over the source/drain features **118**P through the source/drain trenches **920**, with minimal etching (or substantially no etching) of the semiconductor layers **908**B, the gate spacers **114**, and the inner spacers **116**. The selective etching process may be a dry etching process, a wet etching process, other suitable etching process, or combinations thereof.

[0150] After removing the cover spacers **924**, the ILD layer **122** is then formed over the substrate **102**, the isolation structures **104**, and the source/drain features **118**P and between the spaces between the source/drain features **118**P. In some embodiments, the ILD layer **122** may be formed by CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, or other suitable methods. Then, the ILD layer **122** over the source/drain features **118**P are recessed by performing one or more photolithography and etching processes, so that the sidewalls of the semiconductor layers **908**B over the source/drain features **118**P are exposed.

[0151] In some embodiments, the ILD layer **122** may include tetraethylorthosilicate (TEOS) formed oxide, undoped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fluoride-doped silica glass (FSG), PSG, BSG, low-k dielectric materials, other suitable dielectric materials, or combinations thereof. Exemplary low-k dielectric materials include carbon doped silicon oxide, Black Diamond® (Applied Materials of Santa Clara, California), xerogel, aerogel, amorphous fluorinated carbon, parylene, BCB-based dielectric material, SiLK (Dow Chemical, Midland, Michigan), polyimide, other low-k dielectric materials, or combinations thereof.

[0152] Referring to FIGS. **19**A and **19**B, the source/drain features **118**N are formed in the source/drain trenches **920**, in accordance with some embodiments. In some embodiments, the source/drain features **118**N are formed over the ILD layer **122** in the source/drain trenches **920**. The source/drain features **118**N are also formed on opposite sides of the dummy gate structure **914** in the X-direction. In some embodiments, source/drain features **118**N**1** and **118**N**2** are formed on opposite sides of the dummy gate structure **914**, and over the source/drain features **118**P**1** and **118**P**2** respectively, as shown in FIG. **19**B. In other words, the source/drain features **118**N**1** and **118**N**2** vertically overlaps the source/drain features **118**P**1** and **118**P**2**, respectively. In some embodiments, the source/drain features **118**N are connected to and in contact with the semiconductor layers **908**B. In other words, the source/drain features **118**N are attached to opposite sides of the second group of semiconductor layers **908** (i.e., the semiconductor layers **908**B). In some embodiments, the semiconductor layers **908**B connect one source/drain feature **118**N to another source/drain feature **118**N.

[0153] In some embodiments, the source/drain features **118**N may have top surfaces that extend higher than top surfaces of the topmost semiconductor layers **908**B (e.g., in the Z-direction). In other embodiments, the top surfaces of the source/drain features **118**N are substantially level with the top surfaces of the topmost semiconductor layers **908**B. In some embodiments, the bottom surfaces of the source/drain features **118**N are lower than bottom surfaces of the bottommost semiconductor layers **908**B. In other embodiments, the bottom surfaces of the source/drain features **118**N are substantially level with the bottom surfaces of the bottommost semiconductor layers **908**B.

[0154] In some embodiments, the source/drain features **118**N may be formed by using epitaxial growth process such as VPE, MOCVD, MBE, although other deposition processes, such as CVD, LPCVD, PECVD, ALD, UHVCVD, RPCVD, a combination thereof, or the like, may also be utilized. The source/drain features **118**N are grown from the semiconductor layers **908**B.

[0155] In some embodiments, the source/drain features **118**N may include epitaxially-grown material selected from a group consisting of SiP, SiC, SiPC, SiAs, Si, or a combination thereof. In some embodiments, the epitaxially-grown material of the source/drain features **118**N may be doped with n-type dopants (e.g., phosphorus (P), arsenic (As), other n-type dopant, or a combination thereof) and have a doping concentration in a range from about $2\times10^{19}/cm^3$ to $3\times10^{21}/cm^3$. In some embodiments, the source/drain features **118**N used for the NFETs of the CFETs (e.g., the NFETs **100**N of the CFETs **100**A and **100**B shown in FIGS. **1**A to **8**B) may be referred to as n-type source/drain features. The source/drain features **118**N may be doped in-situ or ex-situ. One or more annealing processes may be performed to activate the dopants in the source/drain features **118**N. The annealing processes may include RTA and/or laser annealing processes.

[0156] Still referring to FIGS. **19**A and **19**B, the CESLs **124** are formed over the source/drain features **118**N and the ILD layer **122**, and the ILD layer **126** is formed over the CESLs **124** to fill the space between the source/drain features **118**N, in accordance with some embodiments. In some embodiments, the CESLs **124** are conformally formed on the sidewalls of the gate spacers **114**. In some embodiments, the CESLs **124** are also conformally formed on the top surfaces and the sidewalls of the source/drain features **118**N. In some embodiments, the CESLs **124** may include La_2O_3, Al_2O_3, SiOCN, SiOC, SiCN, SiO_2, SiC, ZnO, ZrN, $Zr_2Al_3O_9$, TiO_2, TaO_2, ZrO_2, HfO_2, Si_3N_4, Y_2O_3, AlON, TaCN, ZrSi, or other suitable materials. The CESLs **124** may be formed by CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, or other suitable methods.

[0157] The ILD layer **126** is formed over and between the CESLs **124** to fill the space between the CESLs **124** and in the source/drain trenches **920**. In some embodiments, the ILD layer **126** includes a material that is different than the CESLs **124**. In some embodiments, the ILD layer **126** may include TEOS formed oxide, un-doped silicate glass, or doped silicon oxide such as BPSG, FSG, PSG, BSG, low-k dielectric materials, other suitable dielectric materials, or combinations thereof. The ILD layer **126** may be formed by

CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, or other suitable methods. After forming the CESLs **124** and the ILD layer **126**, a CMP process is performed to reduce heights of the CESLs **124** and the ILD layer **126** until top surface of the dummy gate electrode **918** of the dummy gate structure **914** is exposed.

[0158] Referring to FIGS. **20**A and **20**B, the dummy gate structure **914** are selectively removed through any suitable photolithography and etching processes, in accordance with some embodiments. In some embodiments, the photolithography process may include forming a photoresist layer (resist), exposing the resist to a pattern, performing a post-exposure bake process, and developing the resist to form a masking element, which exposes a region including the dummy gate structure **914**. Then, the dummy gate structure **914** is selectively etched through the masking element. The gate spacers **114** may be used as the masking element or a part thereof. Etch selectivity may be achieved by selecting appropriate etching chemicals, and the dummy gate structure **914** may be removed without substantially affecting the CESLs **124** and the ILD layer **126**. The removal of the dummy gate structure **914** creates a gate trench **926**. The gate trench **926** exposes the top surfaces of the topmost semiconductor layers **908**B that underlie the dummy gate structure **914**.

[0159] Still referring to FIGS. **20**A and **20**B, the semiconductor layers **906** are selectively removed through the gate trench **926**, using a wet or dry etching process for example, in accordance with some embodiments. After the semiconductor layers **906** are selectively removed, the semiconductor layers **908**A and **908**B are exposed in the gate trench **926** to form the nanostructures **106**P and **106**N stacked on top of each other. Specifically, the nanostructures **106**P (the semiconductor layers **908**A) are stacked vertically in the Z-direction, and the nanostructures **106**N (the semiconductor layers **908**B) are directly over the nanostructures **106**P and are stacked vertically in the Z-direction. This process may be referred to as a wire release process, a nanowire release process, a nanosheet release process, a nanowire formation process, a nanosheet formation process, or a wire formation process.

[0160] In some embodiments, the thicknesses of nanostructures **106** in the NFET and the PFET (e.g., the NFETs **100**N and the PFETs **100**P of the CFETs **100**A and **100**B shown in FIGS. **1**A to **8**B) in the Z-direction are different. For example, a thickness of the nanostructure **106**P may be greater than a thickness of the nanostructure **106**N in the Z-direction, alternatively, the thickness of the nanostructure **106**P may be smaller than that of the nanostructure **106**N in the Z-direction. In some embodiments, the difference in thickness between the nanostructure **106**P and the nanostructure **106**N is in a range from about 0.5 nm to about 5 nm.

[0161] In some embodiments, the pitch between the nanostructures **106**P (i.e., the distance from the top surface of the lower nanostructure **106**P to the bottom surface of the upper nanostructure **106**P) is different from the pitch between the nanostructures **106**N (i.e., the distance from the top surface of the lower nanostructure **106**N to the bottom surface of the upper nanostructure **106**N). In some embodiments, the difference between the pitch of the nanostructures **106**P and the pitch of the nanostructures **106**N is in a range from about 0.5 nm to about 5 nm. In some embodiments, the length of the nanostructures **106**P is greater than the length of the nanostructures **106**N in the X-direction. In some embodiments, the difference between the length of the nanostructures **106**P and the length of the nanostructures **106**N is in a range from about 0.5 nm to about 5 nm.

[0162] Referring to FIGS. **21**A and **21**B, the gate structure **108** including the gate dielectric layers **110**P, **110**N and the gate electrode layers **112**P, **112**N is formed, in accordance with some embodiments. In some embodiments, the gate dielectric layer **110**P and the gate electrode layer **112**P are formed in the gate trench **926** to wrap around each of the nanostructures **106**P and **106**N (the semiconductor layers **908**A and **908**B). In some embodiments, the gate dielectric layer **110**P is wrapped around each of the nanostructures **106**P and **106**N, and the gate electrode layer **112**P is wrapped around the gate dielectric layer **110**P and each of the nanostructures **106**P and **106**N. Additionally, the gate dielectric layer **110**P is also formed on the sidewalls of the inner spacers **116** and the gate spacers **114**, as well as over the top surfaces of the substrate **102** and the isolation structures **104**.

[0163] Next, in some embodiments, the gate dielectric layer **110**P and the gate electrode layer **112**P in the gate trench **926** are etched back to expose the nanostructures **106**N (the semiconductor layers **908**B). In some embodiments, portions of the gate dielectric layer **110**P and gate electrode layer **112**P that are wrapped around the nanostructures **106**N are removed by one or more etching processes. The etching processes may be selective etching processes that selectively etch the gate dielectric layer **110**P and the gate electrode layer **112**P, with minimal etching (or substantially no etching) of the nanostructures **106**N, the gate spacers **114**, and the inner spacers **116**. The selective etching process may be a dry etching process, a wet etching process, other suitable etching process, or combinations thereof. In some embodiments, after the etching processes, the top surface of the gate electrode layer **112**P is lower than the bottommost surfaces of the nanostructures **106**N.

[0164] Then, in some embodiments, the gate dielectric layer **110**N and the gate electrode layer **112**N are formed in the gate trench **926** and over the gate dielectric layer **110**P and the gate electrode layer **112**P to wrap around the nanostructures **106**N (the semiconductor layers **908**B). In some embodiments, the gate dielectric layer **110**N wraps around each of the nanostructures **106**N, and the gate electrode layer **112**N wraps around the gate dielectric layer **110**N and each of the nanostructures **106**N. Moreover, the gate dielectric layer **110**N is also formed on the sidewalls of the inner spacers **116** and the gate spacers **114**, as well as over the top surface of the gate electrode layer **112**P. In some embodiments, the gate dielectric layer **110**N is further in contact with the gate dielectric layer **110**P, as shown in FIG. **21**A. Therefore, the gate dielectric layer **110**P, the gate electrode layer **112**P, the gate dielectric layer **110**N, and the gate electrode layer **112**N may be together referred to as the gate structure **108** to replace the dummy gate structure **914**.

[0165] In some embodiments, the gate dielectric layers **110**P and **110**N may include oxide with nitrogen doped dielectric material (initial layer) combined with metal content high-k dielectric material (e.g., k value (dielectric constant)>7.9). For example, the gate dielectric layers **110**P and **110**N may include hafnium oxide (HfO_2), which has a dielectric constant in a range from about 18 to about 40. Alternatively, the gate dielectric layers **110**P and **110**N may include other high-k dielectrics, such as TiO_2, HfZrO,

Ta_2O_3, $HfSiO_4$, ZrO_2, $ZrSiO_2$, LaO, AlO, ZrO, TiO, Ta_2O_5, Y_2O_3, SrTiO3 (STO), $BaTiO_3$ (BTO), BaZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfTiO, (Ba, Sr) TiO_3 (BST), Al_2O_3, Si_3N_4, oxynitrides (SiON), other suitable materials, or combinations thereof. The gate dielectric layers **110**P and **110**N may be formed by chemical oxidation, thermal oxidation, CVD, physical vapor deposition (PVD), ALD, high-density plasma CVD (HDPCVD), MOCVD, RPCVD, PECVD, LPCVD, atomic layer CVD (ALCVD), atmospheric pressure CVD (APCVD), flowable CVD (FCVD), other suitable methods, or combinations thereof.

[0166] In some embodiments, the gate electrode layer **112**P may include one or more p-type work function metal layers for PFETs **100**P, and the gate electrode layer **112**N may include one or more n-type work function metal layers for NFETs **100**N. In other embodiments, the gate electrode layer **112**P and the gate electrode layer **112**N may include the same work function metal layer.

[0167] In some embodiments, the n-type and p-type work function metal layers may include a material such as such as W, Al, Cu, TIN, Ti, TiAlN, TiAl, Pt, Ta, TaN, Co, Ni, TaC, TaCN, TaSiN, $TaSi_2$, $NiSi_2$, Mn, Zr, $ZrSi_2$, Ru, AlCu, Mo, $MoSi_2$, WN, other suitable work function materials, or combinations thereof. In some embodiments, the n-type and p-type work function metal layers may be deposited utilizing CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, or the like. However, any suitable materials and processes may be utilized to form the n-type work function metal layer and the p-type work function metal layer.

[0168] In some embodiments, each of the gate electrode layers **112**P and **112**N may include a single layer or alternatively a multi-layer structure. In some embodiments, each of the gate electrode layers **112**P and **112**N may include a capping layer, a barrier layer, and a fill material (not shown). The capping layer may be formed adjacent to the gate dielectric layers **110**P and **110**N, and may be formed from a metallic material such as TaN, Ti, TiAlN, TiAl, Pt, TaC, TaCN, TaSiN, Mn, Zr, TiN, Ru, Mo, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like.

[0169] The barrier layer may be formed adjacent to the capping layer, and may be formed of a different material than the capping layer. For example, the barrier layer may be formed of one or more layers of a metallic material such as TIN, TaN, Ti, TiAlN, TiAl, Pt, TaC, TaCN, TaSiN, Mn, Zr, Ru, Mo, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like.

[0170] In some embodiments, the fill material may include a suitable conductive material, such as Al, W, and/or Cu. In some embodiments, the capping layer, the barrier layer, and the fill material may be deposited using a deposition process such as CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, or the like, although any suitable deposition process may be used.

[0171] Referring to FIGS. **22**A and **22**B, the source/drain contacts **130** and **132** and the corresponding silicide layers **140** and **142** are formed on the frontside of the workpiece **900**, in accordance with some embodiments. In some embodiments, contact openings are formed to pass through the ILD layer **126**, the CESLs **124**, and portions of the source/drain features **118**N, so as to expose the source/drain features **118**N. In some embodiments, one or more photolithography and etching processes are performed to etch the ILD layer **126**, the CESLs **124**, and the source/drain features **118**N, so as to form the contact openings that exposes the source/drain features **118**N.

[0172] Next, in some embodiments, the silicide layers **140** and **142** are formed on the exposed surfaces of the source/drain features **118**N in the contact openings. In some embodiments, the silicide layers **140** and **142** are formed by depositing metal layers on the source/drain features **118**N, and heating the workpiece **900** to cause constituents of the source/drain features **118**N to react with metal constituents of the metal layers. Then, in some embodiments, a conductive material is deposited in the contact openings and on the silicide layers **140** and **142** by a deposition process, so as to form the source/drain contacts **130** and **132**. That is, the contact openings are filled with the conductive material to form the source/drain contacts **130** and **132**.

[0173] Referring to FIGS. **23**A and **23**B, a trench **928** is formed on the frontside of the workpiece **900** to expose both the source/drain features **118**N and **118**P, in accordance with some embodiments. In some embodiments, the trench **928** is formed to pass through the ILD layer **126**, the CESLs **124**, the source/drain feature **118**N, the ILD layer **122**, and portions of the source/drain features **118**P, so as to expose the source/drain features **118**N and **118**P. In some embodiments, one or more photolithography and etching processes are performed to etch the ILD layer **126**, the CESLs **124**, the source/drain feature **118**N, and the ILD layer **122**, and partially etch the source/drain feature **118**P, so as to form the trench **928** that exposes the source/drain features **118**N and **118**P. In some embodiments, the trench **928** exposes the source/drain features **118N1** and **118P1** of the CFET **100**A, as shown in FIGS. **23**A and **23**B.

[0174] Referring to FIGS. **24**A and **24**B, the source/drain contact **134** and the corresponding silicide layers **144**N and **144**P are formed on the frontside of the workpiece **900**, in accordance with some embodiments. In some embodiments, the silicide layers **144**N and **144**P are formed on the exposed surfaces of the source/drain features **118**N and **118**P in the trench **928**, respectively. In some embodiments, the silicide layers **144**N and **144**P are formed by depositing metal layers on the source/drain features **118**N and **118**P, and heating the workpiece **900** to cause constituents of the source/drain features **118**N and **118**P to react with metal constituents of the metal layers. In some embodiments, the silicide layers **144**N and **144**P may include titanium silicide (TiSi), nickel silicide (NiSi), tungsten silicide (WSi), nickel-platinum silicide (NiPtSi), nickel-platinum-germanium silicide (NiPtGeSi), nickel-germanium silicide (NiGeSi), ytterbium silicide (YbSi), platinum silicide (PtSi), iridium silicide (IrSi), erbium silicide (ErSi), cobalt silicide (CoSi), or other suitable compounds. In some embodiments, the material of the silicide layer **144**N is different from the material of the silicide layer **144**P. For example, the silicide layer **144**N may include a compound with metal(s) and Si, and the silicide layer **144**P may include a compound with metal(s), Si, and Ge.

[0175] Then, in some embodiments, a conductive material is deposited in the trench **928** and on the silicide layers **144**N and **144**P by a deposition process, so as to form the source/

drain contact **134**. That is, the trench **928** is filled with the conductive material to form the source/drain contact **134**. In some embodiments, the source/drain contact **134** is in contact with and electrically connected to the source/drain features **118N1** and **118P1** of the CFET **100**A, as shown in FIGS. **24**A and **24**B. In some embodiments, the silicide layer **144**N is formed on sidewalls of a portion of the source/drain contact **134** inside the source/drain feature **118N1** of the CFET **100**A, and the silicide layer **144**P is formed on sidewalls and the bottom surface of a portion of the source/drain contact **134** inside the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **24**A and **24**B.

[0176] Referring to FIGS. **25**A and **25**B, a portion of the substrate **102** is removed from the backside of the workpiece **900**, and the ILD layer **128** is formed on backside surface of the substrate **102**, in accordance with some embodiments. In other words, the substrate is thinned and the ILD layer **128** is formed under and in contact with the thinned substrate **102**. Before thinning the substrate **102** and forming the ILD layer **128**, the workpiece **900** may be flipped. For the purpose of simplicity, the sequent figures are shown without being flipped. In some embodiments, a carrier wafer may be bonded to the frontside of the workpiece **900** before flipping. In some embodiments, the substrate **102** is thinned (or partially removed) by a selective etching process or a CMP process.

[0177] After thinning the substrate **102**, the ILD layer **128** may be formed under the thinned substrate **102**, as shown in FIGS. **25**A and **25**B. In some embodiments, the ILD layer **128** may include TEOS formed oxide, un-doped silicate glass, or doped silicon oxide such as BPSG, FSG, PSG, BSG, low-k dielectric materials, other suitable dielectric materials, or combinations thereof. The ILD layer **128** may be formed by CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, or other suitable methods.

[0178] Still referring to FIGS. **25**A and **25**B, trenches **930**, **932**, and **934** are formed on the backside of the workpiece **900** to expose both the source/drain features **118**P, in accordance with some embodiments. In some embodiments, the trenches **930**, **932**, and **934** are formed to pass through the ILD layer **128**, the substrate **102**, the bottom isolation layers **120**, and portions of the source/drain features **118**P, so as to expose the source/drain features **118**P. In some embodiments, one or more photolithography and etching processes are performed to etch the ILD layer **128**, the substrate **102**, and the bottom isolation layers **120**, and partially etch the source/drain features **118**P, so as to form the trenches **930**, **932**, and **934** that expose the source/drain features **118**P. In some embodiments, the trench **930** exposes the source/drain feature **118P1** of the CFET **100**B, the trench **932** exposes the source/drain feature **118P1** of the CFET **100**A, and the trench **934** exposes the source/drain feature **118P2** of the CFET **100**A, as shown in FIGS. **25**A and **25**B.

[0179] Referring back to FIGS. **1**A and **1**B, the source/drain contacts **136**, **137**, **138** and the corresponding silicide layers **146**, **147**, **148** are formed on the backside of the workpiece **900**, in accordance with some embodiments. In these embodiments, the resultant device of the workpiece **900** may be fabricated to as the semiconductor device **100** shown in FIGS. **1**A and **1**B. In some embodiments, the silicide layers **146**, **147**, and **148** are formed on the exposed surfaces of the source/drain features **118**P in the trenches **930**, **932**, and **934**, respectively. In some embodiments, the silicide layers **146**, **147**, and **148** are formed by depositing metal layers on the source/drain features **118**P, and heating the workpiece **900** to cause constituents of the source/drain features **118**P to react with metal constituents of the metal layers. Then, in some embodiments, a conductive material is deposited in the trenches **930**, **932**, and **934** and on the silicide layers **146**, **147**, and **148** by a deposition process, so as to form the source/drain contacts **136**, **137**, and **138**. That is, the trenches **930**, **932**, and **934** are filled with the conductive material to form the source/drain contacts **136**, **137**, and **138**.

[0180] In some embodiments, the source/drain contact **137** is in contact with and electrically connected to the source/drain feature **118P1** of the CFET **100**A, and the silicide layer **147** is formed on sidewalls and the top surface of a portion of the source/drain contact **137** inside the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **1**A and **1**B.

[0181] Referring to FIGS. **26**A and **26**B, the fabrication stage shown in FIGS. **26**A and **26**B follows the fabrication stage shown in FIGS. **22**A and **22**B. In some embodiments, a trench **1028** is formed to pass through the ILD layer **126**, the CESLs **124**, the source/drain feature **118**N, the ILD layer **122**, and portions of the source/drain features **118**P, so as to expose the source/drain features **118**N and **118**P. In some embodiments, one or more photolithography and etching processes are performed to etch the ILD layer **126**, the CESLs **124**, the ILD layer **122**, the source/drain features **118**N and **118**P, so as to form the trench **1028** that exposes the source/drain features **118**N and **118**P. In some embodiments, the trench **1028** exposes the source/drain features **118N1** and **118P1** of the CFET **100**A, as shown in FIGS. **26**A and **26**B. In some embodiments, in the source/drain feature **118P1** of the CFET **100**A, the trench **1028** extends downward more than the trench **928**. That is, the trench **1028** is deeper than the trench **928** in the Z-direction.

[0182] Referring to FIGS. **27**A and **27**B, the source/drain contact **234** and the corresponding silicide layers **244**N and **244**P are formed on the frontside of the workpiece **900**, in accordance with some embodiments. In some embodiments, the silicide layers **244**N and **244**P are formed on the exposed surfaces of the source/drain features **118**N and **118**P in the contact openings, respectively. In some embodiments, the materials and methods used in forming the silicide layers **244**N and **244**P are the same as or similar to those of the silicide layers **144**N and **144**P, and are not repeated herein. In some embodiments, the material of the silicide layer **244**N is different from the material of the silicide layer **244**P.

[0183] Then, in some embodiments, a conductive material is deposited in the trench **1028** and on the silicide layers **244**N and **244**P by a deposition process, so as to form the source/drain contact **234**. That is, the trench **1028** is filled with the conductive material to form the source/drain contact **234**. In some embodiments, the source/drain contact **234** is in contact with and electrically connected to the source/drain features **118N1** and **118P1** of the CFET **100**A, as shown in FIGS. **27**A and **27**B. In some embodiments, the silicide layer **244**N is formed on sidewalls of a portion of the source/drain contact **234** inside the source/drain feature **118N1** of the CFET **100**A, and the silicide layer **244**P is formed on sidewalls and the bottom surface of a portion of the source/drain contact **234** inside the source/drain feature **118P1** of the CFET **100**A, as shown in FIGS. **27**A and **27**B.

[0184] Referring to FIGS. **28**A and **28**B, the ILD layer **128** is formed on backside surface of the substrate **102**, in accordance with some embodiments. The material and method used in forming the ILD layer **128** has been discussed above with reference to FIGS. **25**A and **25**B, and are not repeated herein.

[0185] Still referring to FIGS. **28**A and **28**B, trenches **1030**, **1032**, and **1034** are formed on the backside of the workpiece **900** to expose both the source/drain features **118**P, in accordance with some embodiments. In some embodiments, the trenches **1030**, **1032**, and **1034** are formed to pass through the ILD layer **128**, the substrate **102**, the bottom isolation layers **120**, and portions of the source/drain features **118**P, so as to expose the source/drain features **118**P. In some embodiments, one or more photolithography and etching processes are performed to etch the ILD layer **128**, the substrate **102**, and the bottom isolation layers **120**, and partially etch the source/drain features **118**P, so as to form the trenches **1030**, **1032**, and **1034** that expose the source/drain features **118**P. In some embodiments, the trench **1030** exposes the source/drain feature **118**P**1** of the CFET **100**B, the trench **1032** exposes the source/drain feature **118**P**1** of the CFET **100**A, and the trench **1034** exposes the source/drain feature **118**P**2** of the CFET **100**A, as shown in FIGS. **28**A and **28**B.

[0186] In some embodiments, the remaining portion **1040** of the source/drain feature **118**P**1** of the CFET **100**A is left between the silicide layer **244**P and the trench **1032**. That is, the remaining portion **1040** is left after forming the trench **1032**, and it separates the top surface of the trench **1032** from the bottom surface of the silicide layer **244**P, as shown in FIGS. **28**A and **28**B. In some embodiments, the thickness of the remaining portion **1040** has been configured to make the remaining portion **1040** be completely converted into silicide during the subsequent fabrication stage.

[0187] Referring back to FIGS. **2**A and **2**B, the source/drain contacts **136**, **237**, **138** and the corresponding silicide layers **146**, **247**, **148** are formed on the backside of the workpiece **900**, in accordance with some embodiments. In these embodiments, the resultant device of the workpiece **900** may be fabricated to as the semiconductor device **200** shown in FIGS. **2**A and **2**B. In some embodiments, the silicide layers **146**, **247**, and **148** are formed on the exposed surfaces of the source/drain features **118**P in the trenches **1030**, **1032**, and **1034**, respectively. In some embodiments, the material and method used in forming the silicide layer **247** are the same as or similar to those of the silicide layer **147**, and are not repeated herein.

[0188] In some embodiments, the source/drain contact **237** is in contact with and electrically connected to the source/drain feature **118**P**1** of the CFET **100**A, and the silicide layer **247** is formed on sidewalls and the top surface of a portion of the source/drain contact **137** inside the source/drain feature **118**P**1** of the CFET **100**A, as shown in FIGS. **2**A and **2**B. In some embodiments, the remaining portion **1040** of the source/drain feature **118**P**1** of the CFET **100**A is completely converted into the silicide layer **247** during the silicide process of the silicide layer **247**. That is, after forming the silicide layer **247**, the silicide layer **247** is in direct contact with the silicide layer **244**P without material of the source/drain feature **118**P**1** of the CFET **100**A locating between the silicide layers **247** and **244**P, as shown in FIGS. **2**A and **2**B. In these embodiments, a horizontal portion of the silicide layer **244**P formed on the bottom surface of the source/drain contact **234** is in direct contact with a horizontal portion of the silicide layer **247** formed on the top surface of the source/drain contact **237** in the source/drain feature **118**P**1** of the CFET **100**A.

[0189] In some embodiments, the thickness of the remaining portion **1040** can be controlled by adjusting the depth of the trench **1028** in the Z-direction. In other embodiments, the thickness of the remaining portion **1040** can be controlled by adjusting the depth of the trench **1032** in the Z-direction. In certain embodiments, the thickness of the remaining portion **1040** can be controlled by simultaneously adjusting the depths of the trenches **1028** and **1032** in the Z-direction.

[0190] In some embodiments, the resultant device of the workpiece **900** may be fabricated to as the semiconductor device **300** shown in FIGS. **3**A and **3**B. In these embodiments, in order to form the silicide layer **344**P and the source/drain contact **337** that are in direct contact, the trench **1028** may extend downward more in the Z-direction and/or the trench **1032** may extend upward more in the Z-direction. As a result, the silicide layer **344**P (formed in a manner similar to the silicide layer **244**P) on the source/drain contact **334** formed in the trench **1028** may be exposed in the trench **1032**. In this way, after forming the silicide layer **347** and the source/drain contact **337** in the trench **1032**, since the silicide layer **347** cannot be formed on the silicide layer **344**P, the source/drain contact **337** is in direct contact with the silicide layer **344**P. That is, a horizontal portion of the silicide layer **344**P formed on the bottom surface of the source/drain contact **334** is in direct contact with the top surface of the source/drain contact **337** in the source/drain feature **118**P**1** of the CFET **100**A, as shown in FIGS. **3**A and **3**B.

[0191] In some embodiments, the resultant device of the workpiece **900** may be fabricated to as the semiconductor device **400** shown in FIGS. **4**A and **4**B. In these embodiments, in order to form the source/drain contacts **434** and **437** that are in direct contact, the trench **1028** may extend downward more in the Z-direction and/or the trench **1032** may extend upward more in the Z-direction, and the formation of the trench **1032** may further include removing a portion of the silicide layer **444**P (formed in a manner similar to the silicide layer **244**P) formed on the source/drain contact **434**. As a result, the bottom surface of the source/drain contact **334** formed in the trench **1028** may be exposed in the trench **1032**. In this way, after forming the silicide layer **447** and the source/drain contact **437** in the trench **1032**, since the silicide layer **447** cannot be formed on the source/drain contact **434**, the source/drain contact **437** is in direct contact with the source/drain contact **434**. That is, the bottom surface of the source/drain contact **434** is in direct contact with the top surface of the source/drain contact **437** in the source/drain feature **118**P**1** of the CFET **100**A, as shown in FIGS. **4**A and **4**B.

[0192] Referring to FIGS. **29**A and **29**B, the fabrication stage shown in FIGS. **29**A and **29**B follows the fabrication stage shown in FIGS. **21**A and **21**B. In some embodiments, the source/drain contacts **130**, **534**, and **132** and the corresponding silicide layers **140**, **544**, and **142** are formed on the frontside of the workpiece **900**. In some embodiments, one or more photolithography and etching processes are performed to etch the ILD layer **126**, the CESLs **124**, and the source/drain features **118**N, so as to form the contact openings that exposes the source/drain features **118**N.

[0193] Next, in some embodiments, the silicide layers **140**, **544**, and **142** and the source/drain contacts **130**, **534**, and **132** are formed in the contact openings. In some embodiments, the materials and methods used in forming the silicide layer **544** and the source/drain contact **534** are the same as or similar to those of the silicide layers **140**, **142** and the source/drain contacts **130**, **132**, and are not repeated herein. In some embodiments, the source/drain contact **534** is in contact with and electrically connected to the source/drain feature **118N1** of the CFET **100**A, and the silicide layer **544** is formed on sidewalls and bottom surface of a portion of the source/drain contact **534** inside the source/drain feature **118**P1 of the CFET **100**A, as shown in FIGS. **29**A and **29**B.

[0194] Referring to FIGS. **30**A and **30**B, the ILD layer **128** is formed on backside surface of the substrate **102**, in accordance with some embodiments. The material and method used in forming the ILD layer **128** has been discussed above with reference to FIGS. **25**A and **25**B, and are not repeated herein.

[0195] Still referring to FIGS. **30**A and **30**B, the source/drain contacts **136** and **138** and the corresponding silicide layers **146** and **148** are formed on the backside of the workpiece **900**, in accordance with some embodiments. The materials and methods used in forming the source/drain contacts **136** and **138** and the silicide layers **146** and **148** have been discussed above with reference to FIGS. 1A, 1B, **25**A and **25**B, and are not repeated herein.

[0196] Referring to FIGS. **31**A and **31**B, a trench **1132** is formed on the backside of the workpiece **900** to expose both of the source/drain features **118**N and **118**P, in accordance with some embodiments. In some embodiments, the trench **1132** is formed to pass through the ILD layer **128**, the substrate **102**, the bottom isolation layer **120**, the source/drain feature **118**P, the ILD layer **122**, and portions of the source/drain features **118**N, so as to expose the source/drain features **118**N and **118**P. In some embodiments, one or more photolithography and etching processes are performed to etch the ILD layer **128**, the substrate **102**, the bottom isolation layer **120**, the source/drain feature **118**P, and the ILD layer **122**, and partially etch the source/drain feature **118**N, so as to form the trench **1132** that exposes the source/drain features **118**N and **118**P. In some embodiments, the trench **1132** exposes the source/drain features **118N1** and **118**P1 of the CFET **100**A, as shown in FIGS. **31**A and **31**B.

[0197] Referring back to FIGS. **5**A and **5**B, the source/drain contact **537** and the corresponding silicide layers **547**N and **547**P are formed on the backside of the workpiece **900**, in accordance with some embodiments. In some embodiments, the silicide layers **547**N and **547**P are formed on the exposed surfaces of the source/drain features **118**N and **118**P in the trench **1132**, respectively. In some embodiments, the materials and methods used in forming the silicide layers **547**N and **547**P are the same as or similar to those of the silicide layers **144**N and **144**P, and are not repeated herein. In some embodiments, the material of the silicide layer **547**N is different from the material of the silicide layer **547**P. In some embodiments, the material of the silicide layer **544** is different from the materials of the silicide layers **547**N and **547**P.

[0198] Then, in some embodiments, a conductive material is deposited in the trench **1132** and on the silicide layers **547**N and **547**P by a deposition process, so as to form the source/drain contact **537**. That is, the trench **1132** is filled with the conductive material to form the source/drain contact **537**. In some embodiments, the source/drain contact **537** is in contact with and electrically connected to the source/drain features **118N1** and **118**P1 of the CFET **100**A, as shown in FIGS. **5**A and **5**B. In some embodiments, the silicide layer **547**N is formed on sidewalls and the top surface of a portion of the source/drain contact **537** inside the source/drain feature **118N1** of the CFET **100**A, and the silicide layer **547**P is formed on sidewalls of a portion of the source/drain contact **537** inside the source/drain feature **118**P1 of the CFET **100**A, as shown in FIGS. **5**A and **5**B.

[0199] Referring to FIGS. **32**A and **32**B, the fabrication stage shown in FIGS. **32**A and **32**B follows the fabrication stage shown in FIGS. **30**A and **30**B. For the purpose of clarity, in FIGS. **32**A and **32**B, the source/drain contact **534** and the silicide layer **544** shown in FIGS. **30**A and **30**B are replaced by the source/drain contact **634** and the silicide layer **644**, respectively. In some embodiments, a trench **1232** is formed to pass through the ILD layer **128**, the substrate **102**, the bottom isolation layer **120**, the source/drain feature **118**P, the ILD layer **122**, and portions of the source/drain features **118**N, so as to expose the source/drain features **118**N and **118**P, so as to expose the source/drain features **118**N and **118**P. In some embodiments, one or more photolithography and etching processes are performed to etch the ILD layer **128**, the substrate **102**, the bottom isolation layer **120**, the source/drain feature **118**P, and the ILD layer **122**, and partially etch the source/drain feature **118**N, so as to form the trench **1232** that exposes the source/drain features **118**N and **118**P. In some embodiments, the trench **1232** exposes the source/drain features **118N1** and **118**P1 of the CFET **100**A, as shown in FIGS. **32**A and **32**B. In some embodiments, in the source/drain feature **118N1** of the CFET **100**A, the trench **1232** extends upward more than the trench **1132**. That is, the trench **1232** is deeper than the trench **1132** in the Z-direction.

[0200] In some embodiments, the remaining portion **1240** of the source/drain feature **118N1** of the CFET **100**A is left between the silicide layer **644** and the trench **1232**. That is, the remaining portion **1240** is left after forming the trench **1232**, and it separates the top surface of the trench **1232** from the bottom surface of the silicide layer **644**, as shown in FIGS. **32**A and **32**B. In some embodiments, the thickness of the remaining portion **1240** has been configured to make the remaining portion **1240** be completely converted into silicide during the subsequent fabrication stage.

[0201] Referring back to FIGS. **6**A and **6**B, the source/drain contact **637** and the corresponding silicide layers **647**N and **647**P are formed on the backside of the workpiece **900**, in accordance with some embodiments. In these embodiments, the resultant device of the workpiece **900** may be fabricated to as the semiconductor device **600** shown in FIGS. **6**A and **6**B. In some embodiments, the silicide layers **647**N and **647**P are formed on the exposed surfaces of the source/drain features **118**N and **118**P in the trench **1232**, respectively. In some embodiments, the material and method used in forming the silicide layers **647**N and **647**P are the same as or similar to those of the silicide layers **144**N and **144**P, and are not repeated herein.

[0202] Then, in some embodiments, a conductive material is deposited in the trench **1232** and on the silicide layers **647**N and **647**P by a deposition process, so as to form the source/drain contact **637**. That is, the trench **1232** is filled with the conductive material to form the source/drain contact **637**. In some embodiments, the source/drain contact **637** is in contact with and electrically connected to the source/drain features **118N1** and **118P1** of the CFET **100A**. In some embodiments, the silicide layer **647N** is formed on sidewalls and the top surface of a portion of the source/drain contact **637** inside the source/drain feature **118N1** of the CFET **100A**, and the silicide layer **647P** is formed on sidewalls of a portion of the source/drain contact **637** inside the source/drain feature **118P1** of the CFET **100A**, as shown in FIGS. **6A** and **6B**. In some embodiments, the remaining portion **1240** of the source/drain feature **118N1** of the CFET **100A** is completely converted into the silicide layer **647N** during the silicide process of the silicide layer **647N**. That is, after forming the silicide layer **647N**, the silicide layer **647N** is in direct contact with the silicide layer **644** without material of the source/drain feature **118N1** of the CFET **100A** locating between the silicide layers **647N** and **644**, as shown in FIGS. **6A** and **6B**. In these embodiments, in the source/drain feature **118N1** of the CFET **100A**, a horizontal portion of the silicide layer **644** formed on the bottom surface of the source/drain contact **634** is in direct contact with a horizontal portion of the silicide layer **647N** formed on the top surface of the source/drain contact **637**.

[0203] In some embodiments, the thickness of the remaining portion **1240** can be controlled by adjusting the depth of the trench for forming the source/drain contact **634** in the Z-direction. In other embodiments, the thickness of the remaining portion **1240** can be controlled by adjusting the depth of the trench **1232** in the Z-direction. In certain embodiments, the thickness of the remaining portion **1240** can be controlled by simultaneously adjusting the depths of the trench **1232** and the trench for forming the source/drain contact **634** in the Z-direction.

[0204] In some embodiments, the resultant device of the workpiece **900** may be fabricated to as the semiconductor device **700** shown in FIGS. **7A** and **7B**. In these embodiments, in order to form the silicide layer **744** and the source/drain contact **737** that are in direct contact, the trench for forming the source/drain contact **734** may extend downward more in the Z-direction and/or the trench **1232** may extend upward more in the Z-direction. As a result, the silicide layer **744** (formed in a manner similar to the silicide layer **544**) on the source/drain contact **734** may be exposed in the trench **1232**. In this way, after forming the silicide layers **747N** and **747P** and the source/drain contact **737** in the trench **1232**, since the silicide layer **747N** cannot be formed on the silicide layer **744**, the source/drain contact **737** is in direct contact with the silicide layer **744**. That is, a horizontal portion of the silicide layer **744** formed on the bottom surface of the source/drain contact **734** is in direct contact with the top surface of the source/drain contact **737** in the source/drain feature **118N1** of the CFET **100A**, as shown in FIGS. **7A** and **7B**.

[0205] In some embodiments, the resultant device of the workpiece **900** may be fabricated to as the semiconductor device **800** shown in FIGS. **8A** and **8B**. In these embodiments, in order to form the source/drain contacts **834** and **837** that are in direct contact, the trench for forming the source/drain contact **834** may extend downward more in the Z-direction and/or the trench **1232** may extend upward more in the Z-direction, and the formation of the trench **1232** may further include removing a portion of the silicide layer **844** (formed in a manner similar to the silicide layer **544**) formed on the source/drain contact **834**. As a result, the bottom surface of the source/drain contact **834** may be exposed in the trench **1232**. In this way, after forming the silicide layers **847N** and **847P** and the source/drain contact **837** in the trench **1232**, since the silicide layer **847N** cannot be formed on the source/drain contact **834**, the source/drain contact **837** is in direct contact with the source/drain contact **834**. That is, the bottom surface of the source/drain contact **834** is in direct contact with the top surface of the source/drain contact **837** in the source/drain feature **118N1** of the CFET **100A**, as shown in FIGS. **8A** and **8B**.

[0206] The source/drain contacts **130**, **132**, **134**, **136**, **137**, **138**, **234**, **237**, **334**, **337**, **434**, **437**, **534**, **537**, **634**, **637**, **734**, **737**, **834**, and **837** may each include the conductive material such as Al, Cu, W, Co, Ti, Ta, Ru, Rh, Ir, Pt, Mo, TiN, TiAl, TiAlN, TaN, TaC, combinations of these, or the like, although any suitable material may be deposited using a deposition process such as sputtering, CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, FCVD, electroplating, electroless plating, or the like. In some embodiments, the source/drain contacts **130**, **132**, **134**, **136**, **137**, **138**, **234**, **237**, **334**, **337**, **434**, **437**, **534**, **537**, **634**, **637**, **734**, **737**, **834**, and **837** may each include a single conductive material layer or multiple conductive layers.

[0207] In some embodiments, the source/drain contacts **130**, **132**, **134**, **136**, **137**, **138**, **234**, **237**, **334**, **337**, **434**, **437**, **534**, **537**, **634**, **637**, **734**, **737**, **834**, and **837** may each include a sidewall dielectric layer formed on the sidewalls. The sidewall dielectric layer may be used to improve the isolation between the source/drain contacts and surrounding components. In some embodiments, the material of the sidewall dielectric layer may include silicon nitride (Si_3N_4), silicon oxide (SiO_2), SiC, SiOC, SiON, SiCN, SiOCN, high-k dielectrics, other suitable materials, or combinations thereof. In some embodiments, the sidewall dielectric layer **250** may be deposited by CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, other suitable methods, or combinations thereof.

[0208] In some embodiments, the workpiece **900** may further include a dielectric gate structure to isolate the CFET device (e.g., CFETs **100A** and **100B** discussed above) from other devices. For example, the dielectric gate structure may be formed in a non-functional channel region adjacent to the CFET device. In some embodiments, one dielectric gate structure is formed on right side or left side of the CFET **100A** in the X-direction, alternatively, two dielectric gate structures are formed on opposite sides of the CFET **100A** in the X-direction. In some embodiments, the material of the dielectric gate structure may include silicon nitride (Si_3N_4), silicon oxide (SiO_2), SiC, SiOC, SION, SiCN, SiOCN, high-k dielectrics, other suitable materials, or combinations thereof. In some embodiments, the dielectric gate structure may be deposited by CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, other suitable methods, or combinations thereof.

[0209] The embodiments disclosed herein relate to semiconductor structures and their forming methods, and more particularly to methods and semiconductor structures that include forming a frontside S/D LI contact and a corresponding backside S/D contact in a CFET device. In this way, signals could be transmitted between the frontside S/D LI contact and the backside S/D contact. As a result, the signal routing can be arranged on the frontside, the backside of the device, or on both the frontside and the backside of the device. Therefore, the flexibility for the signal routing is provided. Similarly, alternative embodiments discussed herein include forming a frontside S/D contact and a corresponding backside S/D LI contact in a CFET device. It also provides flexibility for the signal routing. On the other hand, the embodiments provided by the present disclosure are not limited to the signal routing, they can further be applied to any device having a S/D LI contact to provide the flexibility for the routing of the interconnection structure.

[0210] In one exemplary aspect, the present disclosure is directed to a semiconductor structure. The semiconductor structure includes a first transistor and a second transistor vertically overlapping the first transistor. The first transistor includes first nanostructures that are spaced apart from each other in a Z-direction, and a first source/drain feature and a second source/drain feature attached to opposite sides of the first nanostructures in an X-direction. The second transistor includes second nanostructures vertically overlapping the first nanostructures, wherein the second nanostructures are spaced apart from each other in the Z-direction; and a third source/drain feature and a fourth source/drain feature, attached to opposite sides of the second nanostructures in the X-direction and vertically overlapping the first source/drain feature and the second source/drain feature, respectively. The semiconductor structure further includes a gate structure that is wrapped around the first nanostructures and the second nanostructures. The semiconductor structure further includes a first source/drain contact, extending through the third source/drain feature and partially extending into the first source/drain feature; and a second source/drain contact extending into the first source/drain feature.

[0211] In another exemplary aspect, the present disclosure is directed to a semiconductor structure. The semiconductor structure includes a first transistor and a second transistor over the first transistor. The first transistor includes first nanostructures that are over a substrate and spaced apart from each other in a Z-direction, and a first source/drain feature and a second source/drain feature attached to opposite sides of the first nanostructures in an X-direction. The second transistor includes second nanostructures that are over the first nanostructures and spaced apart from each other in the Z-direction; and a third source/drain feature and a fourth source/drain feature, attached to opposite sides of the second nanostructures in the X-direction and over the first source/drain feature and the second source/drain feature, respectively. The semiconductor structure further includes a gate structure that is wrapped around the first nanostructures and the second nanostructures. The semiconductor structure further includes a first source/drain contact partially extending into the third source/drain feature; and a second source/drain contact, extending through the substrate and the first source/drain feature and partially extending into the third source/drain feature.

[0212] In yet another exemplary aspect, the present disclosure is directed to a method of forming a semiconductor structure. The method includes forming a fin structure extending in an X-direction over a substrate, wherein the fin structure includes first semiconductor layers and second semiconductor layers alternately stacked in a Z-direction; forming a dummy gate structure over the fin structure and extending in a Y-direction; and forming a first source/drain feature and a second source/drain feature on opposite sides of the dummy gate structure in the X-direction, wherein the first source/drain feature and the second source/drain feature are attached to a first group of the second semiconductor layers. The method further includes forming a third source/drain feature and a fourth source/drain feature over the first source/drain feature and the second source/drain feature, respectively, wherein the third source/drain feature and the fourth source/drain feature are attached to a second group of the second semiconductor layers; forming a first trench extending through the third source/drain feature and partially extending into the first source/drain feature; and filling the first trench with a first conductive material to form a first source/drain contact. The method further includes forming a second trench extending through the substrate and partially extending into the first source/drain feature; and filling the second trench with a second conductive material to form a second source/drain contact.

[0213] In yet another exemplary aspect, the present disclosure is directed to a semiconductor structure. The semiconductor structure includes a first transistor and a second transistor over the first transistor. The first transistor includes first nanostructures over a substrate, wherein the first nanostructures are spaced apart from each other in a Z-direction, and a first source/drain feature and a second source/drain feature attached to opposite sides of the first nanostructures in an X-direction. The second transistor includes second nanostructures over the first nanostructures, wherein the second nanostructures are spaced apart from each other in the Z-direction; and a third source/drain feature and a fourth source/drain feature, attached to opposite sides of the second nanostructures in the X-direction and over the first source/drain feature and the second source/drain feature, respectively. The semiconductor structure further includes a contact etch stop layer over the third source/drain feature and the fourth source/drain feature, and a first interlayer dielectric (ILD) layer over the contact etch stop layer. The semiconductor structure further includes a first source/drain contact, extending through the first ILD layer, the contact etch stop layer, and the third source/drain feature, and partially extending into the first source/drain feature; and a second source/drain contact, extending through the substrate and partially extending into the first source/drain feature.

[0214] In some embodiments, the semiconductor structure further includes a first silicide layer, formed on an interface between the first source/drain contact and the first source/drain feature; and a second silicide layer, formed on an interface between the second source/drain contact and the first source/drain feature.

[0215] In some embodiments, a horizontal portion of the first silicide layer formed on a bottom surface of the first source/drain contact is in direct contact with a horizontal portion of the second silicide layer formed on a top surface of the second source/drain contact.

[0216] In some embodiments, a horizontal portion of the first silicide layer formed on a bottom surface of the first source/drain contact is in direct contact with a top surface of the second source/drain contact.

[0217] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:

a first transistor, comprising:

first nanostructures, wherein the first nanostructures are spaced apart from each other in a Z-direction; and

a first source/drain feature and a second source/drain feature, attached to opposite sides of the first nanostructures in an X-direction;

a second transistor vertically overlapping the first transistor, wherein the second transistor comprises:

second nanostructures vertically overlapping the first nanostructures, wherein the second nanostructures are spaced apart from each other in the Z-direction; and

a third source/drain feature and a fourth source/drain feature, attached to opposite sides of the second nanostructures in the X-direction and vertically overlapping the first source/drain feature and the second source/drain feature, respectively;

a gate structure wrapped around the first nanostructures and the second nanostructures;

a first source/drain contact, extending through the third source/drain feature and partially extending into the first source/drain feature; and

a second source/drain contact extending into the first source/drain feature.

2. The semiconductor structure of claim **1**, further comprising:

a first interlayer dielectric (ILD) layer, surrounding the first source/drain feature and the second source/drain feature, and separating the first source/drain feature and the second source/drain feature from the third source/drain feature and the fourth source/drain feature,

wherein the first source/drain contact further extends through the first ILD layer.

3. The semiconductor structure of claim **1**, further comprising:

a first silicide layer, formed on an interface between the first source/drain contact and the third source/drain feature; and

a second silicide layer, formed on an interface between the first source/drain contact and the first source/drain feature.

4. The semiconductor structure of claim **3**, further comprising:

a third silicide layer, formed on an interface between the second source/drain contact and the first source/drain feature.

5. The semiconductor structure of claim **4**, wherein a horizontal portion of the second silicide layer formed on a bottom surface of the first source/drain contact is separated from a horizontal portion of the third silicide layer formed on a top surface of the second source/drain contact by a portion of the first source/drain feature.

6. The semiconductor structure of claim **4**, wherein a horizontal portion of the second silicide layer formed on a bottom surface of the first source/drain contact is in direct contact with a horizontal portion of the third silicide layer formed on a top surface of the second source/drain contact.

7. The semiconductor structure of claim **4**, wherein a horizontal portion of the second silicide layer formed on a bottom surface of the first source/drain contact is in direct contact with a top surface of the second source/drain contact.

8. The semiconductor structure of claim **4**, wherein a bottom surface of the first source/drain contact is in direct contact with a top surface of the second source/drain contact.

9. A semiconductor structure, comprising:

a first transistor, comprising:

first nanostructures over a substrate, wherein the first nanostructures are spaced apart from each other in a Z-direction; and

a first source/drain feature and a second source/drain feature, attached to opposite sides of the first nanostructures in an X-direction;

a second transistor over the first transistor, wherein the second transistor comprises:

second nanostructures over the first nanostructures, wherein the second nanostructures are spaced apart from each other in the Z-direction; and

a third source/drain feature and a fourth source/drain feature, attached to opposite sides of the second nanostructures in the X-direction and being over the first source/drain feature and the second source/drain feature, respectively;

a gate structure wrapped around the first nanostructures and the second nanostructures;

a first source/drain contact, partially extending into the third source/drain feature; and

a second source/drain contact, extending through the substrate and the first source/drain feature, and partially extending into the third source/drain feature.

10. The semiconductor structure of claim **9**, further comprising:

a first interlayer dielectric (ILD) layer, formed on a backside of the substrate,

wherein the second source/drain contact further extends through the first ILD layer.

11. The semiconductor structure of claim **9**, further comprising:

a first silicide layer, formed on an interface between the first source/drain contact and the third source/drain feature.

12. The semiconductor structure of claim **11**, further comprising:

a second silicide layer, formed on an interface between the second source/drain contact and the first source/drain feature; and

a third silicide layer, formed on an interface between the second source/drain contact and the third source/drain feature.

13. The semiconductor structure of claim **12**, wherein a horizontal portion of the first silicide layer formed on a bottom surface of the first source/drain contact is separated from a horizontal portion of the third silicide layer formed on a top surface of the second source/drain contact by a portion of the third source/drain feature.

14. The semiconductor structure of claim **12**, wherein a horizontal portion of the first silicide layer formed on a bottom surface of the first source/drain contact is in direct contact with a horizontal portion of the third silicide layer formed on a top surface of the second source/drain contact.

15. The semiconductor structure of claim **12**, wherein a horizontal portion of the first silicide layer formed on a bottom surface of the first source/drain contact is in direct contact with a top surface of the second source/drain contact.

16. The semiconductor structure of claim **12**, wherein a bottom surface of the first source/drain contact is in direct contact with a top surface of the second source/drain contact.

17. A method of forming semiconductor structure, comprising:

forming a fin structure extending in an X-direction over a substrate, wherein the fin structure comprises first semiconductor layers and second semiconductor layers alternately stacked in a Z-direction;

forming a dummy gate structure over the fin structure and extending in a Y-direction;

forming a first source/drain feature and a second source/drain feature on opposite sides of the dummy gate structure in the X-direction, wherein the first source/drain feature and the second source/drain feature are attached to a first group of the second semiconductor layers;

forming a third source/drain feature and a fourth source/drain feature over the first source/drain feature and the second source/drain feature, respectively, wherein the third source/drain feature and the fourth source/drain feature are attached to a second group of the second semiconductor layers;

forming a first trench extending through the third source/drain feature and partially extending into the first source/drain feature;

filling the first trench with a first conductive material to form a first source/drain contact;

forming a second trench extending through the substrate and partially extending into the first source/drain feature; and

filling the second trench with a second conductive material to form a second source/drain contact.

18. The method of claim **17**, further comprising:

before filling the first trench with the first conductive material, forming a first silicide layer on a surface of the first source/drain feature exposed by the first trench; and

before filling the second trench with the second conductive material, forming a second silicide layer on a surface of the third source/drain feature exposed by the first trench.

19. The method of claim **18**, wherein the forming the second trench comprises:

etching through the substrate and partially etching the first source/drain feature to form the second trench,

wherein after forming the second trench, a remaining portion of the first source/drain feature remains and separates a top surface of the second trench from a horizontal portion of the first silicide layer formed on a bottom surface of the first source/drain contact.

20. The method of claim **19**, further comprising:

forming a third silicide layer on a surface of the first source/drain feature exposed by the second trench,

wherein the forming the third silicide layer converts the remaining portion of the first source/drain feature into a horizontal portion of the third silicide layer formed on the top surface of the second trench, such that the horizontal portion of the third silicide layer formed on the top surface of the second trench is in direct contact with the horizontal portion of the first silicide layer formed on the bottom surface of the first source/drain contact.

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